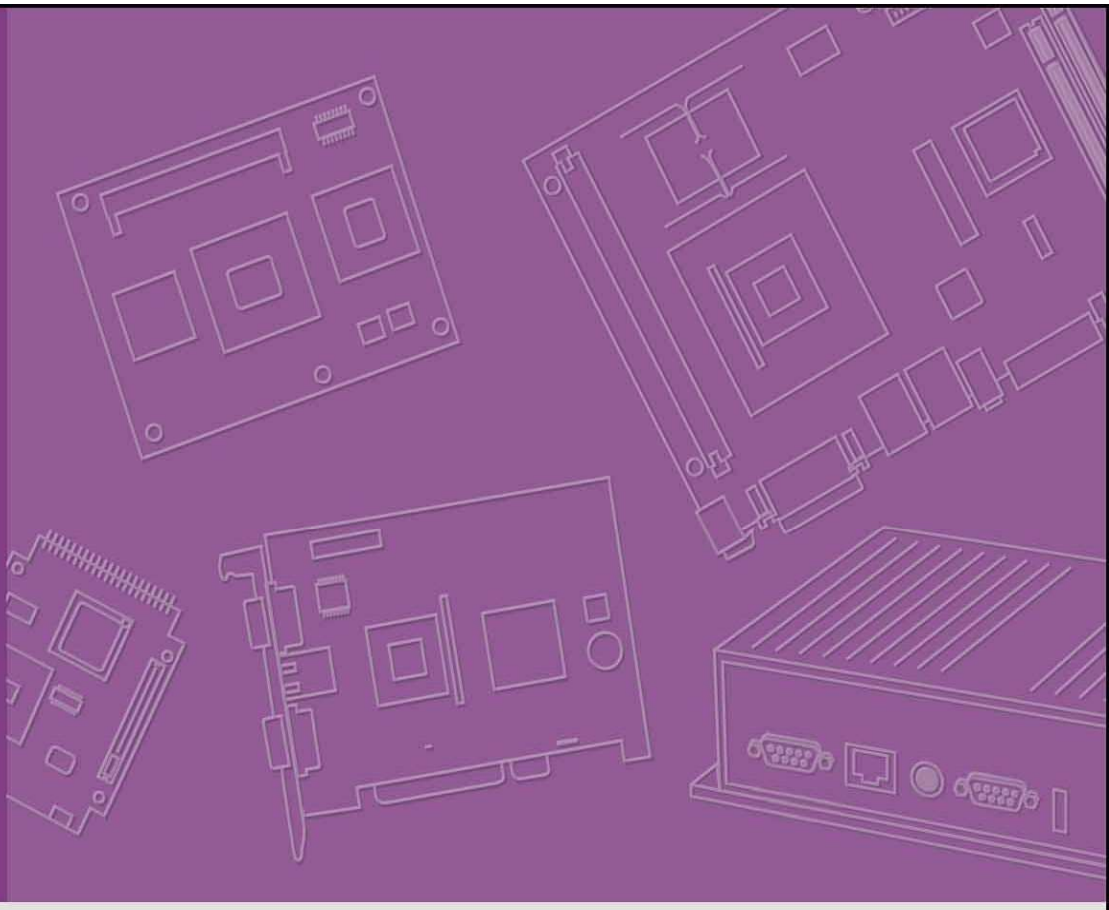
Embedded - IoT



CarrierBoard Design Guide

COM Express®

SOM-5871 COMe R3.0 TYPE6

R110 2022'07'08

ADVANTECH

Enabling an Intelligent Planet

Contents

1 Introduction 8
1.1 About This Document 8
1.2 Signal Table Terminology 8
1.3 Terminology 9
1.4 Reference Documents 13
1.5 Revision History 13
1.6 SOM-5871 Block Diagram 14
2 COM Express Type 6 Interfaces 15
2.1 Module Pin-out Types 6 - Required and Optional Features 15
2.2 COM Express Type 6 Connector Pin-out 17
2.3 PCI Express 25
2.3.1 COM Express A-B Connector and C-D Connector PCIe Groups 25
2.3.2 General Purpose PCIe Signal Definitions 25
2.3.3 PCI Express Lane Configurations – Per COM Express Spec 29
2.3.4 PCI Express* General Routing Guidelines 29
2.3.4.1 PCI Express Insertion Loss Budget with Carrier Board PCIe Device 30
2.3.4.2 PCI Express Differential Transitional Via Recommendations 32
2.3.5 PCI Express* Trace Length Guidelines 33
2.4 PEG (PCI Express Graphics) 36
2.4.1 PEG Signal Definitions 36
2.4.2. PEG* General Routing Guidelines 41
2.4.2.1 PEG Differential Transitional Via Recommendations 42
2.4.3 PEG* Trace Length Guidelines 42
2.5 Digital Display Interfaces (DDI) 45
2.5.1 DDI Signal Definitions 45
2.5.2 Digital Display Interfaces (DDI) Routing Guidelines 51
2.5.2.1 DisplayPort Routing Guidelines 52
2.5.2.2 HDMI / DVI Routing Guidelines 53
2.5.3 HDMI / DVI Level Shifter Requirements 55
2.5.4 ESD Protection 56
2.6 LAN Interface 57
2.6.1 LAN Signal Definitions 57
2.6.2 SDP Pins 60
2.6.3 LAN Routing Guidelines 61

2.6.4 LAN Trace Length Guidelines 6 2
2.6.5 Reference Ground Isolation and Coupling 6 3
2.7 USB2.0 Ports 6 4
2.7.1 USB2.0 Signal Definitions 6 4
2.7.1.1 USB Over-Current Protection (USB_x_y_OC#) 6 7
2.7.1.2 Powering USB devices during S5 ..... 6 8
2.7.2 USB2.0 Routing Guidelines 6 8
2.7.2.1 USB 2.0 General Design Considerations and Optimization 6 9
2.7.2.2 USB 2.0 Port Power Delivery 7 0
2.7.2.3 USB 2.0 Common Mode Chokes 7 1
2.7.2.4 EMI / ESD Protection 7 2
2.7.2.5 Depicts the schematic and layout requirements for the USB 2.0 / 1.1 interface 7 3
2.7.3 USB2.0 Trace Length Guidelines 7 4
2.8 USB3.0/USB3.1 Ports 7 5
2.8.1 USB3.0/USB3.1 Signal Definitions 7 5
2.8.1.1 USB Over-Current Protection (USB_x_y_OC#) 7 7
2.8.1.2 EMI / ESD Protection 7 8
2.8.2 USB3.0/USB3.1 Routing Guidelines 7 8
2.8.2.1 USB3.0/USB3.1 Differential Transitional Via Recommendations 7 9
2.8.2.2 USB3.1 General Guidelines 7 9
2.8.3 USB3.0/USB3.1 Trace Length Guidelines 8 1
2.9 SATA 8 2
2.9.1 SATA Signal Definitions 8 2
2.9.2 SATA Routing Guidelines 8 4
2.9.2.1 General SATA Routing Guidelines 8 5
2.9.2.2 SATA Differential Transitional Via Recommendations 8 5
2.9.3 SATA Trace Length Guidelines 8 6
2.10 LVDS 8 7
2.10.1 Signal Definitions 8 7
2.10.1.1 Display Timing Configuration 9 0
2.10.1.2 Backlight Control 9 0
2.10.2 LVDS Routing Guidelines 9 1
2.10.3 LVDS Trace Length Guidelines 9 1
2.11 Embedded DisplayPort (eDP) *SOM-5871 is Option function 9 2
2.11.1 eDP Signal Definitions 9 3
2.11.2 eDP Trace Length Guidelines 9 4

2.12 VGA 95
2.12.1 VGA Signal Definitions 95
2.12.2 VGA Routing Guidelines 97
2.12.2.1 RGB Analog Signals 97
2.12.2.2 HSYNC and VSYNC Signals 98
2.12.2.3 DDC Interface 98
2.12.2.4 ESD Protection/EMI 98
2.12.3 VGA Trace Length Guidelines 98
2.13 Digital Audio Interfaces 99
2.13.1 Audio Codec Signal Descriptions 100
2.13.2 Audio Routing Guidelines 101
2.13.3 Audio Trace Length Guidelines 102
2.14 LPC and eSPI Interface *SOM-5871 is not support eSPI. 103
2.14.1 LPC /eSPI Signal Definition 104
2.14.2 LPC Routing Guidelines 106
2.14.2.1 General Signals 106
2.14.2.2 Bus Clock Routing 107
2.14.2.3 Carrier Board LPC Devices 107
2.14.2.4 eSPI Devices *SOM-5871 is not support eSPI Interface 107
2.14.3 LPC Trace Length Guidelines 108
2.15 SPI – Serial Peripheral Interface Bus 109
2.15.1 SPI Signal Definition 110
2.15.2 BIOS Boot Selection 112
2.15.3 SPI Routing Guidelines 116
2.15.4 SPI Trace Length Guidelines 116
2.16 General Purpose I2C Bus Interface 117
2.16.1 Signal Definitions 118
2.16.2 I2C Routing Guidelines 119
2.16.3 I2C Trace Length Guidelines 119
2.16.4 Connectivity Considerations 120
2.17 System Management Bus (SMBus) 120
2.17.1 SMB Signal Definitions 121
2.17.2 SMB Routing Guidelines 122
2.17.3 SMB Trace Length Guidelines 123
2.18 General Purpose Serial Interface 123
2.18.1 Serial interface Signal Definitions 124
2.18.2 Serial interface Routing Guidelines 125

2.18.3 Serial interface Trace Length Guidelines 125
2.19 CAN Interface 126
2.19.1 CAN interface Signal Definitions 127
2.19.2 CAN interface Routing Guidelines 128
2.19.3 CAN interface Trace Length Guidelines 128
2.20 Miscellaneous Signals 129
2.20.1 Miscellaneous Signals 129
2.20.1.1 Logic Level Signals on Pins Reclaimed from VCC_12 132
2.20.2 Power Management Signals 133
2.20.3 Thermal Protection Interface 137
2.20.4 Miscellaneous Signals Routing Guidelines 137
2.20.5 Miscellaneous Signals Trace Length Guidelines 137
2.21 Rapid Shutdown 138
2.21.1 Rapid Shutdown Signal 138
2.21.2 Rapid Shutdown Signal Routing Guidelines 138
2.21.3 Rapid Shutdown Signal Trace Length Guidelines 138
2.22 General Purpose Input Output (GPIO) 139
2.22.1 GPIO Signals 139
2.22.2 Rapid Shutdown Signal Routing Guidelines 139
2.22.3 Rapid Shutdown Signal Trace Length Guidelines 139
2.23 SDIO *SOM-5871 is not support SDIO Interface 139
2.23.1 SDIO Signals 140
2.23.2 SDIO Topology Guidelines 141
2.23.3 SDIO Signals Trace Length Guidelines 142
3 Power 143
3.1 General Power requirements 143
3.2 ATX and AT Power Sequencing Diagrams 143
3.3 Design Considerations for Carrier Boards containing FPGAs/CPLDs 146
4 Electrical Characteristics 147
4.1 Absolute Maximum Ratings 147
4.2 DC Characteristics 148
4.3 Inrush Current 148

List of Tables

Table 1: Signal Table Terminology Descriptions 8
Table 2: Conventions and Terminology 9
Table 3: Module Pin-out - Required and Optional Features 15
Table 4: COM Express Type6 Pin-out 17
Table 5: General Purpose PCI Express Signal Descriptions 25
Table 6: PCI Express Insertion Loss Budget, 1.25 GHz with Carrier Board PCIe Device 30
Table 7: PCI Express Insertion Loss Budget, 2.5 GHz with Carrier Board PCIe Device 30
Table 8: Differential Transitional Via Layout Recommendations 32
Table 9: PCI Express Gen2 * Slot Card / Device Down Trace Length Guidelines 34
Table 10: PCI Express Gen3 * Slot Card / Device Down Trace Length Guidelines 35
Table 11: PEG Signal Description 36
Table 12: PCI Express* Slot Card / Device Down Trace Length Guidelines 43
Table 13: Port / HDMI / DVI Pin-out of Type 6 45
Table 14: DDI1 Signal Description 46
Table 15: DDI2 Signal Description 47
Table 16: DDI3 Signal Description 49
Table 17: DisplayPort Insertion Loss Budget 52
Table 18: DisplayPort Connector / Device Down Trace Length Guidelines 53
Table 19: HDMI / DVI (TMDS) Insertion Loss Budget 54
Table 20: HDMI / DVI Connector / Device Down Trace Length Guidelines 55
Table 21: LAN Interface Signal Descriptions 57
Table 22: The MDI can operate in 1000, 100 and 10 Mbit / sec modes 59
Table 23: 10/100/1000 Ethernet Insertion Loss Budget, 100 MHz 61
Table 24: Ethernet Trace Length Guidelines 62
Table 25: USB Signal Descriptions 64
Table 26: USB2.0 Insertion Loss Budget, 400 MHz 69
Table 27: USB2.0 Trace Length Guidelines 74
Table 28: USB3.0/USB3.1 Signal Definitions 75
Table 29: USB3.0 Insertion Loss Budget 79
Table 30: USB3.0 Trace Length Guidelines 81
Table 31: SATA Signal Definitions 82
Table 32: SATA Insertion Loss Budge 84
Table 33: SATA Trace Length Guidelines 86

Table 34: LVDS Signal Definitions 87
Table 35: LVDS Trace Length Guidelines 92
Table 36: Edp Signal Definitions 93
Table 37: DisplayPort Connector / Device Down Trace Length Guidelines 95
Table 38: VGA Signal Definitions 96
Table 39: VGA Trace Length Guidelines 99
Table 40: Audio Codec Signal Descriptions 100
Table 41: Audio Trace Length Guidelines 103
Table 42: LPC/eSPI Interface Signal Definition 104
Table 43: LPC Trace Length Guidelines 109
Table 44: SPI Interface Signal Definition 110
Table 45: BIOS Selection Straps 115
Table 46: SPI Trace Length Guidelines 117
Table 47: General Purpose I2C Interface Signal Descriptions 118
Table 48: I2C Trace Length Guidelines 120
Table 49: SMB Signal Definitions 121
Table 50: SMB Trace Length Guidelines 123
Table 51: Serial interface Signal Definitions 124
Table 52: Serial interface Trace Length Guidelines 126
Table 53: CAN interface Signal Definitions 127
Table 54: CAN interface Trace Length Guidelines 129
Table 55: Miscellaneous Signal Definitions 129
Table 56: Power Management Signal Definitions 133
Table 57: Thermal Management Signal Definitions 137
Table 58: Rapid Shutdown Signal 138
Table 59: GPIO Signals, Pin Types, and Descriptions 139
Table 60: SDIO Signals, Pin Types, and Descriptions 140
Table 61: SDIO Trace Length Guidelines 142
Table 62: Power Management Timings 145
Table 63: Absolute Maximum Ratings 147
Table 64: Input Power Consumption of 8.5V / 12V / 20V 148
Table 65: Inrush Current 148

List of Figures

Figure 1: SOM-5871 Block Diagram 14
Figure 2: PCI Express Insertion Loss Budget with Carrier Board PCIe Device 30
Figure 3: Differential Transitional Via Layout 32
Figure 4: Topology for PCI Express Slot Card. 33
Figure 5: Topology for PCI Express Device Down. 33
Figure 6: Topology for PCI Express Slot Card. 42
Figure 7: Topology for PCI Express Device Down. 43
Figure 8: DisplayPort Loss Budget 52
Figure 9: Topology for DisplayPort 52
Figure 10: HDMI / DVI (TMDS) Loss Budget 54
Figure 11: Topology for HDMI / DVI 54
Figure 12: 10/100/1000 Ethernet Insertion Loss Budget 61
Figure 13: Topology for Ethernet Jack 62
Figure 14: USB2.0 Insertion Loss Budget 69
Figure 15: USB 2.0 Good Downstream Power Connection 70
Figure 16: USB 2.0 A Common Mode Choke 71
Figure 17: USB 2.0/1.1 Interface—Schematic and Routing Model 73
Figure 18: Topology for USB2.0 74
Figure 19: USB3.0 Insertion Loss Budget 79
Figure 20: Topology for USB3.0/USB3.1 81
Figure 21: SATA Insertion Loss Budge 84
Figure 22: Topology for SATA 86
Figure 23: Topology for LVDS 91
Figure 24: Topology for eDP 94
Figure 25: Topology for VGA 98
Figure 26: Topology for Audio 102
Figure 27: Typical Routing Topology for a Module LPC Device 107
Figure 28: Topology for LPC 108
Figure 29: BIOS Selection LPC Mode 113
Figure 30: BIOS Selection eSPI Mode 114
Figure 31: Topology for SPI 116
Figure 32: Topology for I2C 119
Figure 33: Topology for SMB 123

Figure 34: Topology for Serial interface 1 2 5
Figure 35: Topology for CAN interface 1 2 8
Figure 36: Protecting Logic Level Signals on Pins Reclaimed from VCC_12V 1 3 3
Figure 37: Topology for SDIO 1 4 2
Figure 38: ATX Style Power Up Boot – Controlled by Power Button 1 4 3
Figure 39: AT Style Power Up Boot 1 4 4

1 Introduction

1.1 About This Document

This design guide provides information for designing a custom system Carrier Board for COM Express Type 6 Module. It includes Signal Descriptions, Routing Guidelines and Trace Length Guidelines. The main purpose is designing Carrier Board for helping customers fast and easy using the module of Advantech to be designed.

1.2 Signal Table Terminology

Table 1 below describes the terminology used in this section for the Signal Description tables.

The "#" symbol at the end of the signal name indicates that the active or asserted state occurs when the signal is at a low voltage level. When "#" is not present, the signal is asserted when at a high voltage level.

The terms "Input" and "Output" and their abbreviations in Table 1 below refer to the Module's view, i.e. an input is an input for the Module and not for the Carrier-Board.

Table 1: Signal Table Terminology Descriptions

Term	*Description*
I/O 3.3V	Bi-directional signal 3.3V tolerant
I/O 5V	Bi-directional signal 5V tolerant
I 3.3V	Input 3.3V tolerant
I 5V	Input 5V tolerant
I/O 3V3_SBY	Bi-directional 3.3V tolerant active during Suspend and running state.
O 3.3V	Output 3.3V signal level
O 5V	Output 5V signal level
OD	Open drain output
P	Power input/output
*_S0	Signal active during running state.
PCIE	In compliance with PCI Express Base Specification
USB	In compliance with the Universal Serial Bus Specification
GbE	In compliance with IEEE 802.3ab 1000BASE-T Gigabit Ethernet
SATA	In compliance with Serial ATA specification
REF	Reference voltage output. May be sourced from a Module power plane.
PDS	Pull-down strap. A Module output pin that is either tied to GND or is not connected. Used to signal Module capabilities (pin-out type) to the Carrier Board.

1.3 Terminology

Table 2: Conventions and Terminology

Terminology	*Description*
10GBASE-KR	10 Gbit internal copper interface. Operates over a single lane and uses the same physical layer coding (defined in IEEE 802.3 Clause 49) as 10GBASE-LR (Single Mode Fiber 1310 nm) /ER (Single Mode Fiber 1550 nm) /SR (Multi Mode Fiber 850 nm)
ACPI	Advanced Configuration Power Interface – standard to implement power saving modes in PCAT systems
Basic Module	COM Express 125mm x 95mm Module form factor.
BIOS	Basic Input Output System – firmware in PC-AT system that is used to initialize system components before handing control over to the operating system.
CAN	Controller-area network (CAN or CAN-bus) is a vehicle bus standard designed to allow microcontrollers to communicate with each other within a vehicle without a host computer.
Carrier Board	An application specific circuit board that accepts a COM Express Module.
Compact Module	COM Express 95mm x 95mm Module form factor
CVBS	Composite Video Baseband Signal
DDC	Display Data Control – VESA (Video Electronics Standards Association) standard to allow identification of the capabilities of a VGA monitor
DDI	Digital Display Interface– containing DisplayPort, HDMI/DVI and SDVO
DIMM	Dual In-line Memory Module
DisplayPort DP	DisplayPort is a digital display interface standard put forth by the Video Electronics Standards Association (VESA). It defines a new license free, royalty free, digital audio/video interconnect, intended to be used primarily between a computer and its display monitor.
DRAM	Dynamic Random Access Memory
DP	DisplayPort is a digital display interface standard put forth by the Video Electronics Standards Association (VESA). It defines a new license free, royalty free, digital audio / videointer connect, intended to be used primarily between a computer and its display monitor.
DVI	Digital Visual Interface - a Digital Display Working Group (DDWG) standard that defines a standard video interface supporting both digital and analog video signals. The digital signals use TMDS.

Terminology	*Description*
EAPI	Embedded Application Programming Interface Software interface for COM Express specific industrial functions • System information • Watchdog timer • I2C Bus • Flat Panel brightness control • User storage area • GPIO
EDID	Extended Display Identification Data
eDP	Embedded DisplayPort
EEPROM	Electrically Erasable Programmable Read-Only Memory
eSPI	Enhanced Serial Peripheral Interface
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge
Embedded DisplayPort eDP	Embedded Display Port (eDP) is a digital display interface standard produced by the Video Electronics Standards Association (VESA) for digital interconnect of Audio and Video.
Extended Module	COM ExpressR 155mm x 110mm Module form factor.
FR4	A type of fiber-glass laminate commonly used for printed circuit boards.
Gb	Gigabit
GbE	Gigabit Ethernet
GPI	General Purpose Input
GPIO	General Purpose Input Output
GPO	General Purpose Output
HDA	Intel High Definition Audio (HD Audio) refers to the specification released by Intel in 2004 for delivering high definition audio.
HDMI	High Definition Multimedia Interface
I2C	Inter Integrated Circuit – 2 wire (clock and data) signaling scheme allowing communication between integrated circuits, primarily used to read and load register values.
DE	Integrated Device Electronics – parallel interface for hard disk drives – also known as PATA
Legacy Device	Relics from the PC-AT computer that are not in use in contemporary PC systems: primarily the ISA bus, UART-based serial ports, parallel printer ports, PS-2 keyboards, and mice. Definitions vary as to what constitutes a legacy device. Some definitions include IDE as a legacy device.

Terminology	*Description*
LAN	Local Area Network
LPC	Low Pin-Count Interface: a low speed interface used for peripheral circuits such as Super I/O controllers, which typically combine legacy-device support into a single IC.
LS	Least Significant
LVDS	Low-Voltage Differential Signaling – widely used as a physical interface for TFT flat panels. LVDS can be used for many high-speed signaling applications. In this document, it refers only to TFT flat-panel applications.
MAFS	Term for Master Attached Flash Sharing where the Flash component is attached to the processor interface.
MDIO	Management Data Input/Output, or MDIO, is a 2-wire serial bus that is used to manage PHYs or physical layer devices in media access controllers (MACs).
ME	Management Engine
Mini Module	COM ExpressR 84x55mm Module form factor
MS	Most Significant
NA	Not available
NC	Not connected
OEM	Original Equipment Manufacturer
PATA	Parallel AT Attachment – parallel interface standard for hard-disk drives – also known as IDE,AT Attachment, and as ATA
PC-AT	"Personal Computer – Advanced Technology" – an IBM trademark term used to refer to Intel x86 based personal computers in the 1990s
PCB	Printed Circuit Board
PCI Express (PCIe)	Peripheral Component Interface Express – next-generation high speed Serialized I/O bus
PCI Express Lane	One PCI Express Lane is a set of 4 signals that contains two differential lines for Transmitter and two differential lines for Receiver. Clocking information is embedded into the data stream.
PD	Pull Down
PEG	PCI Express Graphics
PHY	Ethernet controller physical layer device
Pin-out Type	A reference to one of seven COM ExpressR definitions for the signals that appear on the COM ExpressR Module connector pins.

Terminology	*Description*
PU	Pull Up
ROM	Read Only Memory – a legacy term – often the device referred to as a ROM can actually be written to, in a special mode. Such writable ROMs are sometimes called Flash ROMs. BIOS is stored in ROM or Flash ROM.
RTC	Real Time Clock – battery backed circuit in PC-AT systems that keeps system time and dateas well as certain system setup parameters
S0, S1, S2, S3, S4, S5	Sleep States defined by the ACPI specificationS0 Full power, all devices powered S1Sleep State, all context maintained S2 Sleep State, CPU and Cache context lost S3 Suspend to RAM System context stored in RAM; RAM is in standby S4 Suspend to Disk System context stored on disk S5 Soft Off Main power rail off, only standby power rail present
SATA	Serial AT Attachment: serial-interface standard for hard disks
SDVO	Serial Digital Video Out is a proprietary technology introduced by IntelR to add additional video signaling interfaces to a system. Being phased out
SMBus	System Management Bus
SO-DIMM	Small Outline Dual In-line Memory Module
SPI	Serial Peripheral Interface
TBD	To be determined
TMDS	Transition Minimized Differential Signaling - a digital signaling protocol between the graphics subsystem and display. TMDS is used for the DVI digital signals. DC coupled
TPM	Trusted Platform Module, chip to enhance the security features of a computer system.
UIM	User Identity Module
USB	Universal Serial Bus
VESA	Video Electronics Standards Association
WDT	Watch Dog Timer

1.4 Reference Documents

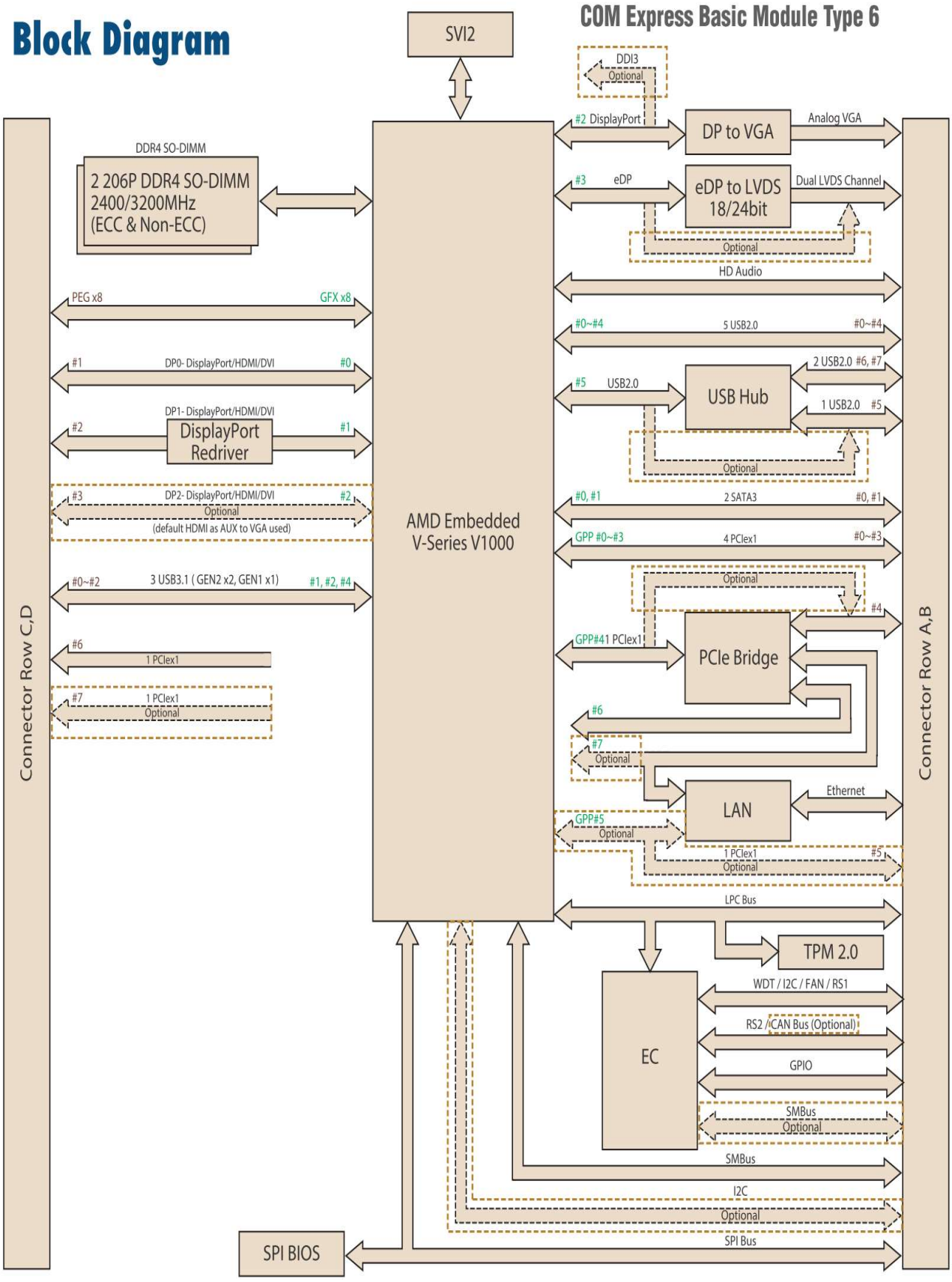
Document
PICMGR COM.0 Revision 3.0 COM Express Base Specification, 2017'03'31 Final
AMD Document
AMD Layout Guide Document
ATX12V Power Supply Design Guide Rev. 2.01

1.5 Revision History

Revision	Date	PCB Rev.	Changes
0.10	09 21, 2018	A101-3	1. 2.7.2.5 Depicts the schematic and layout requirements for the USB 2.0 / 1.1 interface
1.00	02 22,2019	A101-3	Release
1.10	07 08,2022	A101-3	Add power consumption

1.6 SOM-5871 Block Diagram

Figure 1: SOM-5871 Block Diagram



2 COM Express Type 6 Interfaces

2.1 Module Pin-out Types 6 - Required and Optional Features

COM Express Required and Optional features are summarized in the following table. The features identified as Minimum (Min.) ***shall*** be implemented by all Modules. Features identified up to Maximum (Max) ***may*** be additionally implemented by a Module.

Table 3: Module Pin-out - Required and Optional Features

Feature	**Type 6 Min / Max**	SOM-5871
System I/O		
PCI Express Lanes 0 - 5	1 / 6	6
PCI Express Lanes 6 - 15	0 / 2	2
PCI Express Lanes 16 - 31	0 / 16	0
PCI Express Graphics (PEG)	0 / 1	1, Only support X8
1Gb LAN Port 0	1 / 1	1
DDIs 1 - 3	0 / 3	3
LVDS Channel A	0 / 1	1
LVDS Channel B	0 / 1	1
eDP on LVDS CH A pins	0 / 1	1
VGA Port	0 / 1	1
Serial Ports 1 - 2	0 / 2	2
CAN interface on SER1	0 / 1	1
SATA Ports	1 / 4	2
HDA Digital Interface	0 / 1	1
USB 2.0 Ports	4 / 8	8
USB0 Client	0 / 1	0
USB 3.0 Ports	0 / 4	3
LPC Bus or eSPI	1 / 1	1
SPI (Devices)	1 / 2	1
Rapid Shutdown	0 / 1	1
System Management		
SDIO (muxed on GPIO)	0 / 1	1
General Purpose I/O	8 / 8	8
SMBus	1 / 1	1
I2C	1 / 1	1
Watchdog Timer	0 / 1	1
Speaker Out	1 / 1	1
Carrier Board BIOS Flash Support	0 / 1	1
Reset Functions	1 / 1	1
Trusted Platform Module (TPM_PP)	0 / 1	1

Feature	Type 6 Min / Max	SOM-5871
Power Management		
Thermal Protection	0 / 1	1
Battery Low Alarm	0 / 1	1
Suspend/Wake Signals	0 / 3	2
Power Button Support	1 / 1	1
Power Good	1 / 1	1
VCC_5V_SBY Contacts	4 / 4	4
Sleep Input	0 / 1	1
Lid Input	0 / 1	1
Carrier Board Fan Control	0 / 1	1
Power		
VCC_12V Contacts	24 / 24	24

2.2 COM Express Type 6 Connector Pin-out

Table 4: COM Express Type6 Pin-out

Connector Rows A and B

Pin#	Type 6 Description	SOM-5871 Difference	Pin#	Type 6 Description	SOM-5871 Difference
A1	GND(FIXED)		B1	GND(FIXED)	
A2	GBE0_MDI3-		B2	GBE0_ACT#	
A3	GBE0_MDI3+		B3	LPC_FRAME# / ESPI_CS0#	LPC_FRAME#
A4	GBE0_LINK100#		B4	LPC_AD0 / ESPI_IO_0	LPC_AD0
A5	GBE0_LINK1000#		B5	LPC_AD1 / ESPI_ IO_1	LPC_AD1
A6	GBE0_MDI2-		B6	LPC_AD2 / ESPI_ IO_2	LPC_AD2
A7	GBE0_MDI2+		B7	LPC_AD3 / ESPI_ IO_3	LPC_AD3
A8	GBE0_LINK#		B8	LPC_DRQ0#/ESPI_ALERT0#	NC
A9	GBE0_MDI1-		B9	LPC_DRQ1#/ESPI_ALERT1#	NC
A10	GBE0_MDI1+		B10	LPC_CLK/ESPI_CK	LPC_CLK
A11	GND(FIXED)		B11	GND(FIXED)	
A12	GBE0_MDI0-		B12	PWRBTN#	
A13	GBE0_MDI0+		B13	SMB_CK	
A14	GBE0_CTREF	NC	B14	SMB_DAT	
A15	SUS_S3#		B15	SMB_ALERT#	
A16	SATA0_TX+		B16	SATA1_TX+	
A17	SATA0_TX		B17	SATA1_TX	
A18	SUS_S4#	SUS_S5#	B18	SUS_STAT#/ESPI_RESET#	SUS_STAT#
A19	SATA0_RX+		B19	SATA1_RX+	
A20	SATA0_RX		B20	SATA1_RX-	
A21	GND(FIXED)		B21	GND(FIXED)	
A22	SATA2_TX+	NC	B22	SATA3_TX+	NC
A23	SATA2_TX--	NC	B23	SATA3_TX--	NC
A24	SUS_S5#		B24	PWR_OK	
A25	SATA2_RX+	NC	B25	SATA3_RX+	NC
A26	SATA2_RX-	NC	B26	SATA3_RX-	NC
A27	BATLOW#		B27	WDT	
A28	(S)ATA_ACT#		B28	HDA_SDIN2	
A29	HDA_SYNC		B29	HDA_SDIN1	
A30	HDA_RST#		B30	HDA_SDIN1	
A31	GND(FIXED)		B31	GND(FIXED)	
A32	HDA_BITCLK		B32	SPKR	

Connector Rows A and B

Pin#	Type 6 Description	SOM-5871 Difference	Pin#	Type 6 Description	SOM-5871 Difference
A33	HDA_SDOUT		B33	I2C_CK	
A34	BIOS_DIS0#/ESPI_SAFS	BIOS_DIS0#	B34	I2C_DAT	
A35	THRMTRIP#		B35	THRM#	
A36	USB6-		B36	USB7-	
A37	USB6+		B37	USB7+	
A38	USB_6_7_OC#		B38	USB_4_5_OC#	
A39	USB4-		B39	USB5-	
A40	USB4+		B40	USB5+	
A41	GND(FIXED)		B41	GND(FIXED)	
A42	USB2-		B42	USB3-	
A43	USB2+		B43	USB3+	
A44	USB_2_3_OC#		B44	USB_0_1_OC#	
A45	USB0-		B45	USB1-	
A46	USB0+		B46	USB1+	
A47	VCC_RTC		B47	ESPI_EN#	NC
A48	RSVD		B48	USB0_HOST_PRSNT	NC
A49	GBE0_SDP		B49	SYS_RESET#	
A50	LPC_SERIRQ/ESPI_CS1#	LPC_SERIRQ	B50	CB_RESET#	
A51	GND(FIXED)		B51	GND(FIXED)	
A52	PCIE_TX5+	PCIE is optional function	B52	PCIE_RX5+	PCIE is optional function
A53	PCIE_TX5-	PCIE is optional function	B53	PCIE_RX5-	PCIE is optional function
A54	GPI0 / SDIO_DAT0	GPIO	B54	GPO1 / SDIO_CMD	GPIO
A55	PCIE_TX4+		B55	PCIE_RX4+	
A56	PCIE_TX4-		B56	PCIE_RX4-	
A57	GND		B57	GPO2/SDIO_WP	GPIO.
A58	PCIE_TX3+		B58	PCIE_RX3+	
A59	PCIE_TX3-		B59	PCIE_RX3-	
A60	GND(FIXED)		B60	GND(FIXED)	
A61	PCIE_TX2+		B61	PCIE_RX2+	
A62	PCIE_TX2-		B62	PCIE_RX2-	
A63	GPI1 / SDIO_DAT1	GPIO	B63	GPO3 / SDIO_CD#	GPIO
A64	PCIE_TX1+		B64	PCIE_RX1+	
A65	PCIE_TX1-		B65	PCIE_RX1-	
A66	GND		B66	WAKE0#	

Connector Rows A and B

Pin#	Type 6 Description	SOM-5871 Difference	Pin#	Type 6 Description	SOM-5871 Difference
A67	GPI2 / SDIO_DAT2	GPIO	B67	WAKE1#	
A68	PCIE_TX0+		B68	PCIE_RX0+	
A69	PCIE_TX0-		B69	PCIE_RX0-	
A70	GND(FIXED)		B70	GND(FIXED)	
A71	LVDS_A0+ / eDP_TX2+	eDP is option function.	B71	LVDS_B0+	
A72	LVDS_A0- / eDP_TX2-	eDP is option function.	B72	LVDS_B0-	
A73	LVDS_A1+ / eDP_TX1+	eDP is option function.	B73	LVDS_B1+	
A74	LVDS_A1- / eDP_TX1-	eDP is option function.	B74	LVDS_B1-	
A75	LVDS_A2+ / eDP_TX0+	eDP is option function.	B75	LVDS_B2+	
A76	LVDS_A2- / eDP_TX01	eDP is option function.	B76	LVDS_B2-	
A77	LVDS_VDD_EN/ eDP_VDD_EN	eDP is option function.	B77	LVDS_B3+	
A78	LVDS_A3+		B78	LVDS_B3-	
A79	LVDS_A3-		B79	LVDS_BKLT_EN/ eDP_BKLT_EN	eDP is option function.
A80	GND(FIXED)		B80	GND(FIXED)	
A81	LVDS_A_CK+/ eDP_TX3+	eDP is option function.	B81	LVDS_B_CK+	
A82	LVDS_A_CK-/ eDP_TX3-	eDP is option function.	B82	LVDS_B_CK-	
A83	LVDS_I2C_CK/ eDP_AUX+	eDP is option function.	B83	LVDS_BKLT_CTRL/ eDP_BKLT_CTRL	eDP is option function.
A84	LVDS_I2C_DAT/ eDP_AUX-	eDP is option function.	B84	VCC_5V_SBY	
A85	GPI3 / SDIO_DAT3	GPIO	B85	VCC_5V_SBY	
A86	RSVD	NC	B86	VCC_5V_SBY	
A87	eDP_HPD	eDP is option function.	B87	VCC_5V_SBY	
A88	PCIE_CLK_REF+		B88	BIOS_DIS1#	
A89	PCIE_CLK_REF-		B89	VGA_RED	
A90	GND(FIXED)		B90	GND(FIXED)	
A91	SPI_POWER		B91	VGA_GRN	
A92	SPI_MISO		B92	VGA_BLU	
A93	GPO0 / SDIO_CLK	GPIO	B93	VGA_HSYNC	

Connector Rows A and B

Pin#	Type 6 Description	SOM-5871 Difference	Pin#	Type 6 Description	SOM-5871 Difference
A94	SPI_CLK		B94	VGA_VSYNC	
A95	SPI_MOSI		B95	VGA_I2C_CK	
A96	TPM_PP		B96	VGA_I2C_DAT	
A97	TYPE10#	NC	B97	SPI_CS#	
A98	SER0_TX		B98	RSVD	
A99	SER0_RX		B99	RSVD	
A100	GND(FIXED)		B100	GND(FIXED)	
A101	SER1_TX/CAN_TX	CAN bus is option function.	B101	FAN_PWMOUT	
A102	SER1_RX/CAN_RX	CAN bus is option function	B102	FAN_TACHIN	
A103	LID#		B103	SLEEP#	
A104	VCC_12V		B104	VCC_12V	
A105	VCC_12V		B105	VCC_12V	
A106	VCC_12V		B106	VCC_12V	
A107	VCC_12V		B107	VCC_12V	
A108	VCC_12V		B108	VCC_12V	
A109	VCC_12V		B109	VCC_12V	
A110	GND(FIXED)		B110	GND(FIXED)	

Connector Rows C and D

Pin#	Type 6 Description	SOM-5871 Difference	Pin#	Type 6 Description	SOM-5871 Difference
C1	GND(FIXED)		D1	GND(FIXED)	
C2	GND		D2	GND	
C3	USB_SSRX0-		D3	USB_SSTX0-	
C4	USB_SSRX0+		D4	USB_SSTX0+	
C5	GND		D5	GND	
C6	USB_SSRX1-		D6	USB_SSTX1-	
C7	USB_SSRX1+		D7	USB_SSTX1+	
C8	GND		D8	GND	
C9	USB_SSRX2-		D9	USB_SSTX2-	
C10	USB_SSRX2+		D10	USB_SSTX2+	
C11	GND(FIXED)		D11	GND(FIXED)	
C12	USB_SSRX3-	NC	D12	USB_SSTX3-	NC
C13	USB_SSRX3+	NC	D13	USB_SSTX3+	NC
C14	GND		D14	GND	
C15	DDI1_PAIR6+	NC	D15	DDI1_CTRLCLK_AUX+	
C16	DDI1_PAIR6-	NC	D16	DDI1_CTRLCLK_AUX-	
C17	RSVD		D17	RSVD	
C18	RSVD		D18	RSVD	
C19	PCIE_RX6+		D19	PCIE_TX6+	
C20	PCIE_RX6-		D20	PCIE_TX6-	
C21	GND(FIXED)		D21	GND(FIXED)	
C22	PCIE_RX7+	PCIE is optional function	D22	PCIE_TX7+	PCIE is optional function
C23	PCIE_RX7-	PCIE is optional function	D23	PCIE_TX7-	PCIE is optional function
C24	DDI1_HPD		D24	RSVD	NC
C25	DDI1_PAIR4+	NC	D25	RSVD	NC
C26	DDI1_PAIR4-	NC	D26	DDI1_PAIR0+	
C27	RSVD		D27	DDI1_PAIR0-	
C28	RSVD		D28	RSVD	
C29	DDI1_PAIR5+	NC	D29	DDI1_PAIR1+	
C30	DDI1_PAIR5-	NC	D30	DDI1_PAIR1-	
C31	GND(FIXED)		D31	GND(FIXED)	
C32	DDI2_CTRLCLK_AUX+		D32	DDI1_PAIR2+	
C33	DDI2_CTRLDATA_AUX-		D33	DDI1_PAIR2-	

Connector Rows C and D

Pin#	Type 6 Description	SOM-5871	Pin#	Type 6 Description	SOM-5871

		Difference			Difference
C34	DDI2_DDC_AUX_SEL		D34	DDI1_DDC_AUX_SEL	
C35	RSVD		D35	RSVD	
C36	DDI3_CTRLCLK_AUX+	DDI3 is optional function	D36	DDI1_PAIR3+	
C37	DDI3_CTRLCLK_AUX-	DDI3 is optional function	D37	DDI1_PAIR3-	
C38	DDI3_DDC_AUX_SEL	DDI3 is optional function	D38	RSVD	
C39	DDI3_PAIR0+	DDI3 is optional function	D39	DDI2_PAIR0+	
C40	DDI3_PAIR0-	DDI3 is optional function	D40	DDI2_PAIR0-	
C41	GND(FIXED)		D41	GND(FIXED)	
C42	DDI3_PAIR1+	DDI3 is optional function	D42	DDI2_PAIR1+	
C43	DDI3_PAIR1-	DDI3 is optional function	D43	DDI2_PAIR1-	
C44	DDI3_HPD	DDI3 is optional function	D44	DDI2_HPD	
C45	RSVD		D45	RSVD	
C46	DDI3_PAIR2+	DDI3 is optional function	D46	DDI2_PAIR2+	
C47	DDI3_PAIR2-	DDI3 is optional function	D47	DDI2_PAIR2-	
C48	RSVD		D48	RSVD	
C49	DDI3_PAIR3+	DDI3 is optional function	D49	DDI2_PAIR3+	
C50	DDI3_PAIR3-	DDI3 is optional function	D50	DDI2_PAIR3-	
C51	GND(FIXED)		D51	GND(FIXED)	
C52	PEG_RX0+		D52	PEG_TX0+	
C53	PEG_RX0-		D53	PEG_TX0-	
C54	TYPE0#	NC	D54	PEG_LANE_RV#	
C55	PEG_RX1+		D55	PEG_TX1+	
C56	PEG_RX1-		D56	PEG_TX1-	
C57	TYPE1#	NC	D57	TYPE2#	GND
C58	PEG_RX2+		D58	PEG_TX2+	
C59	PEG_RX2-		D59	PEG_TX2-	
C60	GND(FIXED)		D60	GND(FIXED)	
C61	PEG_RX3+		D61	PEG_TX3+	
C62	PEG_RX3-		D62	PEG_TX3-	
C63	RSVD		D63	RSVD	
C64	RSVD		D64	RSVD	

Connector Rows C and D

Pin#	Type 6 Description	SOM-5871	Pin#	Type 6 Description	SOM-5871

		Difference			Difference
C65	PEG_RX4+		D65	PEG_TX4+	
C66	PEG_RX4-		D66	PEG_TX4-	
C67	RAPID_SHUTDOWN		D67	GND	
C68	PEG_RX5+		D68	PEG_TX0+	
C69	PEG_RX5-		D69	PEG_TX0-	
C70	GND(FIXED)		D70	GND(FIXED)	
C71	PEG_RX6+		D71	PEG_TX6+	
C72	PEG_RX6-		D72	PEG_TX6-	
C73	GND		D73	GND	
C74	PEG_RX7+		D74	PEG_TX7+	
C75	PEG_RX7-		D75	PEG_TX7-	
C76	GND		D76	GND	
C77	RSVD		D77	RSVD	
C78	PEG_RX8+	NC	D78	PEG_TX8+	NC
C79	PEG_RX8-	NC	D79	PEG_TX8-	NC
C80	GND(FIXED)		D80	GND(FIXED)	
C81	PEG_RX9+	NC	D81	PEG_TX9+	NC
C82	PEG_RX9-	NC	D82	PEG_TX9-	NC
C83	RSVD		D83	RSVD	
C84	GND		D84	GND	
C85	PEG_RX10+	NC	D85	PEG_TX10+	NC
C86	PEG_RX10-	NC	D86	PEG_TX10-	NC
C87	GND		D87	GND	
C88	PEG_RX11+	NC	D88	PEG_TX11+	NC
C89	PEG_RX11-	NC	D89	PEG_TX11-	NC
C90	GND(FIXED)		D90	GND(FIXED)	
C91	PEG_RX12+	NC	D91	PEG_TX12+	NC
C92	PEG_RX12-	NC	D92	PEG_TX12-	NC
C93	GND		D93	GND	
C94	PEG_RX13+	NC	D94	PEG_TX13+	NC
C95	PEG_RX13-	NC	D95	PEG_TX13-	NC
C96	GND		D96	GND	
C97	RSVD		D97	RSVD	
C98	PEG_RX14+	NC	D98	PEG_TX14+	NC
C99	PEG_RX14-	NC	D99	PEG_TX14-	NC

Connector Rows C and D

Pin#	Type 7 Description	SOM-5871	Pin#	Type 7 Description	SOM-5871

		Difference			Difference
C100	GND(FIXED)		D100	GND(FIXED)	
C101	PEG_RX15+	NC	D101	PEG_TX15+	NC
C102	PEG_RX15-	NC	D102	PEG_TX15-	NC
C103	GND		D103	GND	
C104	VCC_12V		D104	VCC_12V	
C105	VCC_12V		D105	VCC_12V	
C106	VCC_12V		D106	VCC_12V	
C107	VCC_12V		D107	VCC_12V	
C108	VCC_12V		D108	VCC_12V	
C109	VCC_12V		D109	VCC_12V	
C110	GND(FIXED)		D110	GND(FIXED)	

2.3 PCI Express

2.3.1 COM Express A-B Connector and C-D Connector PCIe Groups

COM Express Type 6 Modules have two groups of PCIe lanes. There is a group of up to eight lanes; six are located on COM Express A-B connector and two on C-D connector that are intended for general purpose use, such as interfacing the COM Express Module to Carrier Board PCIe peripherals. A second group of PCIe lanes is defined on the COM Express C-D connector.

This group is intended primarily for the PCIe Graphics interfaces (also referred to as the PEG interface), and is typically 16 PCIe lanes wide. For some Modules, the PEG lanes may be used for general purpose PCIe lanes if the external graphics interface is not in use. This usage is Module and Module chipset dependent.

2.3.2 General Purpose PCIe Signal Definitions

The general purpose PCI Express interface of the COM Express Type 6 Module on the COM Express A-B connector consists of up to 6 lanes plus 2 lanes on connector C-D, each with a receive and transmit differential signal pair designated from PCIE_RX0 (+ and -) to PCIE_RX7 (+ and -) and correspondingly from PCIE_TX0 (+ and -) to PCIE_TX7 (+ and -). The 8 lanes may be grouped into various link widths as defined in the COM Express spec.

Table 5: General Purpose PCI Express Signal Descriptions

Signal	Pin#	Description	I/O	Note
PCIE_RX0+ PCIE_RX0-	B68 B69	PCIe channel 0. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.22uF near COME to PCIE0 x1 device PETp/n0. Slot - Connect to PCIE0 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	
PCIE_TX0+ PCIE_TX0-	A68 A69	PCIe channel 0. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE0 x1 device PERp/n0. Slot - Connect to PCIE0 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Note
PCIE_RX1+ PCIE_RX1-	B64 B65	PCIe channel 1. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.22uF near to PCIE1 x1 device PETp/n0. Slot - Connect to PCIE1 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	
PCIE_TX1+ PCIE_TX1-	A64 A65	PCIe channel 1. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE1 x1 device PERp/n0. Slot - Connect to PCIE1 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	
PCIE_RX2+ PCIE_RX2-	B61 B62	PCIe channel 2. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.22uF near COME to PCIE2 x1 device PETp/n0. Slot - Connect to PCIE2 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	
PCIE_TX2+ PCIE_TX2-	A61 A62	PCIe channel 2. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE2 x1 device PERp/n0. Slot - Connect to PCIE2 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	
PCIE_RX3+ PCIE_RX3-	B58 B59	PCIe channel 3. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.22uF near to PCIE3 x1 device PETp/n0. Slot - Connect to PCIE3 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	

Signal	Pin#	Description	I/O	Note

PCIE_TX3+ PCIE_TX3-	A58 A59	PCIe channel 3. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE3 x1 device PERp/n0. Slot - Connect to PCIE3 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	
PCIE_RX4+ PCIE_RX4-	B55 B56	PCIe channel 4. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.1uF near to PCIE4 x1 device PETp/n0. Slot - Connect to PCIE4 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	2
PCIE_TX4+ PCIE_TX4-	A55 A56	PCIe channel 4. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE4 x1 device PERp/n0. Slot - Connect to PCIE4 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	2
PCIE_RX5+ PCIE_RX5-	B52 B53	PCIe channel 5. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.22uF near to PCIE5 x1 device PETp/n0. Slot - Connect to PCIE5 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	
PCIE_TX5+ PCIE_TX5-	A52 A53	PCIe channel 5. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE5 x1 device PERp/n0. Slot - Connect to PCIE5 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Note

PCIE_RX6+ PCIE_RX6-	C19 C20	PCIe channel 6. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.1uF near to PCIE5 x1 device PETp/n0. Slot - Connect to PCIE5 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	1
PCIE_TX6+ PCIE_TX6-	D19 D20	PCIe channel 6. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE6 x1 device PERp/n0. Slot - Connect to PCIE6 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	1
PCIE_RX7+ PCIE_RX7-	C22 C23	PCIe channel 7. Receive Input differential pair. Carrier Board: Device - Connect AC Coupling cap 0.1uF near to PCIE6 x1 device PETp/n0. Slot - Connect to PCIE6 x1 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	1
PCIE_TX7+ PCIE_TX7-	D22 D23	PCIe channel 7. Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Carrier Board: Device - Connect to PCIE7 x1 device PERp/n0. Slot - Connect to PCIE7 x1 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	1
PCIE_CLK_REF+ PCIE_CLK_REF-	A88 A89	PCIe Reference Clock for all COM Express PCIe lanes, and for PEG lanes. Carrier Board: Connect 0Ω in series to Device - PCIE device REFCLK+, REFCLK-. Slot - PCIE Conn pin A13 REFCLK+, A14 REFCLK-. *Connect to PCIE Clock Buffer input to provide PCIE clocks output for more than one PCIE devices or slots. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Note

CB_RESET#	B50	Reset output from Module to Carrier Board. Active low. Issued by Module chipset and may result from a low SYS_RESET# input, a low PWR_OK input, a VCC_12V power input that falls below the minimum specification, a watchdog timeout, or may be initiated by the Module software. Module has integrated 3.3V buffer and series resistor. Connect to reset pin of devices except PCI slots or devices. N/C if not used.	O 3.3V Suspend CMOS	
WAKE0#	B66	PCI Express wake up event signal. Module has integrated PU resistor 1K ohm to 3.3V suspend power. Device - Connect to WAKE# pin of PCIE device. Slot - Connect to WAKE# pin B11 of PCIE slot. Express Card - Connect to WAKE# pin 11 of Express Card socket. N/C if not used.	I 3.3V Suspend CMOS	

Notes:

1. Support only PCIe GEN2 under default function.
2. Connect AC Coupling cap 0.22uF near to PCIe device when PCIe signal comes from APU instead of PCIe bridge.

2.3.3 PCI Express Lane Configurations – Per COM Express Spec

According to the COM Express specification, the general purpose PCIe lanes on the A-B connector can be configured as up to eight PCI Express x1 links or may be combined into various combinations of x8, x4, x2 and x1 links that add up to a total of 8 lanes. These configuration possibilities are based on the COM Express Module's chip-set capabilities.

The COM Express specification defines a "fill order" from mapping PCIe links that are wider than x1 onto the COM Express pins. For example, the spec requires that a x4 PCI Express link be mapped to COM Express PCI Express lanes 0,1,2 and 3. Refer to the COM Express specification for details.

Note: All PCI Express devices are required to work in x1 mode as well as at their full capability. A x4 PCIe card for example is required by the PCI Express specification to be usable in x4 and / or x1 mode.

2.3.4 PCI Express* General Routing Guidelines

2.3.4.1 PCI Express Insertion Loss Budget with Carrier Board PCIe Device

The insertion losses previously allowed for the slot card and slot card connector are reallocated for use on the Carrier Board, allowing longer Carrier Board trace lengths and more Carrier Board design flexibility. The Module and COM Express connector loss budgets remain the same.

Figure 2: PCI Express Insertion Loss Budget with Carrier Board PCIe Device

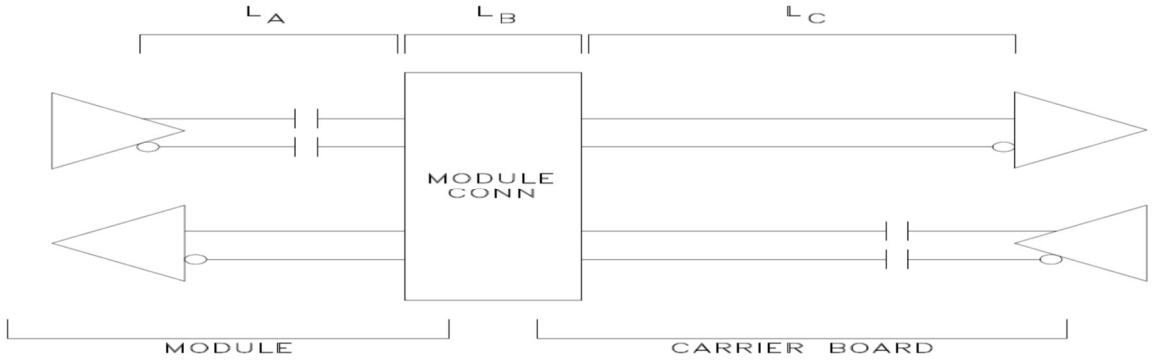


Table 6: PCI Express Insertion Loss Budget, 1.25 GHz with Carrier Board PCIe Device

Segment	**Loss (dB) max. Length [mm/inches]**	Notes
L_A	130/5.15	Allowance for 3.46 dB loss @ 0.28 dB / GHz / inch and 1.66 dB crosstalk allowance. Coupling caps not included.
Coupling Caps		1.19 dB loss. From PCI Express Card Electromechanical Spec., Rev. 1.1, parameters (LST – LSR). Includes crosstalk allowance of 0.79 dB.
L_B		COM Express connector at 1.25 GHz measured value: 0.25 dB loss.
L_C	402/15.85	Allowance for 8.30 dB loss @ 0.28 dB / GHz / inch and a 2.75 dB crosstalk allowance.
Total		13.20 dB loss.

The Module transmit and receive insertion loss budgets are different due to the presence of the coupling caps in the Module transmit path. The Module transmit path insertion loss budget shall be 4.65 dB (3.46 dB + 1.19 dB). The Module receive path insertion loss budget shall be 3.46 dB. COM Express connector loss is accounted for separately.

The Carrier Board transmit and receive insertion loss budgets are different due to the presence of the coupling caps in the Carrier Board transmit path. The Carrier Board transmit path insertion loss budget shall be 9.49 dB (8.30 dB + 1.19 dB). The Carrier Board receive path insertion loss shall be 8.30 dB. COM Express connector loss is accounted for separately.

Table 7: PCI Express Insertion Loss Budget, 2.5 GHz with Carrier Board PCIe Device

Segment	**max. Length [mm/inches]**	Notes
L_A	127/5.0	Allowance for module trace. Coupling cap effects included within simulation.
L_B		COM Express™ connector simulated at 2.5 GHz.
L_C	203/8.0	Allowance for Carrier Board.
Total	320/13.0	PCIe GEN2 Data clocked architecture

PCI Express Insertion Loss Budget, 2.5 GHz with Carrier Board PCIE Device

For "device down" PCIe Gen 2 operation, the Module PCIe maximum trace length is restricted to 5.0 inches and the Carrier Board maximum trace to 8.0 inches. Shorter lengths will yield additional margin and are encouraged where possible. Results assumed FR4 dielectrics. Other dielectrics with lower losses could be considered, but were not simulated.

2.3.4.2 PCI Express Differential Transitional Via Recommendations

Transitional vias will use oval-shapes anti-pads on all plane layers. This can be created using a rectangular-shaped void to overlap with the usual round-shaped via anti-pad. The vias must also have a symmetrical trace entry.

Figure 3 and Table 8 provides the transitional differential via pad stack details.

Figure 3: Differential Transitional Via Layout

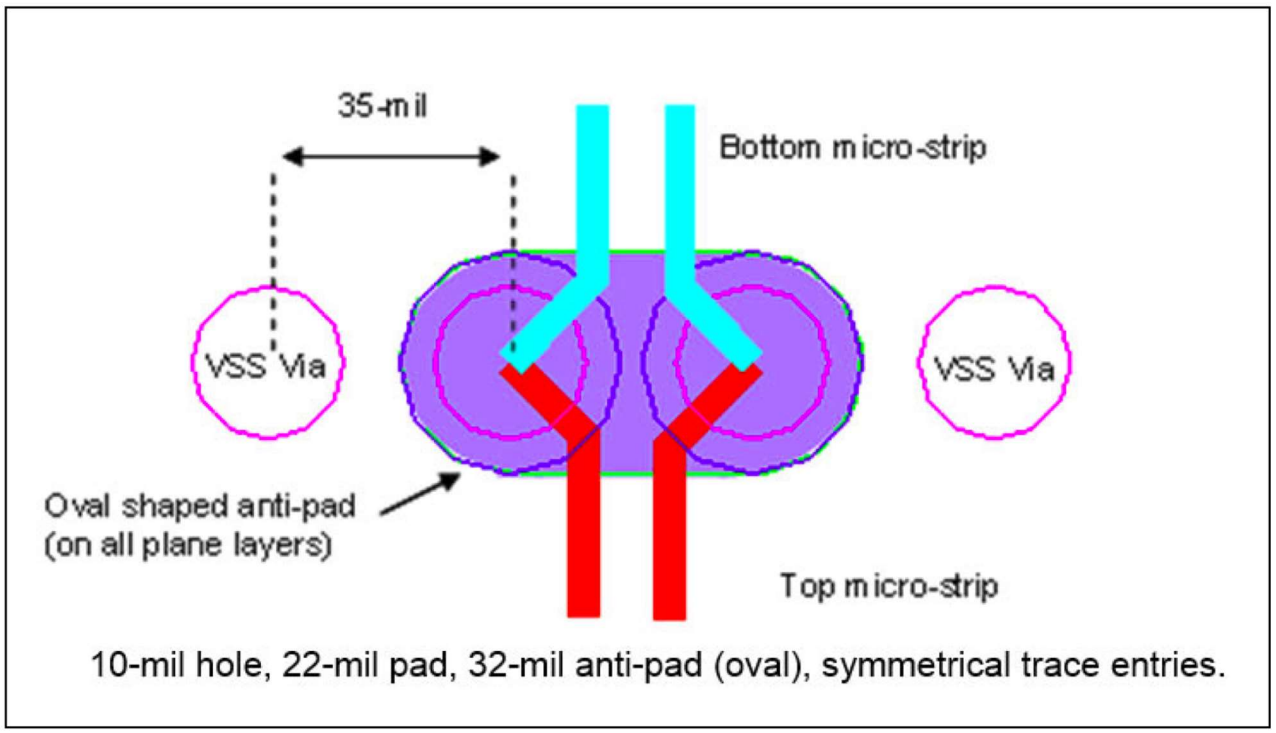


Table 8: Differential Transitional Via Layout Recommendations

Parameter	Units	Recommendation
Via Diameter	mils	10
Via Pad Size	mils	22
Oval-Shaped Anti-Pad Size	mils	32
Via to via Distance(centered)	mils	35

2.3.5 PCI Express* Trace Length Guidelines

Figure 4: Topology for PCI Express Slot Card.

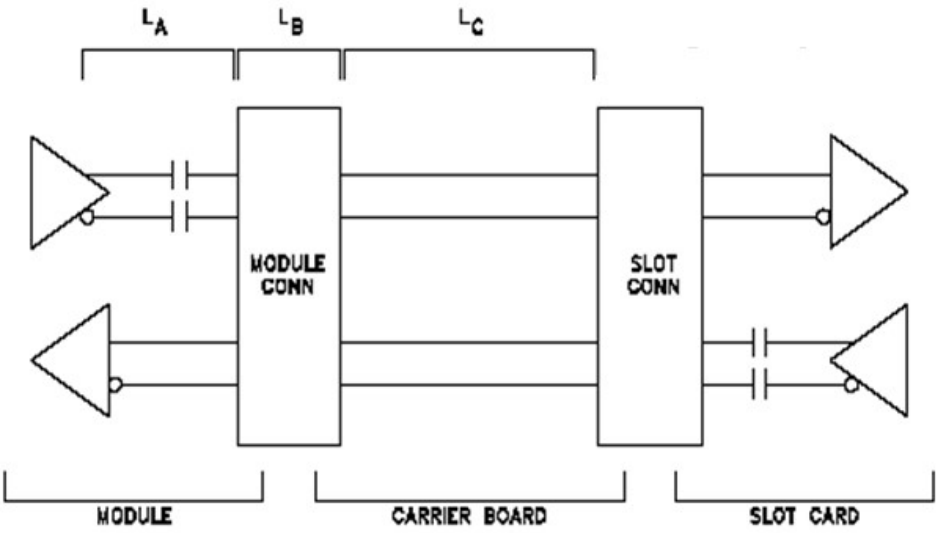


Figure 5: Topology for PCI Express Device Down.

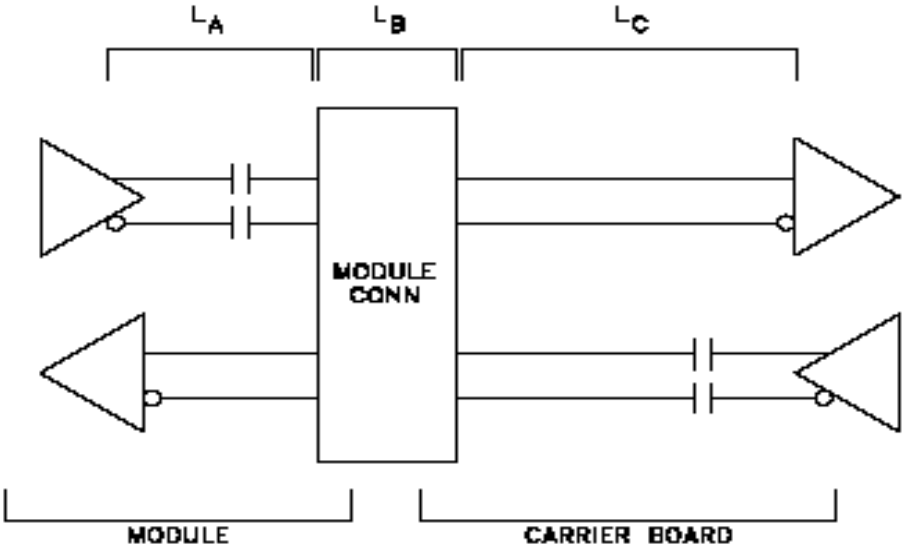


Table 9: PCI Express Gen2 * Slot Card / Device Down Trace Length Guidelines

Parameter	PCI Express Gen2	Notes
Symbol Rate / PCIe Lane	5.0 G Symbols/s	
Differential Impedance Target	85Ω±10%	
Single End	50Ω±10%	
Spacing between differential pairs and high-speed periodic signals	Min. 50mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20mils	
Spacing between RX and TX pairs (inter-pair) (s)	Min. 20mils	
Trace Spacing	Min. 15mils	
Spacing from edge of plane	Min. 15mils	
LA + LB	Please see the SOM-5871 Layout Checklist	
Lc	Carrier Board Length	
Max length of LA+LB+LC	Slot Card: 9" Device Down: 12"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils REFCLK+ and REFCLK- (intra-pair):Max. ±5mils	
Reference Plane	GND referencing preferred Min 15-mil trace edge-to-major plane edge spacing GND stitching vias required next to signal vias if transitioning layers between GND layers Power referencing acceptable if stitching caps are used	
Carrier Board Via Usage	Max. 2 vias per TX trace, Max. 4 vias per RX trace	
AC coupling	The AC coupling capacitors for the TX lines are incorporated on the COM Express Module. The AC coupling capacitors for RX signal lines have to be implemented on the customer COM Express Carrier Board. Capacitor type: X7R, 100nF ±10%, 16V, shape 0402.	1

Notes:

1. AC caps are recommended to be placed close to PCIe device side (avoid placing AC cpas on mid-bus).

Table 10: PCI Express Gen3 * Slot Card / Device Down Trace Length Guidelines

Parameter	PCI Express Gen3	Notes
Symbol Rate / PCIe Lane	8.0 G Symbols/s	
Differential Impedance Target	85Ω±10%	
Single End	50Ω±10%	
Spacing between differential pairs and high-speed periodic signals	Min. 50mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20mils	
Spacing between RX and TX pairs (inter-pair) (s)	Min. 20mils	
Trace Spacing	Min. 15mils	
Spacing from edge of plane	Min. 15mils	
LA + LB	Please see the SOM-5871 Layout Checklist	
Lc	Carrier Board Length	
Max length of LA+LB+LC	Slot Card: 9" Device Down: 12"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils REFCLK+ and REFCLK- (intra-pair):Max. ±5mils	
Reference Plane	GND referencing preferred Min 15-mil trace edge-to-major plane edge spacing GND stitching vias required next to signal vias if transitioning layers between GND layers Power referencing acceptable if stitching caps are used	
Carrier Board Via Usage	Max. 2 vias (near AC-coupling capacitors)	
AC coupling	The AC coupling capacitors for the TX lines are incorporated on the COM Express Module. The AC coupling capacitors for RX signal lines have to be implemented on the customer COM Express Carrier Board. Capacitor type: X7R, 220nF ±10%, 16V, shape 0402.	1

Notes:

AC caps are recommended to be placed close to PCIe device side (avoid placing AC cpas on mid-bus).

2.4 PEG (PCI Express Graphics)

2.4.1 PEG Signal Definitions

The PEG Port can utilize COM Express PCIe lanes 16-31 and is suitable to drive a link for an external high-performance PCI Express Graphics card, if implemented on the COM Express Module. Graphics Cards implemented as x16 use COM Express PCIe lanes 16-31; Graphics Cards implemented as x8 lanes should use COM Express PCIe lanes 16-23. Each lane of the PEG Port consists of a receive and transmit differential signal pair designated 'PEG_RX0' (+ and -) to 'PEG_RX15' (+ and -) and correspondingly from 'PEG_TX0' (+ and -) to 'PEG_TX15' (+ and -). The corresponding signals can be found on the Module connector rows C and D.

Table 11: PEG Signal Description

Signal	Pin#	Description	I/O	Notes
PEG_RX0+ PEG_RX0-	C52 C53	PEG channel 0, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n0. Slot - Connect to PCIE x16 Conn pin A16, A17 PERp/n0. N/C if not used.	I PCIE	
PEG_TX0+ PEG_TX0-	D52 D53	PEG channel 0, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n0. Slot - Connect to PCIE x16 Conn pin B14, B15 PETp/n0. N/C if not used.	O PCIE	
PEG_RX1+ PEG_RX1-	C55 C56	PEG channel 1, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n1. Slot - Connect to PCIE x16 Conn pin A21, A22 PERp/n1. N/C if not used	I PCIE	
PEG_TX1+ PEG_TX1-	D55 D56	PEG channel 1, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n1. Slot - Connect to PCIE x16 Conn pin B14, B15 PETp/n1. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Notes

PEG_RX2+ PEG_RX2-	C58 C59	PEG channel 2, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n2. Slot - Connect to PCIE x16 Conn pin A25, A26 PERp/n2. N/C if not used	I PCIE	
PEG_TX2+ PEG_TX2-	D58 D59	PEG channel 2, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n2. Slot - Connect to PCIE x16 Conn pin B23, B24 PETp/n2. N/C if not used.	O PCIE	
PEG_RX3+ PEG_RX3-	C61 C62	PEG channel 3, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp3. Slot - Connect to PCIE x16 Conn pin A29, A30 PERp3. N/C if not used	I PCIE	
PEG_TX3+ PEG_TX3-	D61 D62	PEG channel 3, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n3. Slot - Connect to PCIE x16 Conn pin B27, B28 PETp/n3. N/C if not used.	O PCIE	
PEG_RX4+ PEG_RX4-	C65 C66	PEG channel 4, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n4. Slot - Connect to PCIE x16 Conn pin A35, A36 PERp/n4. N/C if not used	I PCIE	
PEG_TX4+ PEG_TX4-	D65 D66	PEG channel 4, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n4. Slot - Connect to PCIE x16 Conn pin B33, B34 PETp/n4. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Notes

PEG_RX5+ PEG_RX5-	C68 C69	PEG channel 5, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n5. Slot - Connect to PCIE x16 Conn pin A39, A40 PERp/n5. N/C if not used	I PCIE	
PEG_TX5+ PEG_TX5-	D68 D69	PEG channel 5, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n5. Slot - Connect to PCIE x16 Conn pin B37, B38 PETp/n5. N/C if not used.	O PCIE	
PEG_RX6+ PEG_RX6-	C71 C72	PEG channel 6, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n6. Slot - Connect to PCIE x16 Conn pin A43, A44 PERp/n6. N/C if not used	I PCIE	
PEG_TX6+ PEG_TX6-	D71 D72	PEG channel 6, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n6. Slot - Connect to PCIE x16 Conn pin B37, B38 PETp/n6. N/C if not used	O PCIE	
PEG_RX7+ PEG_RX7-	C74 C75	PEG channel 7, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n7. Slot - Connect to PCIE x16 Conn pin A47, A48 PERp/n7. N/C if not used	I PCIE	
PEG_TX7+ PEG_TX7-	D74 D75	PEG channel 7, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n7. Slot - Connect to PCIE x16 Conn pin B45, B46 PETp/n7. N/C if not used	O PCIE	

Signal	Pin#	Description	I/O	Notes
PEG_RX8+	C78	PEG channel 8, Receive Input differential pair.	I PCIE	1

PEG_RX8-	C79	Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n8. Slot - Connect to PCIE x16 Conn pin A52, A53 PERp/n8. N/C if not used		
PEG_TX8+ PEG_TX8-	D78 D79	PEG channel 8, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n8. Slot - Connect to PCIE x16 Conn pin B50, B51 PETp/n8. N/C if not used	O PCIE	1
PEG_RX9+ PEG_RX9-	C81 C82	PEG channel 9, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n9. Slot - Connect to PCIE x16 Conn pin A56, A57 PERp/n9. N/C if not used	I PCIE	1
PEG_TX9+ PEG_TX9-	D81 D82	PEG channel 9, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n9. Slot - Connect to PCIE x16 Conn pin B54, B55 PETp/n9. N/C if not used	O PCIE	1
PEG_RX10+ PEG_RX10-	C85 C86	PEG channel 10, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n10. Slot - Connect to PCIE x16 Conn pin A60, A61 PERp/n10. N/C if not used	I PCIE	1
PEG_TX10+ PEG_TX10-	D85 D86	PEG channel 10, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n10. Slot - Connect to PCIE x16 Conn pin B58, B59 PETp/n10. N/C if not used	O PCIE	1

Signal	Pin#	Description	I/O	Note
PEG_RX11+	C88	PEG channel 11, Receive Input differential pair.	I PCIE	1

PEG_RX11-	C89	Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n11. Slot - Connect to PCIE x16 Conn pin A64, A65 PERp/n11. N/C if not used		
PEG_TX11+ PEG_TX11-	D88 D89	PEG channel 11, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n11. Slot - Connect to PCIE x16 Conn pin B62, B63 PETp/n11. N/C if not used	O PCIE	1
PEG_RX12+ PEG_RX12-	C91 C92	PEG channel 12, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n12. Slot - Connect to PCIE x16 Conn pin A68, A69 PERp/n12. N/C if not used	I PCIE	1
PEG_TX12+ PEG_TX12-	D91 D92	PEG channel 12, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n12. Slot - Connect to PCIE x16 Conn pin B66, B67 PETp/n12. N/C if not used	O PCIE	1
PEG_RX13+ PEG_RX13-	C94 C95	PEG channel 13, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n13. Slot - Connect to PCIE x16 Conn pin A72, A73 PERp/n13. N/C if not used	I PCIE	1
PEG_TX13+ PEG_TX13-	D94 D95	PEG channel 13, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n13. Slot - Connect to PCIE x16 Conn pin B70, B71 PETp/n13. N/C if not used	O PCIE	1

Signal	Pin#	Description	I/O	Note
PEG_RX14+	C98	PEG channel 14, Receive Input differential pair.	I PCIE	1

PEG_RX14-	C99	Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n14. Slot - Connect to PCIE x16 Conn pin A76, A77 PERp/n14. N/C if not used		
PEG_TX14+ PEG_TX14-	D98 D99	PEG channel 14, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n14. Slot - Connect to PCIE x16 Conn pin B74, B75 PETp/n14. N/C if not used	O PCIE	1
PEG_RX15+ PEG_RX15-	C101 C102	PEG channel 15, Receive Input differential pair. Device - Connect AC Coupling cap 0.22uF near COME to PCIE x16 device PETp/n15. Slot - Connect to PCIE x16 Conn pin A80, A81 PERp/n15. N/C if not used	I PCIE	1
PEG_TX15+ PEG_TX15-	D101 D102	PEG channel 15, Transmit Output differential pair. Module has integrated AC Coupling Capacitor. Device - Connect to PCIE x16 device PERp/n15. Slot - Connect to PCIE x16 Conn pin B78, B79 PETp/n15. N/C if not used	O PCIE	1
PEG_LANE_RV#	D54	PCI Express Graphics lane reversal input strap. Pull low on the carrier board to reverse lane order. Module has integrated PU resistor to 3.3V. Carrier Board: Normal - N/C Reverse - PD 1KΩ to GND N/C if not used.	I 3.3V CMOS	1

Notes:

1. SOM-5871 supports PEG x8 only.

2.4.2. PEG* General Routing Guidelines

PEG Configuration

The COM Express PCIe Graphics (PEG) Port is comprised of COM Express PCIe lanes 16-31.

The primary use of this set of signals is to interface to off-Module graphics controllers or cards.

If the PEG interface is not used for an external graphics card SDVO, it may be possible to use these PCIe lanes for other Carrier Board PCIe devices. The details of this usage are Module and Module chip-set dependent. Operation in a x1 link is also supported. Wider links (x2, x4, x8, x16) are chip-set dependent. Refer to the Module product documentation for details.

The COM Express specification defines a fill order for this set of PCIe lanes. Larger link widths go to the lower lanes. Refer to the COM Express specification for details.

Using PEG Pins for General Purpose PCIe Lanes

The COM Express PEG lanes may be used for general-purpose use if the PEG port is not being used as an interface to an external graphics device. The characteristics of this usage are Module and chip-set dependent.

Modules that employ desktop and mobile chip-sets with PEG capability can usually be set up to allow the COM Express PEG lanes to be configured as a single general purpose PCIe link, with link width possibilities of x1, x4, x8 or x16. The x1 configuration should always work; the wider links may be Module and chip-set dependent. Check with your vendor.

Modules based on server-class chip-sets may allow multiple links over the PEG lanes – for example, a x8 link on COM Express PCIe lanes 16 through 23 and a x4 link over lanes 24 through 27. This is Module and chip-set dependent.

2.4.2.1 PEG Differential Transitional Via Recommendations

Please refer to 2.3.4.2

2.4.3 PEG* Trace Length Guidelines

Figure 6: Topology for PCI Express Slot Card.

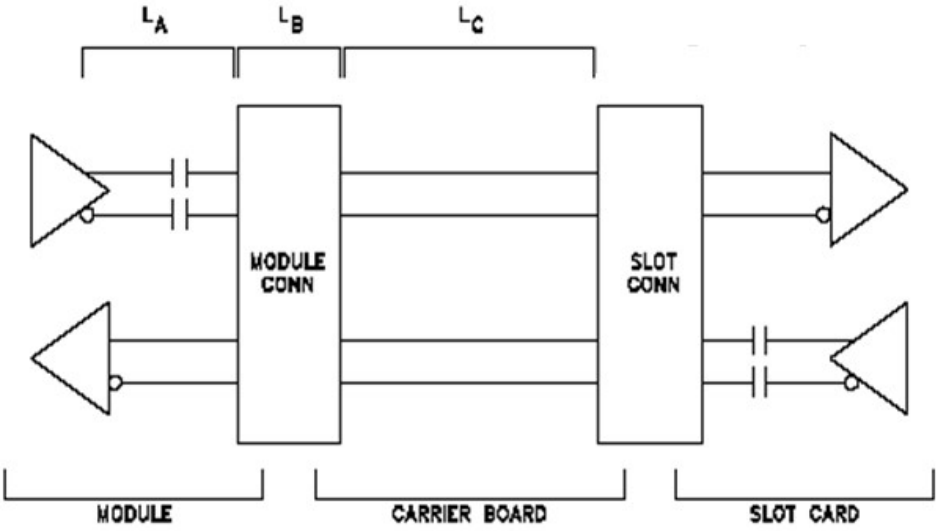


Figure 7: Topology for PCI Express Device Down.

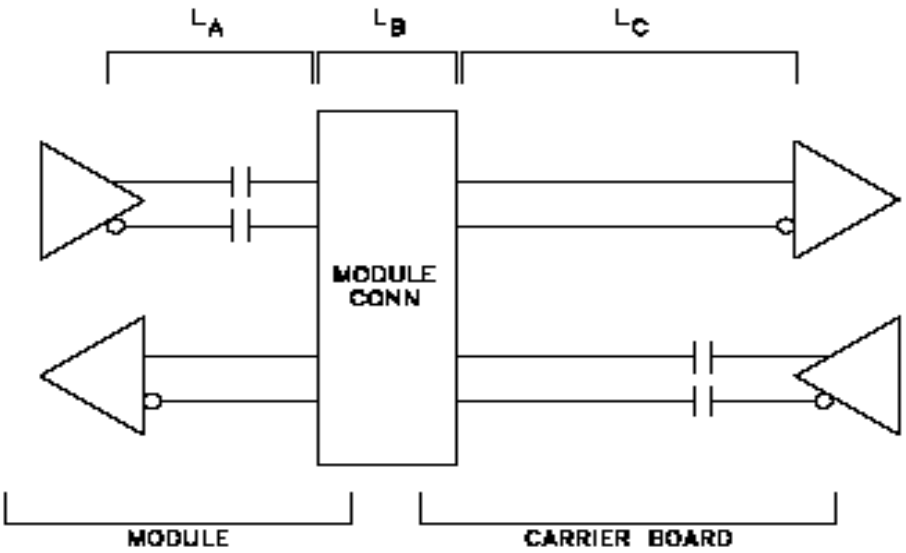


Table 12: PCI Express* Slot Card / Device Down Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	PCI Express* expansion	
Differential Impedance Target	85Ω±10%	

Single End	50Ω±10%	
Isolation to equivalent pairs	20 mils (MS) and 15 mils (DS)	
Isolation to other signal groups	20 mils (MS) and 15 mils (DS)	
Tx/Rx Spacing	20 mils	
Trace spacing	Min. 15mils	
LA + LB	Please see the SOM-5871 Layout Checklist	
Lc	Carrier Board Length	
Max length of LA+LB+LC	Slot Card: 9" Device Down: 12"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils REFCLK+ and REFCLK- (intra-pair):Max. ±5 mils	
Reference Plane	GND referencing preferred Min 15-mil trace edge-to-major plane edge spacing GND stitching vias required next to signal vias if transitioning layers between GND layers Power referencing acceptable if stitching caps are used	
Carrier Board Via Usage	Max. 2 vias per TX/RX trace.	
AC coupling	The AC coupling capacitors for the TX lines are incorporated on the COM Express Module. The AC coupling capacitors for RX signal lines have to be implemented on the customer COM Express Carrier Board. Capacitor type: X7R, 220nF ±10%, 16V, shape 0402.	1

Notes:

1. AC caps are recommended to be placed close to PCIe device side (avoid placing AC cpas on mid-bus).

2.5 Digital Display Interfaces (DDI)

Module Types 6 use Digital Display Interfaces (DDI) to provide DisplayPort, HDMI/DVI, and SDVO interfaces. Type 6 Modules can contain up to 3 DDIs (DDI[1:3]) of which DDI[1:3] can support DisplayPort, HDMI/DVI and DDI[1] can support DisplayPort, HDMI/DVI, and SDVO.
The main difference is that SDVO is only supported DDI[1] for Type 6 Modules.

DisplayPort / HDMI / DVI

DisplayPort was developed by the Video Electronics Standard Association (VESA) in order to create a new digital display port interface to connect a video source to a display device.
DisplayPort can be used to transfer audio and video at the same time, but each one is optional and can be transmitted without the other. A bi-directional, half-duplex auxiliary channel carries device management and device control data for the Main Link, such as VESA EDID.
DisplayPort is nowadays on almost all COM Express Modules available as Dual-mode DisplayPort, that can directly emit single-link HDMI and DVI signals using an adapter, which contains a level shifter to adjust for the lower voltages required by DisplayPort. These adapters can be directly implemented on the Carrier Board to have an easy, simple and future proof implementation of HDMI and/or DVI or an inexpensive cable adapter can be directly connected on the Carrier Board's DisplayPort connector.

2.5.1 DDI Signal Definitions

Type 6 Modules up to 3 DisplayPort interfaces.
Each DisplayPort interface consists of 4 differential lanes, 1 auxiliary lane and 1 hot-plug-detect signal. The DDC_AUX_SEL pin should be routed to pin 13 of the DisplayPort connector, to enable Dual-Mode. When HDMI/DVI is directly done on the Carrier Board, this pin shall be pulled to 3.3V with a 100k Ohm resistor to configure the AUX pairs as DDC channels.

Table 13: Port / HDMI / DVI Pin-out of Type 6

Pin Name	DDI1 Pin#	DDI2 Pin#	DDI3 Pin#	Function (DDIX) DisplayPort	Function (DDIX) HDMI / DVI
DDIX_PAIR0+	D26	D39	C39	DPX_LANE0+	TMDSX_DATA2+
DDIX_PAIR0-	D27	D40	C40	DPX_LANE0-	TMDSX_DATA2-
DDIX_PAIR1+	D29	D42	C42	DPX_LANE1+	TMDSX_DATA1+
DDIX_PAIR1-	D30	D43	C43	DPX_LANE1-	TMDSX_DATA1-
DDIX_PAIR2+	D32	D46	C46	DPX_LANE2+	TMDSX_DATA0+
DDIX_PAIR2-	D33	D47	C47	DPX_LANE2-	TMDSX_DATA0-
DDIX_PAIR3+	D36	D49	C49	DPX_LANE3+	TMDSX_CLK+
DDIX_PAIR3-	D37	D50	C50	DPX_LANE3-	TMDSX_CLK-
DDIX_HPD	C24	D44	C44	DPX_HPD	HDMIX_HPD
DDIX_CTRLCLK_AUX+	D15	C32	C36	DPX_AUX+	HDMIX_CTRLCLK
DDIX_CTRLCLK_AUX-	D16	C33	C37	DPX_AUX-	HDMIX_CTRLDATA
DDIX_DDC_AUX_SEL	D34	C34	C38		

Note: Please verify in the Module's specification if DisplayPort or Dual-Mode DisplayPort is supported.

Table 14: DDI1 Signal Description

Signal	Pin#	Description	I/O	Notes
DDI1_PAIR0+ DDI1_PAIR0-	D26 D27	DDI channel 1, differential pairs 0. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI1_PAIR1+ DDI1_PAIR1-	D29 D30	DDI channel 1, differential pairs 1. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI1_PAIR2+ DDI1_PAIR2-	D32 D33	DDI channel 1, differential pairs 2. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI1_PAIR3+ DDI1_PAIR3-	D36 D37	DDI channel 1, differential pairs 3. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Notes
DDI1_HPD	C24	DDI channel 1, Hot-Plug Detect. Module has integrated current blocking circuit and PD resistor to GND Carrier Board: For DP, connector to device or DP connector HP pin. For HDMI / DVI, connect 3.3V to 5V Level Shifter to device, HDMI or DVI PD ping. NC if not used.	I 3.3V COMS	
DDI1_CTRLCLK_AUX+ DDI1_CTRLCLK_AUX-	D15 D16	DDI channel1, DP AUX function. DDC_AUX_SEL is no connect. Half-duplex bi-directional AUX channel for services such as link configuration or maintenance and EDID access. Module has integrated AC Coupling Capacitor and PD resistor to GND. Carrier Board: Connect to device or DP connector. N/C if not used.	I/O PCIE	
		DDI channel1, HDMI / DVI I^2C function. DDC_AUX_SEL is pulled high. Carrier Board: Connect 3.3V-5V Level Shifter to device, HDMI or DVI connector. N/C if not used.	I/O OD 3.3V COMS	
DDI1_DDC_AUX_SEL	D34	Selects the function of DDI1_CTRLCLK_AUX+ and DDI1_CTRLDATA_AUX-. If this input is floating the AUX pair is used for the DP AUX+/- signals. If pulled-high the AUX pair contains the CRTLCLK and CTRLDATA signals. Module has integrated PD resistor to GND. Carrier Board: DP1 AUX+/- - N/C HDMI1: PU 100K to 3.3V N/C if not used.	I 3.3V COMS	

Table 15: DDI2 Signal Description

Signal	Pin#	Description	I/O	Notes
DDI2_PAIR0+ DDI2_PAIR0-	D39 D40	DDI channel 2, differential pairs 0. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI2_PAIR1+ DDI2_PAIR1-	D42 D43	DDI channel 2, differential pairs 1. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI2_PAIR2+ DDI2_PAIR2-	D46 D47	DDI channel 2, differential pairs 2. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	
DDI2_PAIR3+ DDI2_PAIR3-	D49 D50	DDI channel 2, differential pairs 3. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	

Signal	Pin#	Description	I/O	Notes
DDI2_HPD	D44	DDI channel 2, Hot-Plug Detect.	I 3.3V	

		Module has integrated current blocking circuit and PD resistor to GND Carrier Board: For DP, connector to device or DP connector HP pin. For HDMI / DVI, connect 3.3V to 5V Level Shifter to device, HDMI or DVI PD ping. NC if not used.	COMS	
DDI2_CTRLCLK_AUX+ DDI2_CTRLCLK_AUX-	C32 C33	DDI channel12 DP AUX function. DDC_AUX_SEL is no connect. Half-duplex bi-directional AUX channel for services such as link configuration or maintenance and EDID access. Module has integrated AC Coupling Capacitor and PD resistor to GND. Carrier Board: Connect to device or DP connector. N/C if not used.	I/O PCIE	
		DDI channel2, HDMI / DVI I^2C function. DDC_AUX_SEL is pulled high. Carrier Board: Connect 3.3V-5V Level Shifter to device, HDMI or DVI connector. N/C if not used.	I/O OD 3.3V COMS	
DDI2_DDC_AUX_SEL	C34	Selects the function of DDI2_CTRLCLK_AUX+ and DDI2_CTRLDATA_AUX-. If this input is floating the AUX pair is used for the DP AUX+/- signals. If pulled-high the AUX pair contains the CRTLCLK and CTRLDATA signals. Module has integrated PD resistor to GND. Carrier Board: DP2 AUX+/- - N/C HDMI2: PU 100K to 3.3V N/C if not used.	I 3.3V COMS	

Notes:

Table 16: DDI3 Signal Description

Signal	Pin#	Description	I/O	Notes

DDI3_PAIR0+ DDI3_PAIR0-	C39 C40	DDI channel 3, differential pairs 0. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	1
DDI3_PAIR1+ DDI3_PAIR1-	C42 C43	DDI channel 3, differential pairs 1. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	1
DDI3_PAIR2+ DDI3_PAIR2-	C46 C47	DDI channel 3, differential pairs 2. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	1
DDI3_PAIR3+ DDI3_PAIR3-	C49 C50	DDI channel 3, differential pairs 3. Carrier Board: For DP, connect AC Coupling Capacitors 75~200 nF near COME to device or DP connector. For HDMI / DVI, connect AC Coupling Capacitors 75~200 nF near COME and Level Shifter to HDMI or DVI connector. N/C if not used.	O PCIE	1

Signal	Pin#	Description	I/O	Notes
DDI3_HPD	C44	DDI channel 3, Hot-Plug Detect.	I 3.3V	1

		Module has integrated current blocking circuit and PD resistor to GND Carrier Board: For DP, connector to device or DP connector HP pin. For HDMI / DVI, connect 3.3V to 5V Level Shifter to device, HDMI or DVI PD ping. NC if not used.	COMS	
DDI3_CTRLCLK_AUX+ DDI3_CTRLCLK_AUX-	C36 C37	DDI channel 3, DP AUX function. DDC_AUX_SEL is no connect. Half-duplex bi-directional AUX channel for services such as link configuration or maintenance and EDID access. Module has integrated AC Coupling Capacitor, AUX+ PD resistor to GND and AUX- Pull up to 3.3V. Carrier Board: Connect to device or DP connector. N/C if not used.	I/O PCIE	1
		DDI channel1, HDMI / DVI I^2C function. DDC_AUX_SEL is pulled high. Carrier Board: Connect 3.3V-5V Level Shifter to device, HDMI or DVI connector. N/C if not used.	I/O OD 3.3V COMS	
DDI3_DDC_AUX_SEL	C38	Selects the function of DDI3_CTRLCLK_AUX+ and DDI3_CTRLDATA_AUX-. If this input is floating the AUX pair is used for the DP AUX+/- signals. If pulled-high the AUX pair contains the CRTLCLK and CTRLDATA signals. Module has integrated PD resistor to GND. Carrier Board: DP3 AUX+/- - N/C HDMI3: PU 100K to 3.3V N/C if not used.	I 3.3V COMS	1

Notes:

1. SOM-5871 DDI3, default DP as AUX to VGA used.

2.5.2 Digital Display Interfaces (DDI) Routing Guidelines

2.5.2.1 DisplayPort Routing Guidelines

Carriers that support DisplayPort (DisplayPort only or dual mode):

- DC blocking capacitors shall be placed on the Carrier for the DDI[n]_PAIR[0:3] signals.
- The Carrier shall include a blocking FET on DDI[n]_HPD to prevent back-drive current from damaging the Module.

When implementing DisplayPort on the Carrier Board, the DP_AUX+ line shall have a pulldown resistor to GND. The resistor value should be 100kΩ. The DP_AUX- line shall have a pull-up resistor to 2.5V. The resistor value should be 100kΩ. The resistors shall be placed on the DisplayPort connector side of the AC coupling capacitors. The DP_HP signal shall include a blocking FET to prevent back-drive current damage. The DP_HP signal shall be pulled-down to GND with a 110kΩ resistor.

The DDI signals can be used to support a variety of video interfaces. The circuits required to realize the different video interfaces will be determined by a future PICMG Carrier Design Guide subcommittee. At this time, they only requirement placed on Modules for the DDI signals is the maximum trace length specified below.

Figure 8: DisplayPort Loss Budget

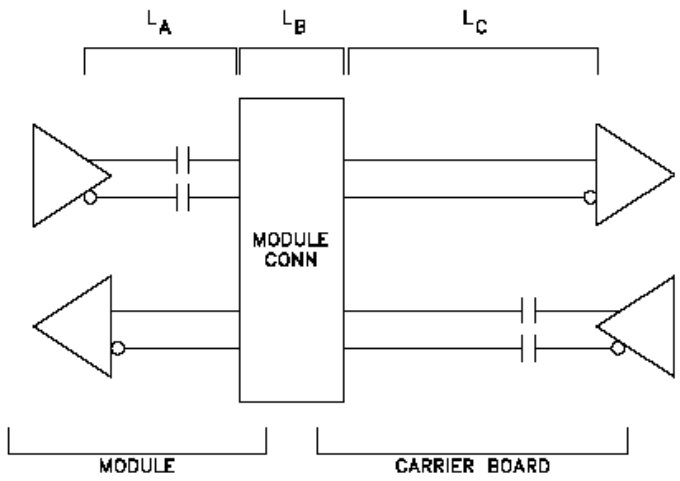


Table 17: DisplayPort Insertion Loss Budget

Segment	max. Length [mm/inches]	Notes
L_A	127/5.0	Allowance for module trace. Coupling cap effects included within simulation.
L_B		COM Express™ connector simulated at 2.5 GHz.
L_C	TBD	Allowance for Carrier Board trace.
Total	TBD	

Figure 9: Topology for DisplayPort

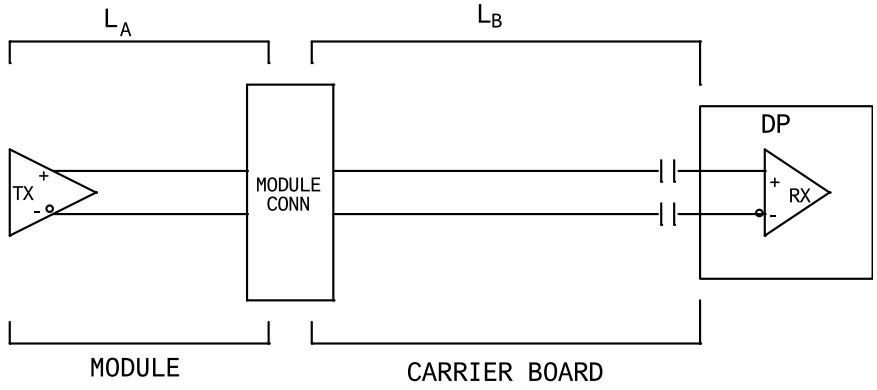


Table 18: DisplayPort Connector / Device Down Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	DisplayPort	
Differential Impedance Target	85 Ω ±10%	
Single End	50Ω ±10%	
Isolation to equivalent pairs	20 mils (MS) and 15 mils (DS)	
Isolation to other signal groups	20 mils (MS) and 15 mils (DS)	
Trace spacing	15 mils (MS) and 12 mils (DS)	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	DDI differential pairs to DP connector:7" DDI differential pairs to DP++ connector:9" DDI differential pairs to Device Down: 7"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils For each channel, match the lengths between differential pairs (Inter-Pair) to be within a 3-inch window (max length – min length < 3 inch) when DisplayPort to DP connector, within 1.8-inch when DisplayPort to DP++ connector.	
Reference Plane	GND referencing preferred. Min 15-mil trace edge-to-major plane edge spacing.	
Carrier Board Via Usage	Max. 2 vias.	
AC coupling	Min = 75 nF Max = 200 nF	1

Notes:

1. AC caps are recommended to be placed close to device side (avoid placing AC cpas on mid-bus).

2.5.2.2 HDMI / DVI Routing Guidelines

When implementing HDMI level shifters shall be used on the TMDS signals. Bi-directional level shifters

shall be used between the 3.3V and 5V CTRLCLK and CTRLDATA signals with 2.2kΩ pull-ups to 3.3V and 5V.

Carriers that support TMDS (DVI/HDMI):

- DDI[n]_DDC_AUX_SEL shall be pulled to 3.3V on the Carrier with a 100K Ohm resistor to configure the DDI[n]_AUX pair as the DDC channel.
- Bi-directional level translators shall be placed on the Carrier DDI[n]_CTRLDATA_AUX- and DDI[n]_CTRLCLK_AUX+ to convert the 3.3V DDC channel on the Module to the 5V DDC channel for the TMDS display.
- Pull-up resistors shall be placed on the Carrier from 3.3V (Module side of level translator) and 5V (display side of level translator) and the [n]_CTRLDATA_AUX- and DI[n]_CTRLCLK_AUX+ signals. The pull-up resistor should be 2k.
- Level translators Shall be placed on the Carrier DDI[n]_PAIR[0:3] signals.
- DC blocking capacitors shall be placed on the Carrier for the DDI[n]_PAIR[0:3] signals.
- The Carrier shall include a blocking FET on DDI[n]_HPD to prevent back-drive current from damaging the module.

The DDI signals can be used to support a variety of video interfaces. The circuits required to realize the different video interfaces will be determined by a future PICMG Carrier Design Guide subcommittee. At this time, they only requirement placed on Modules for the DDI signals is the maximum trace length specified below.

Figure 10: HDMI / DVI (TMDS) Loss Budget

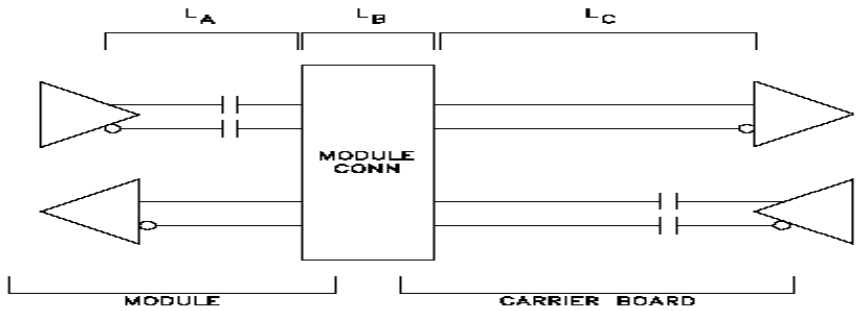


Table 19: HDMI / DVI (TMDS) Insertion Loss Budget

Segment	max. Length [mm/inches]	Notes
L_A	127/5.0	Allowance for module trace. Coupling cap effects included within simulation.
L_B		COM Express™ connector simulated at 2.5 GHz.
L_C	TBD	Allowance for Carrier Board trace.
Total	TBD	

Figure 11: Topology for HDMI / DVI

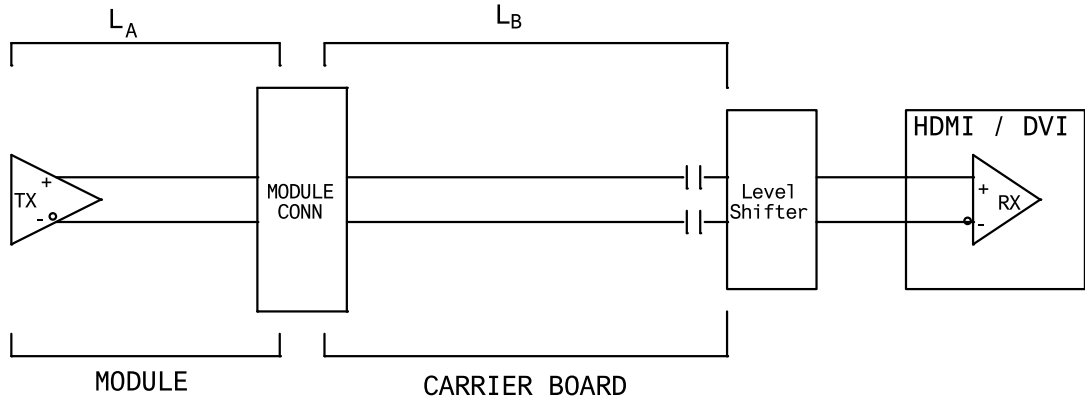


Table 20: HDMI / DVI Connector / Device Down Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	HDMI / DVI (TMDS)	
Differential Impedance Target	85 Ω ±5%	
Single End	50Ω ±5%	
Isolation to equivalent pairs	20 mils (MS) and 15 mils (DS)	
Isolation to other signal groups	20 mils (MS) and 15 mils (DS)	
Trace spacing	20 mils (MS) and 15 mils (DS)	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	DDI differential pairs to HDMI/DVI connector : 6"	2
Length matching	Differential pairs (intra-pair): Max. ±5 mils For each channel, match the lengths of the differential pairs (Inter-Pair) to be within a 1.8-inch window (max length – min length < 1.8 inch), including DDC signal.	
Reference Plane	GND referencing preferred. Min 15-mil trace edge-to-major plane edge spacing.	
Carrier Board Via Usage	Max. 2 vias.	
AC coupling	Min = 75 nF Max = 200 nF	1

Notes:

1. AC caps are recommended to be placed close to device side (avoid placing AC cpas on mid-bus).

2. HDMI 2.0 requires the use of an external retime/redriver. An external retime retimes the input stream to compensate for random jitter. An external redriver provides additional signal drive strength.

2.5.3 HDMI / DVI Level Shifter Requirements

The HDMI specification requires the receiver to be terminated to AVCC (nominally 3.3 V) through Rt (nominally 50 Ω). The HDMI receiver requirements require the native HDMI signals from the SOC to be

level shifted. This prevents electrical overstress of the driver and ensure that the receiver is operational within the receiver specifications defined in the *High Definition Multimedia Interface Specification 1.4a*.

2.5.4 ESD Protection

HDMI signals are subjected to ESD strikes due to plugging in of the devices through the HDMI cable and frequent human contact that can destroy both the HDMI host and devices on the platform. Therefore these ports need to be protected.

There are a wide variety of ESD protection devices and ESD suppressors readily available in the market such as Metal Oxide Varistors (MOVs), Zener Diode, Transient Voltage Suppressor (TVS), Polymer devices and ESD diode arrays. With 1.65 Gbps of data rate, HDMI is very sensitive to parasitic capacitance. Excessive parasitic capacitance can severely degrade the signal integrity and lead to a compliance or operational failure. To maintain signal integrity, Intel recommends to use ESD suppressors or diode arrays having a low junction capacitance.

Recommended characteristics of an ideal ESD Protection Diode for HDMI:

- Able to withstand at least 8 kV of ESD strikes.
- Low capacitance <1 pF to minimize signal distortion at high data rates as higher capacitance degrade the HDMI signal quality.
- Fast response/rise time to protect from the fast rise time of ESD surge pulses.
- Low-leakage current to minimize static power consumption.
- Ensure the selected ESD solution will not violate HDMI Voff spec. In a low power state a power rail ESD diode can become forward biased as a result on the HDMI sink (panel) termination of 50 Ω to 3.3 V
- Some ESD devices may impact the trace impedance. Care should be taken while choosing such devices so that the differential-impedance target in the *HDMI 1.4 Specification* is not violated.

The *HDMI 1.4a Specification* requires that 8 kV of ESD strikes be tolerated.

The ESD protection devices should be placed as close to the HDMI connector as possible so that when ESD strikes occur, the discharges can be quickly absorbed or diverted to the ground/power plane before it is coupled to another signal path nearby.
Footprints for ESD components or diode arrays can be provided on board with no stubs and no more than 750 mils (19.05 mm) from the connector. The ESD components can be stuffed depending on the requirement.

2.6 LAN Interface

All COM Express Modules provide at least one LAN port. The 8-wire 10/100/1000BASE-T Gigabit Ethernet interface compliant to the IEEE 802.3-2005 specification is the preferred interface for this port, with the COM Express Module PHY responsible for implementing auto-negotiation of 10/100BASE-TX vs 10/100/1000BASE-T operation. The carrier may also support a 4-wire 10/100BASE-TX interface from the COM Express Module on an exception basis. Check with your vendor for 10/100 only implementations.

2.6.1 LAN Signal Definitions

The LAN interface of the COM Express Module consists of 4 pairs of low voltage differential pair signals designated from *'GBE0_MDI0'* (+ and -) to *'GBE0_MDI3'* (+ and -) plus additional control signals for link activity indicators. These signals can be used to connect to a 10/100/1000BASE-T RJ45 connector with integrated or external isolation magnetics on the Carrier Board. The corresponding LAN differential pair and control signals can be found on rows A and B of the Module's connector.

Table 21: LAN Interface Signal Descriptions

Signal	Pin#	Description	I/O	Note
GBE0_MDI0+ GBE0_MDI0-	A13 A12	Media Dependent Interface (MDI) differential pair 0. The MDI can operate in 1000, 100, and 10Mbit/sec modes. Module has integrated termination. Carrier Board: Connect to Magnetics Module MDI0+/- N/C if not used.	I/O GBE	
GBE0_MDI1+ GBE0_MDI1-	A10 A9	Media Dependent Interface (MDI) differential pair 1. The MDI can operate in 1000, 100, and 10Mbit/sec modes. Module has integrated termination. Carrier Board: Connect to Magnetics Module MDI0+/- N/C if not used	I/O GBE	
GBE0_MDI2+ GBE0_MDI02	A7 A6	Media Dependent Interface (MDI) differential pair 2. The MDI can operate in 1000, 100, and 10Mbit/sec modes. Module has integrated termination. Carrier Board: Connect to Magnetics Module MDI2+/- N/C if not used.	I/O GBE	

Signal	Pin#	Description	I/O	Note
GBE0_MDI3+	A3	Media Dependent Interface (MDI) differential pair 3.	I/O GBE	

GBE0_MDI3-	A2	The MDI can operate in 1000, 100, and 10Mbit/sec modes. Module has integrated termination. Carrier Board: Connect to Magnetics Module MDI3+/- N/C if not used		
GBE0_CTREF	A14	Reference voltage for Carrier Board Ethernet channel 0 magnetics center tap. The reference voltage is determined by the requirements of the Module PHY and may be as low as 0V and as high as 3.3V. The reference voltage output shall be current limited on the Module. In the case in which the reference is shorted to ground, the current shall be limited to 250 mA or less.. Carrier Board: 0.1uF to ground. N/C if not used.	REF GND min 3.3V max	
GBE0_LINK#	A8	Ethernet controller 0 link indicator, active low.	3.3V Suspend / 3.3V OD CMOS	
GBE0_LINK100#	A4	Ethernet controller 0 100Mbit / sec link indicator, active low.	3.3V Suspend / 3.3V OD CMOS	
GBE0_LINK1000#	A5	Ethernet controller 0 1000Mbit / sec link indicator, active low.	3.3V Suspend / 3.3V OD CMO	
GBE0_ACT#	B2	Ethernet controller 0 activity indicator, active low.	3.3V Suspend / 3.3V OD CMO	
GBE0_SDP	A49	Gigabit Ethernet Controller 0 Software-Definable Pin. Can also be used for IEEE1588 support such as a 1pps signal. See section 2.6.2 for details.	3.3V Suspend / 3.3V I/O	

Note:.

Gigabit Ethernet Controller 0: Media Dependent Interface Differential Pairs 0,1,2,3. The MDI can operate in 1000, 100 and 10 Mbit / sec modes.

Some pairs are unused in some modes, per the following:

Table 22: The MDI can operate in 1000, 100 and 10 Mbit / sec modes

	1000BASE-T	100BASE-TX	10BASE-T
MDI[0]+/-	B1_DA+/-	TX+/-	TX+/-
MDI[1]+/-	B1_DB+/-	RX+/-	RX+/-
MDI[2]+/-	B1_DC/-		
MDI[3]+/-	B1_DD+/-		

2.6.2 SDP Pins

The Software Defined Pins (SDP) can be used to provide a timing communication path between the Module and Carrier. A board level signal that communicates time is a key element that facilitates clock synchronization between elements of a platform. Examples of such elements include, but are not limited to, CPU, Chipset, FPGA and others.

Modules ***should*** connect the SDP signal to a module element pin capable of propagating (transmitting) time, and/or time-stamping (receiving) the signal to extract time information from it. If implemented, the direction of the signal with respect to the module element ***should*** be able to be determined by system software.

Pulse Per Second (PPS):

A PPS signal conveys both frequency and phase and can be used to transfer time information between elements within a platform. It is commonly used because it encapsulates both frequency and time into a single signal. It is preferred over other methods that require more complex implementations of hardware and software. A GPS is probably the most widespread, high-quality, clock source capable of generating a PPS signal.

Platform-level Synchronization Implementation Examples:

Example1: The Network Interface Controller (NIC) on the COM Module is Precision Time Protocol (PTP) capable and the COM designer has connected a software configurable, timing aware, pin on the NIC to the SDP pin on the module/carrier interface. Software can configure the NIC to output a PPS signal onto this pin that connects it to one or more elements on the module and/or carrier board.

Example2: The carrier board has provisions for connecting a PPS output from a GPS to the SDP signal connection to the module. The module element (i.e. NIC, CPU, Chipset) can receive the timing information from the carrier board and adjust its time accordingly.

Precision Time Protocol - Background

Standards such as IEEE 1588, 802.1AS, and Time Sensitive Networking (TSN) provide standards for synchronizing time between nodes on a local area network. Additional benefits of the standards may include lower latency and improved network traffic Quality of Service (QoS). Systems that commonly require synchronization include those made up of distributed nodes that perform measurement, control, and compute functions. These nodes may have clock sources with varying degrees of accuracy and stability.

System-wide time synchronization with sub-microsecond accuracy is supported, by PTP standards, with minimal network and compute resource utilization.

It is the merger of the platform-level synchronization and network level synchronization pieces that enable real-time distributed systems. Additional information regarding the aforementioned standards can be found in their respective specifications and widely available supporting documents.

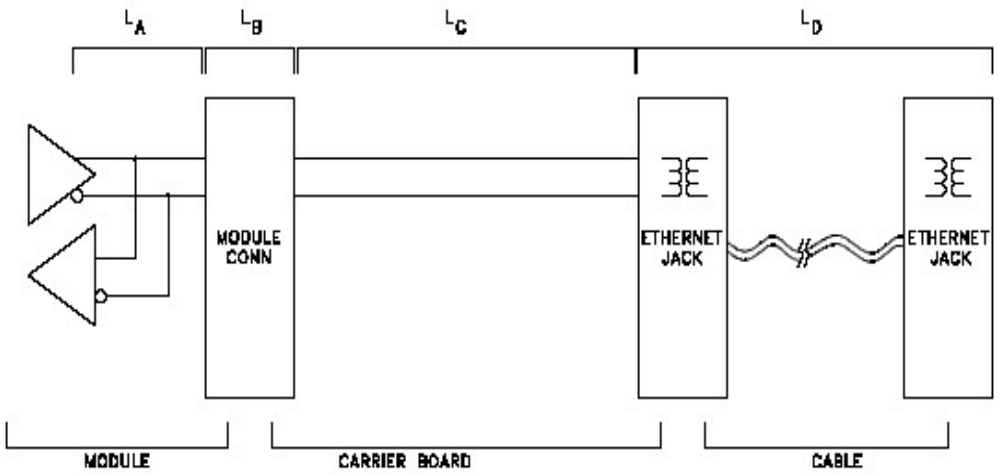
Software Implementation:

The software architecture and features required to support platform and network level synchronization are outside the scope of this specification.

2.6.3 LAN Routing Guidelines

10/100/1000 Ethernet Insertion Loss Budget

Figure 12: 10/100/1000 Ethernet Insertion Loss Budget



COM Express Ethernet implementations should conform to insertion loss values less than or equal to those shown in the table above. The insertion loss values shown account for frequency dependent material losses only. Cross talk losses are separate from material losses in the Gb Ethernet specification.

"Device Down" implementations, in which the Ethernet target device is implemented on the Carrier Board (for instance, an Ethernet switch), may add the insertion loss for the RJ45 Ethernet jack and integrated magnetics to the Carrier Board budget. This insertion loss value is typically 1 dB. The Carrier Board insertion loss budget then becomes LC + 1 dB, or 1.15 dB.

Table 23: 10/100/1000 Ethernet Insertion Loss Budget, 100 MHz

Segment	Loss (dB)	Notes
L_A	0.08	Up to 3 inches of module trace @ 0.28 dB / GHz / inch
L_B	0.02	COM Express™ connector at 100 MHz measured value
L_C	0.15	Up to 5 inches of Carrier Board trace @ 0.28 dB / GHz / inch
L_D	24.00	Cable and cable connectors, integrated magnetics, per source spec.
Total	24.25	

2.6.4 LAN Trace Length Guidelines

Figure 13: Topology for Ethernet Jack

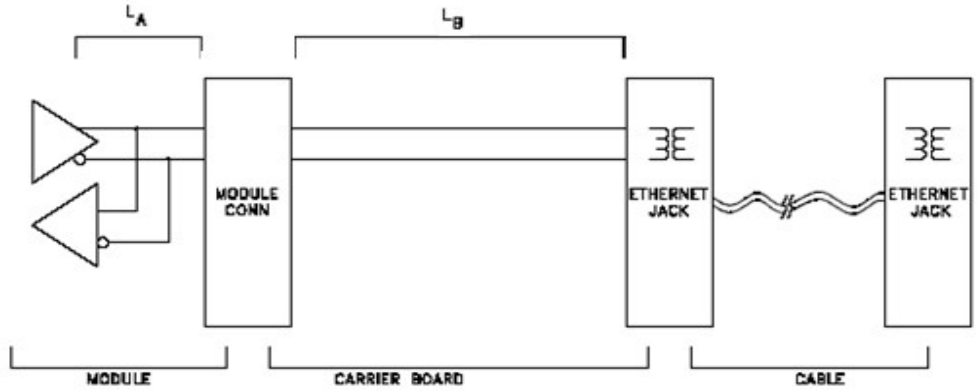


Table 24: Ethernet Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	GBE0_MDIx+, GBE0_MDIx-	
Differential Impedance Target	100 Ω ±10%	
Single End	50Ω ±10%	
Spacing between RX and TX pairs (inter-pair) (s)	Min. 50mils	
Spacing between differential pairs and high-speed periodic signals	Min. 300mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 100mils	
Spacing between digital ground and analog ground plane (between the magnetics Module and RJ45 connector)	Min. 60mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	COM Express Module to the magnetics Module – 5.0 inches. Magnetics Module to RJ45 connector - Max. 1.0 inches.	
Length matching	Differential pairs (intra-pair): Max. ±5 mils RX and TX pairs (inter-pair) - Max. ±15 mils	
Reference Plane	GND referencing preferred.	
Spacing from edge of plane	Min. 40mils	
Carrier Board Via Usage	Max. 2 vias.	

Notes:

2.6.5 Reference Ground Isolation and Coupling

The Carrier Board should maintain a well-designed analog ground plane around the components on the primary side of the transformer between the transformer and the RJ-45 receptacle. The analog ground plane is bonded to the shield of the external cable through the RJ-45 connector housing.

The analog ground plane should be coupled to the carrier's digital logic ground plane using a capacitive coupling circuit that meets the ground plane isolation requirements defined in the 802.3-2005 specification. It is recommended that the Carrier Board PCB design maintain a minimum 30 mil gap between the digital logic ground plane and the analog ground plane.

It's recommended to place an optional GND to SHIELDGND connection near the RJ-45 connector to improve EMI and ESD capabilities.

2.7 USB2.0 Ports

A COM Express Module must support a minimum of 4 USB Ports and can support up to 8 USB Ports. All of the USB Ports must be USB2.0 compliant. There are 4 over-current signals shared by the 8 USB Ports. A Carrier must current limit the USB power source to minimize disruption of the Carrier in the event that a short or over-current condition exists on one of the USB Ports. A Module must fill the USB Ports starting at Port 0. The USB SuperSpeed ports 0, 1, 2 and 3, if used, are to be paired with USB 2.0 ports 0, 1, 2 and 3 in the same order. The USB SuperSpeed ports use the same over current signaling mechanism as the USB 2.0 ports, but USB 3.0 allows up to 1A current per port instead of 500mA allowed in USB 2.0. Although USB 2.0 signals use differential signaling, the USB specification also encodes single ended state information in the differential pair, making EMI filtering somewhat challenging. Ports that are internal to the Carrier do not need EMI filters. A USB Port can be powered from the Carrier Main Power or from the Carrier Suspend Power. Main Power is used for USB devices that are accessed when the system is powered on. Suspend Power (VCC_5V_SBY) is used for devices that need to be powered when the Module is in Sleep-State S5. This would typically be for USB devices that support Wake-on-USB. The amount of current available on VCC_5V_SBY is limited so it should be used sparingly.

2.7.1 USB2.0 Signal Definitions

Table 25: USB Signal Descriptions

Signal	Pin#	Description	I/O	Note
USB0+ USB0-	A46 A45	USB Port 0, data + or D+ USB Port 0, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB1+ USB1-	B46 B45	USB Port 1, data + or D+ USB Port 1, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	

Signal	Pin#	Description	I/O	Note
USB2+ USB2-	A43 A42	USB Port 2, data + or D+ USB Port 2, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB3+ USB3-	B43 B42	USB Port 3, data + or D+ USB Port 3, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB4+ USB4-	A40 A39	USB Port 4, data + or D+ USB Port 4, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB5+ USB5-	B40 B39	USB Port 5, data + or D+ USB Port 5, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB6+ USB6-	A37 A36	USB Port 6, data + or D+ USB Port 6, data + or D- Carrier board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	

Signal	Pin#	Description	I/O	Note
USB7+ USB7-	B37 B36	USB Port 7, data + or D+ USB Port 7, data + or D- Carrier Board: Device - Connect to D+/- Conn. - Connect 90Ω @100MHz Common Choke in series and ESD suppressors to GND to Pin 3 D+ / Pin 2 D- N/C if not used	I/O USB	
USB_0_1_OC#	B44	USB over-current sense, USB channels 0 and 1. A pull-up for this line shall be present on the Module. An open drain driver from a USB current monitor on the Carrier Board may drive this line low. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to Overcurrent of Power Distribution Switch and Bypass 0.1uF to GND N/C if not used	I 3.3V CMOS	
USB_2_3_OC#	A44	USB over-current sense, USB channels 2 and 3. A pull-up for this line shall be present on the Module. An open drain driver from a USB current monitor on the Carrier Board may drive this line low. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to Overcurrent of Power Distribution Switch and Bypass 0.1uF to GND N/C if not used	I 3.3V CMOS	
USB_4_5_OC#	B38	USB over-current sense, USB channels 4 and 5. A pull-up for this line shall be present on the Module. An open drain driver from a USB current monitor on the Carrier Board may drive this line low. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to Overcurrent of Power Distribution Switch and Bypass 0.1uF to GND N/C if not used	I 3.3V CMOS	

Signal	Pin#	Description	I/O	Note
USB_6_7_OC#	A38	USB over-current sense, USB channels 6 and 7. A pull-up for this line shall be present on the Module. An open drain driver from a USB current monitor on the Carrier Board may drive this line low. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to Overcurrent of Power Distribution Switch and Bypass 0.1uF to GND N/C if not used	I 3.3V CMOS	
USB0_HOST_PRSINT	B48	Module USB client may detect the presence of a USB host on USB0. A high value indicates that a host is present. Carrier Board: USB Conn. Pin 1 connects to 1. Clampling diode, 1uF and 100KΩ to GND. 2. 5V to 3.3V level shifter circuit to COME USB0_HOST_PRSNT. N/C if not used	I 3.3V CMOS	1

Notes:

1. SOM-5871 is not support USB client.

2.7.1.1 USB Over-Current Protection (USB_x_y_OC#)

The USB Specification describes power distribution over the USB port, which supplies power for USB

devices that are directly connected to the Carrier Board. Therefore, the host must implement over-current protection on the ports for safety reasons. Should the aggregate current drawn by the downstream ports exceed a permitted value, the over-current protection circuit removes power from all affected downstream ports. The over-current limiting mechanism must be resettable without user mechanical intervention. For more detailed information about this subject, refer to the 'Universal Serial Bus Specifications Revision 2.0', which can be found on the website http://www.usb.org.

Over-current protection for USB ports can be implemented by using power distribution switches on the Carrier Board that monitor the USB port power lines. Power distribution switches usually have a soft-start circuitry that minimizes inrush current in applications where highly capacitive loads are employed. Transient faults are internally filtered.

Additionally, they offer a fault status output that is asserted during over-current and thermal shutdown conditions. These outputs should be connected to the corresponding COM Express Modules USB over-current sense signals. Fault status signaling is an option at the USB specification. If you don't need the popup message in your OS you may leave the signals USB_0_1_OC#, USB_2_3_OC#, USB_4_5_OC# and USB_6_7_OC# unconnected.

Simple resettable PolySwitch devices are capable of fulfilling the requirements of USB overcurrent protection and therefore can be used as a replacement for power distribution switches.
Fault status signals are connected by a pullup resistor to VCC_3V3_SBY on COM Express Module. Please check your tolerance on a USB port with VCC_5V supply.

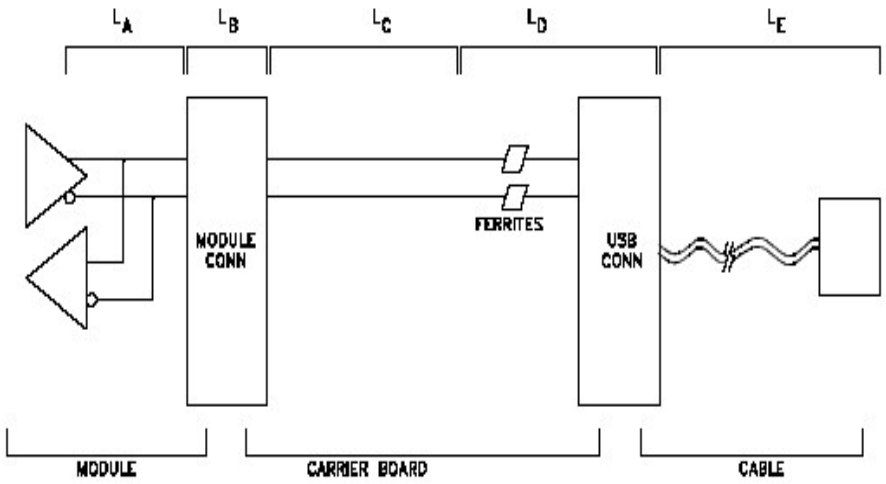
2.7.1.2 Powering USB devices during S5

The power distribution switches and the ESD protection shown in the schematics can be powered from Main Power or Suspend Power (VCC_5V_SBY). Ports powered by Suspend Power are powered during the S3 and S5 system states. This provides the ability for the COM Express Module to generate system wake-up events over the USB interface.

2.7.2 USB2.0 Routing Guidelines

USB Insertion Loss Budget

Figure 14: USB2.0 Insertion Loss Budget



COM Express USB implementations should conform to insertion loss values less than or equal to those shown in the table above. The insertion loss values shown account for frequency dependent material losses only. Cross talk losses are separate from material losses in the USB specification.

"Device Down" implementations, in which the USB target device is implemented on the Carrier Board, may add the ferrite and USB connector insertion loss values to the Carrier Board budget.

The Carrier Board insertion loss budget then becomes LC + LD, or 2.68 dB.

Table 26: USB2.0 Insertion Loss Budget, 400 MHz

Segment	Loss (dB)	Notes
L_A	0.67	Up to 6 inches of module trace @ 0.28 dB / GHz / inch
L_B	0.05	COM ExpressTM connector at 400 MHz measured value
L_C	1.68	Up to 14 inches of Carrier Board trace @ 0.28 dB / GHz / inch
L_D	1.00	USB connector and ferrite loss
L_E	5.80	USB cable and far end connector loss, per source specification
Total	9.20	

2.7.2.1 USB 2.0 General Design Considerations and Optimization

Use the following general routing and placement guidelines when laying out a new design. These

guidelines help minimize signal quality and EMI problems.

- Do not route USB 2.0 traces under crystals, oscillators, clock synthesizers, magnetic devices or ICs that use and/or duplicate clocks.
- Separate signal traces into similar categories, and route similar signal traces together (such as routing differential-pairs together).
- Keep USB 2.0 signals clear of the core logic set. High current transients are produced during internal state transitions and can be very difficult to filter out.
- Follow the 20 x h rule by keeping traces at least [20 x (height above the plane)] mils away from the edge of the plane (VCC or GND). For an example stackup, the height above the plane is 4.5 mils (0.114 mm). This calculates to a 90-mil (2.286-mm) spacing requirement from the edge of the plane. This helps prevent the coupling of the signal onto adjacent wires and also helps prevent free radiation of the signal from the edge of the PCB.
- Avoid stubs on high-speed USB signals because stubs cause signal reflections and affect signal quality. If a stub is unavoidable in the design, the total of all the stubs on a particular line should not be greater than 200 mils (5.08 mm).

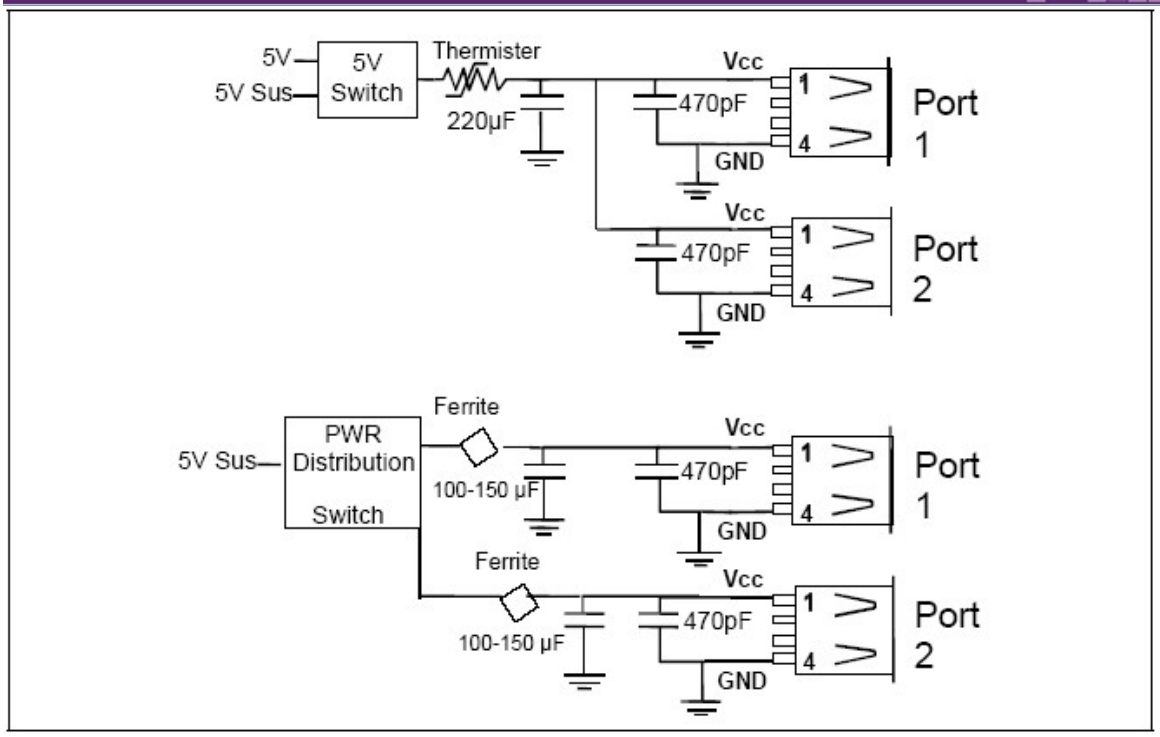
2.7.2.2 USB 2.0 Port Power Delivery

The following is a suggested topology for power distribution of VBUS to USB ports.

These circuits provide two types of protection during dynamic attach and detach situations on the bus: inrush current limiting (droop) and dynamic detach flyback protection. These two types require both bulk capacitance (droop) and filtering capacitance (for dynamic detach flyback voltage filtering). Intel recommends the following:

- Minimize the inductance and resistance between the coupling capacitors and the USB ports.
- Place capacitors as close as possible to the port and the power-carrying traces should be as wide as possible, preferably, a plane.
- Make the power-carrying traces wide enough that the system fuse blows on an over current event. If the system fuse is rated at 1 A, then the power-carrying traces should be wide enough to carry at least 1.5 A.

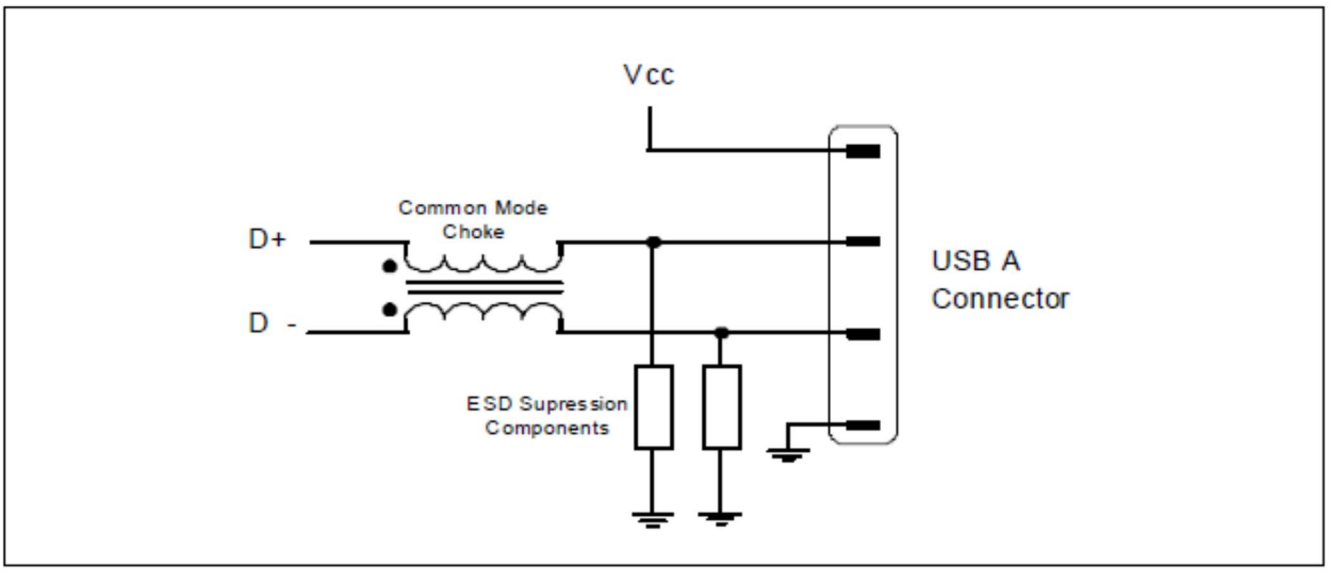
Figure 15: USB 2.0 Good Downstream Power Connection



2.7.2.3 USB 2.0 Common Mode Chokes

Testing has shown that common mode chokes can provide required noise attenuation. A design should include a common mode choke footprint to provide a stuffing option in the event the choke is needed to pass EMI testing. Below figure shows the schematic of a typical common mode choke and ESD suppression components. Place the choke as close as possible to the USB connector signal pins.

Figure 16: USB 2.0 A Common Mode Choke



Common mode chokes distort full-speed and high-speed signal quality. As the common mode

impedance increases the distortion increases, therefore test the effects of the common mode choke on full speed and high-speed signal quality. Common mode chokes with a target impedance of 80 Ω to 90 Ω, at 100 MHz, generally provide adequate noise attenuation.

Finding a common mode choke that meets the designer's needs is a two-step process:

1. Choose a part with the impedance value that provides the required noise attenuation. This is a function of the electrical and mechanical characteristics of the part chosen and the frequency and strength of the noise present on the USB traces that should be suppressed.
2. After obtaining a part that gives passing EMI results, the second step is to test the effect this part has on signal quality. Higher impedance common mode chokes generally have a greater damaging effect on signal quality, so care must be used when increasing the impedance without doing thorough testing. Thorough testing means that the signal quality must be checked for low-speed, full-speed, and highspeed USB operation.

Further common mode choke information can be found on the high-speed USB Platform Design Guides available at www.usb.org.

2.7.2.4 EMI / ESD Protection

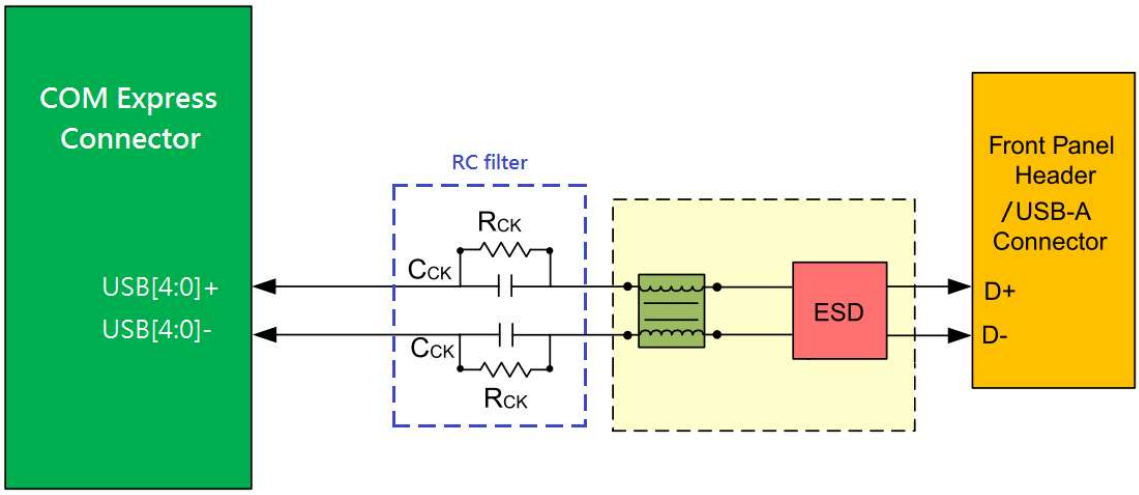
To improve the EMI behavior of the USB interface, a design should include common mode chokes, which have to be placed as close as possible to the USB connector signal pins.
Common mode chokes can provide required noise attenuation but they also distort the signal quality of full-speed and high-speed signaling. Therefore, common mode chokes should be chosen carefully to meet the requirements of the EMI noise filtering while retaining the integrity of the USB signals on the Carrier Board design.

To protect the USB host interface of the Module from over-voltage caused by electrostatic discharge (ESD) and electrical fast transients (EFT), low capacitance steering diodes and transient voltage suppression diodes have to be implemented on the Carrier Board design. In the USB reference schematics Figure 29 above, this is implemented by using 'SR05 RailClampR' surge rated diode arrays from Semtech (http://semtech.com).

2.7.2.5 Depicts the schematic and layout requirements for the USB 2.0 / 1.1 interface

To prevent USB2.0/1.1 (Full Speed Mode) hubs or devices connected to the USB ports via cables longer than 3-meter may not be enumerated correctly, it is recommended to implement RC filter on the USB port D + and D- near EMI / ESD devices. (RCK=15ohm/1%, CCK=470pF/5%)

Figure 17: USB 2.0/1.1 Interface—Schematic and Routing Model



• Common-mode chokes are suggested but not required for all USB D+ and D− signals. Alternatively, a 0 ohm resistor pad can be used.

• ESD-suppression devices are required on the D+ and D− signals. Use NXP BAV99 or equivalent low capacitive-loading ESD-suppression devices.

• Place the ESD devices as close as possible to the USB connector, but no farther than 12.7 mm. If the reference plane under the ESD component is voided, then the rule can be relaxed to 7.62 mm to ease placement requirements.

• If the ESD device has a power pin, decouple it with one 470-nF capacitor or with a capacitor value specified by the device manufacturer.

2.7.3 USB2.0 Trace Length Guidelines

Figure 18: Topology for USB2.0

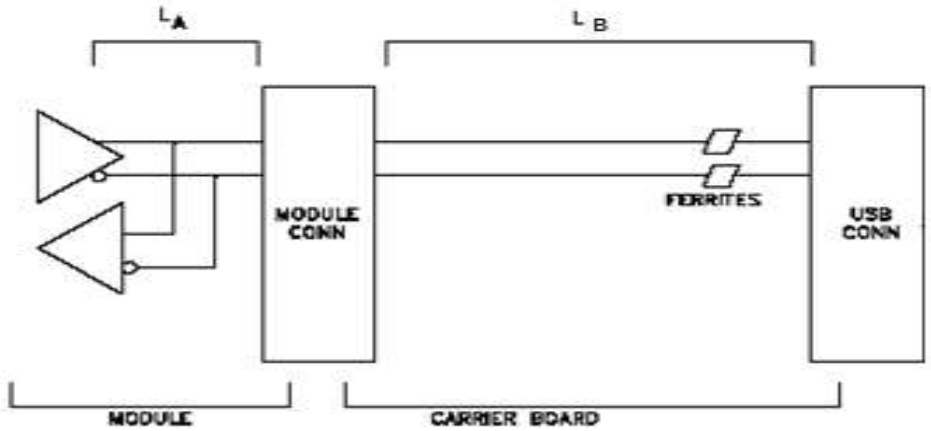


Table 27: USB2.0 Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	USB[7:0]+, USB[7:0]-	
Differential Impedance Target	90 Ω ±10%	
Single End	50Ω ±10%	
Spacing between pairs-to-pairs (inter-pair) (s)	Min. 20 mils	
Spacing between differential pairs and high-speed periodic signals	Min. 50 mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20 mils	
Trace spacing	Total trace length ≤ 5-inch:12 mils (MS) and 9 mils (DS) Total trace length > 5-inch:15 mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	USB[4:0] to USB2.0 connector :18” USB[4:0] to front panel :6” USB[7:5] to USB2.0 connector:6”	
Length matching	Differential pairs (intra-pair): Max. ±60 mils	
Reference Plane	GND referencing preferred	
Spacing from edge of plane	Min. 15mils	
Carrier Board Via Usage	Try to minimize number of vias	

Notes:

2.8 USB3.0/USB3.1 Ports

USB 3.0/USB3.1 is the third major revision of the Universal Serial Bus (USB) standard for computer connectivity. It adds a new transfer speed called SuperSpeed (SS) to the already existing LowSpeed (LS), FullSpeed (FS) and HighSpeed (HS).

USB 3.0/USB3.1 leverages the existing USB 2.0 infrastructure by adding two additional data pair lines to allow a transmission speed up to 5/10 Gbit/s, which is 10 times faster than USB 2.0 with 480 Mbit/s.

The additional data lines are unidirectional instead of the bidirectional USB 2.0 data lines. USB 3.0/USB3.1 is fully backward compatible to USB 2.0. USB 3.0/USB3.1 connectors are different from USB 2.0 connectors. The USB 3.0/USB3.1 connector is a super set of a USB 2.0 connector, with 4 additional pins that are invisible to USB 2.0 connectors. A USB 2.0 Type A plug may be used in a USB 3.0/USB3.1 Type A receptacle, but the USB 3.0/USB3.1 SuperSpeed functions will not be available.

2.8.1 USB3.0/USB3.1 Signal Definitions

Table 28: USB3.0/USB3.1 Signal Definitions

Signal	Pin#	Description	I/O	Note
USB_SSTX0+ USB_SSTX0-	D4 D3	USB Port 0, SuperSpeed TX + USB Port 0, SuperSpeed TX – Module has integrated AC Coupling Capacitors Carrier Board: Device - Connect to StdA_SSRX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 9 StdA_SSTX+ / Pin 8 StdA_SSTX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	O PCIE	
USB_SSTX1+ USB_SSTX1-	D7 D6	USB Port 1, SuperSpeed TX + USB Port 1, SuperSpeed TX – Module has integrated AC Coupling Capacitors Carrier Board: Device - Connect to StdA_SSRX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 9 StdA_SSTX+ / Pin 8 StdA_SSTX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	O PCIE	

Signal	Pin#	Description	I/O	Note
USB_SSTX2+ USB_SSTX2-	D10 D9	USB Port 2, SuperSpeed TX + USB Port 2, SuperSpeed TX – Module has integrated AC Coupling Capacitors Carrier Board: Device - Connect to StdA_SSRX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 9 StdA_SSTX+ / Pin 8 StdA_SSTX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	O PCIE	
USB_SSTX3+ USB_SSTX3-	D13 D12	USB Port 3, SuperSpeed TX + USB Port 3, SuperSpeed TX – Module has integrated AC Coupling Capacitors Carrier Board: Device - Connect to StdA_SSRX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 9 StdA_SSTX+ / Pin 8 StdA_SSTX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	O PCIE	1
USB_SSRX0+ USB_SSRX0-	C4 C3	USB Port 0, SuperSpeed RX + USB Port 0, SuperSpeed RX – Carrier Board: Device - Connect AC Coupling Capacitors 100nF near COME to StdA_SSTX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 6 StdA_SSRX+ / Pin 5 StdA_SSRX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	I PCIE	

Signal	Pin#	Description	I/O	Note
USB_SSRX1+ USB_SSRX1-	C7 C6	USB Port 1, SuperSpeed RX + USB Port 1, SuperSpeed RX – Carrier Board: Device - Connect AC Coupling Capacitors 100nF near COME to StdA_SSTX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 6 StdA_SSRX+ / Pin 5 StdA_SSRX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	I PCIE	
USB_SSRX2+ USB_SSRX2-	C10 C9	USB Port 2, SuperSpeed RX + USB Port 2, SuperSpeed RX – Carrier Board: Device - Connect AC Coupling Capacitors 100nF near COME to StdA_SSTX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 6 StdA_SSRX+ / Pin 5 StdA_SSRX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	I PCIE	
USB_SSRX3+ USB_SSRX3-	C13 C12	USB Port 3, SuperSpeed RX + USB Port 3, SuperSpeed RX – Carrier Board: Device - Connect AC Coupling Capacitors 100nF near COME to StdA_SSTX+/- Conn. - Connect 0Ω and 90Ω @100MHz USB3.0 Common Mode Choke(NL) combined in series and USB3.0 ESD suppressors to GND to Pin 6 StdA_SSRX+ / Pin 5 StdA_SSRX-, the value of CMC depends on EMI and signal integrity performance. N/C if not used	I PCIE	1

Notes:

1. SOM-5871 support 3-port USB3.1, included USB3.1 Gen1 1 port and Gen2 2-port.

2.8.1.1 USB Over-Current Protection (USB_x_y_OC#)

The USB Specification describes power distribution over the USB port, which supplies power for

USB devices that are directly connected to the Carrier Board. Therefore, the host must implement over-current protection on the ports for safety reasons. Should the aggregate current drawn by the downstream ports exceed a permitted value, the over-current protection circuit removes power from all affected downstream ports. The over-current limiting mechanism must be resettable without user mechanical intervention. For more detailed information about this subject, refer to the 'Universal Serial Bus Specifications Revision 2.0', which can be found on the website http://www.usb.org.

Over-current protection for USB ports can be implemented by using power distribution switches on the Carrier Board that monitor the USB port power lines. Power distribution switches usually have a soft-start circuitry that minimizes inrush current in applications where highly capacitive loads are employed. Transient faults are internally filtered.

Additionally, they offer a fault status output that is asserted during over-current and thermal shutdown conditions. These outputs should be connected to the corresponding COM Express Modules USB over-current sense signals. Fault status signaling is an option at the USB specification. If you don't need the popup message in your OS you may leave the signals USB_0_1_OC#, USB_2_3_OC#, USB_4_5_OC# and USB_6_7_OC# unconnected.

Fault status signals are connected by a pullup resistor to VCC_3V3_SBY on COM Express Module. Please check your tolerance on a USB port with VCC_5V supply.

USB 2.0 port's VCC current limit should be set to 500mA. For USB 3.0 implementations, the VCC current limit is raised to 1A. A different, USB 3.0 compatible, power switch is used.

2.8.1.2 EMI / ESD Protection

To improve the EMI behavior of the USB interface, a design should include common mode chokes, which have to be placed as close as possible to the USB connector signal pins.

Common mode chokes can provide required noise attenuation but they also distort the signal quality of FullSpeed, HighSpeed and SuperSpeed signaling. Therefore, common mode chokes should be chosen carefully to meet the requirements of the EMI noise filtering while retaining the integrity of the USB signals on the Carrier Board design.

To protect the USB host interface of the Module from over-voltage caused by electrostatic discharge (ESD) and electrical fast transients (EFT), low capacitance steering diodes and transient voltage suppression diodes have to be implemented on the Carrier Board design.

2.8.2 USB3.0/USB3.1 Routing Guidelines

USB3.0 Insertion Loss Budget

Figure 19: USB3.0 Insertion Loss Budget

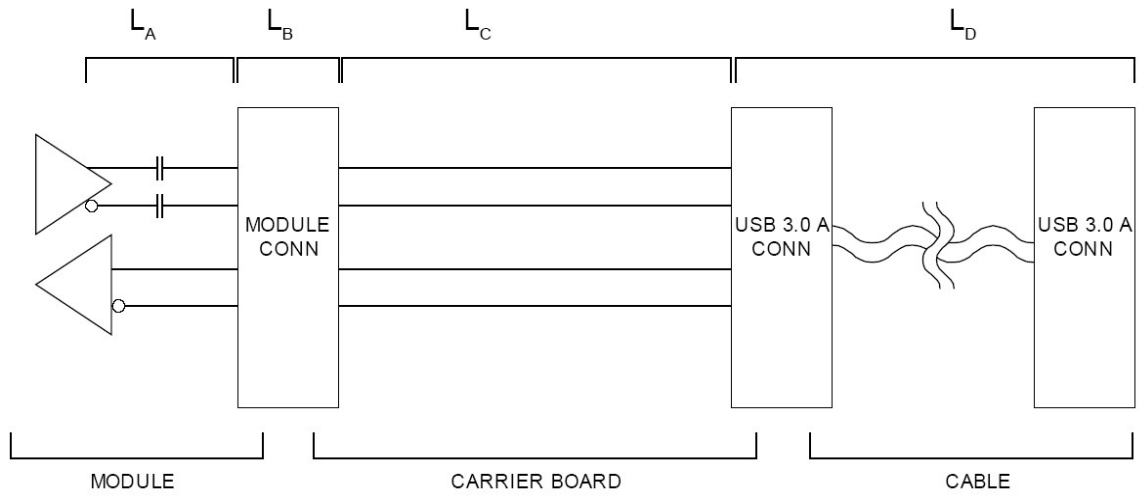


Table 29: USB3.0 Insertion Loss Budget

Segment	Loss (dB)	Notes
L_A	1.94	Up to 3 inches of Module trace @ 2.5 GHz
L_B	1.20	COM Express connector at 2.5 GHz
L_C	3.64	Up to 5 inches of Carrier Board trace @ 2.5 GHz with Common-Mode Component
Total	6.78	

2.8.2.1 USB3.0/USB3.1 Differential Transitional Via Recommendations

Please refer to 2.3.4.2

2.8.2.2 USB3.1 General Guidelines

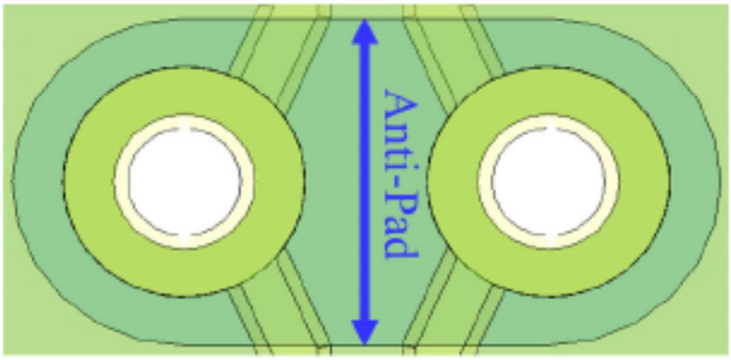
• Trace Geometry: For Gen2, it is recommended to use 85-ohm trace geometry (with larger trace width), primarily to address routing insertion loss at 10Gbps. For Gen1, it is recommended to use 85-ohm trace

geometry (narrower trace width), for PCB real estate saving. It is worth noting that Gen2 with 85-ohm trace geometry is fully functional and backward compatible with Gen1 5Gbps signaling.

• Reference plane: Continuous Ground.

• Via stub: For USB3.1 gen 2, via stub length < 15mils.

• Via anti-pad: Oval anti-pad size of 40mils is required for better impedance matching.



• AC capacitor value: 100nF nominal (75-265nF range).

• It is strongly recommended to use balanced ground vias to reduce introduction of common-mode noise. The distance from the GND to signal vias must remain symmetrical (vias aligned so that a horizontal line could be drawn intersecting the center of all four vias).

• CMC: CMC is not needed for Rx lanes.

• ESD: ESD may/may not be required depending on the 3rd party's device. On the removal of discrete ESD, there are two requirements have to be met:
(a) Mux/Re-driver can handle the ESD at least 8kV
(b) Mux/Re-driver to be placed near to USB-C connector (< 1")
Refer to 3rd party component specification.

• Distance between coils should be greater than 12 mils (recommended is 15 mils).

2.8.3 USB3.0/USB3.1 Trace Length Guidelines

Figure 20: Topology for USB3.0/USB3.1

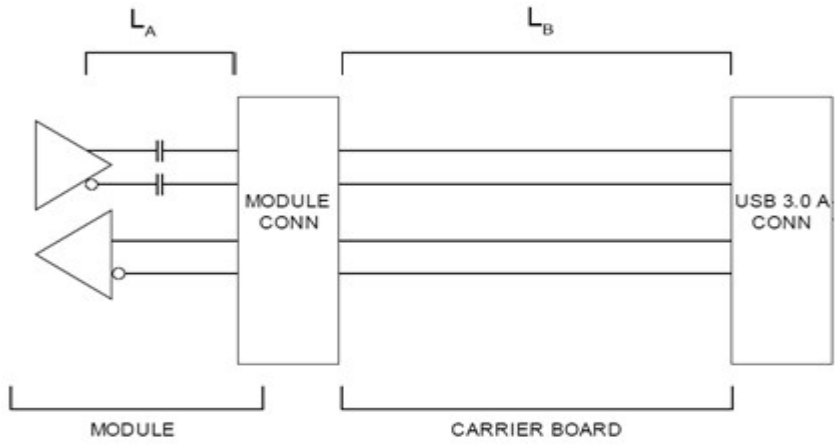


Table 30: USB3.0 Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	USB3.0/USB3.1	
Differential Impedance Target	85 Ω ±10%	
Single End	50Ω ±10%	
Spacing between pairs-to-pairs (inter-pair) (s)	Min. 15 mils	
Spacing between differential pairs and high-speed periodic signals	Min. 15 mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20 mils	
Trace spacing	21 mils (MS) and 15 mils (DS)	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	USB3.1 Gen1 to type-A connector : 8" USB3.1 Gen2 to type-A connector : 5 " USB3.1 Gen1 to front panel connector : 3.5"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils	
Reference Plane	GND referencing preferred	
Spacing from edge of plane	Min. 15mils	
Carrier Board Via Usage	Try to minimize number of vias.	

Notes:

2.9 SATA

Support for up to four SATA ports is defined on the COM Express A-B connector. Support for a minimum of two ports is required for all Module Types. Serial ATA links for support of existing SATA-150 (revision 1.0, 1.5Gb/s), SATA-300 (revision 2.0, 3Gb/s), and SATA-600 (revision 3.0, 6Gb/s) devices. The COM Express Specification addresses both in the section on insertion losses.

SATA devices can be internal to the system or external. The eSATA specification defines the connector used for external SATA devices. The eSATA interface must be designed to prevent damage from ESD, comply with EMI limits, and withstand more insertion/removals cycles than standard SATA. A specific eSATA connector was designed to meet these needs. The Esata connector does not have the "L" shaped key, and because of this, SATA and eSATA cables cannot be interchanged.

2.9.1 SATA Signal Definitions

Table 31: SATA Signal Definitions

Signal	Pin#	Description	I/O	Note
SATA0_RX+ SATA0_RX-	A19 A20	Serial ATA channel 0, Receive input differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA0 Conn pin 6 RX+ Connect to SATA0 Conn pin 5 RX- N/C if not used.	I SATA	
SATA0_TX+ SATA0_TX-	A16 A17	Serial ATA channel 0, Transmit output differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA0 Conn pin 2 TX+ Connect to SATA0 Conn pin 3 TX- N/C if not used.	O SATA	
SATA1_RX+ SATA1_RX-	B19 B20	Serial ATA channel 1, Receive input differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA1 Conn pin 6 RX+ Connect to SATA1 Conn pin 5 RX- N/C if not used.	I SATA	
SATA1_TX+ SATA1_TX-	B16 B17	Serial ATA channel 1, Transmit output differential pair. Module has integrated AC Coupling capacito Carrier Board: Connect to SATA1 Conn pin 2 TX+ Connect to SATA1 Conn pin 3 TX- N/C if not used.	O SATA	

Signal	Pin#	Description	I/O	Note
SATA2_RX+ SATA2_RX-	A25 A26	Serial ATA channel 2, Receive input differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA2 Conn pin 6 RX+ Connect to SATA2 Conn pin 5 RX- N/C if not used.	I SATA	1
SATA2_TX+ SATA2_TX-	A22 A23	Serial ATA channel 2, Transmit output differential pair. Module has integrated AC Coupling capacito Carrier Board: Connect to SATA2 Conn pin 2 TX+ Connect to SATA2 Conn pin 3 TX- N/C if not used.	O SATA	1
SATA3_RX+ SATA3_RX-	B25 B26	Serial ATA channel 3, Receive input differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA3 Conn pin 6 RX+ Connect to SATA3 Conn pin 5 RX- N/C if not used.	I SATA	1
SATA3_TX+ SATA3_TX-	B22 B23	Serial ATA channel 3, Transmit output differential pair. Module has integrated AC Coupling capacitor Carrier Board: Connect to SATA3 Conn pin 2 TX+ Connect to SATA3 Conn pin 3 TX- N/C if not used.	O SATA	1
SATA_ACT#	A28	Serial ATA activity LED. Open collector output pin driven during SATA command activity. Module has integrated PU resistor 10K ohm to 3.3V. Carrier Board: Connect to LED and current limiting resistors 250 to 330 Ω to 3.3V N/C if not used.	I/O 3.3V CMOS OC	Able to drive 10 mA

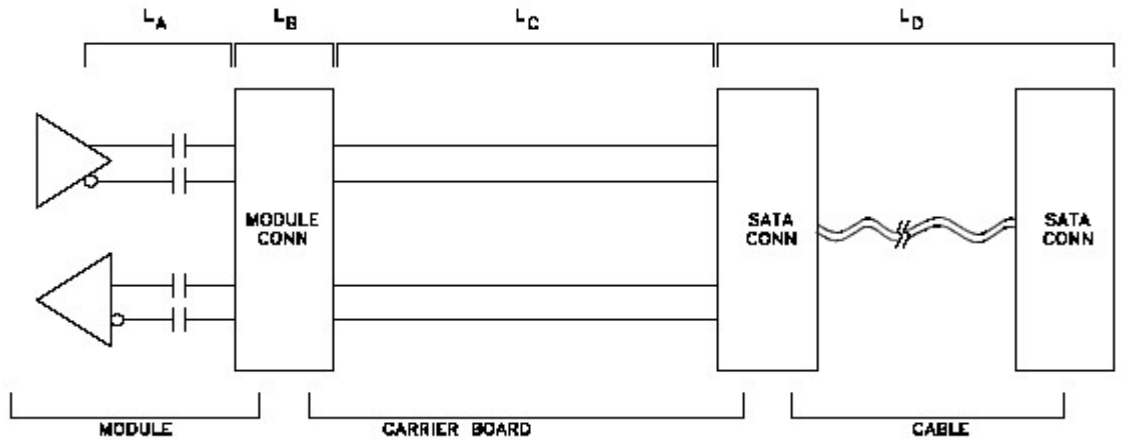
Notes:

1. SOM-5871 support SATA Gen3 2-port only.

2.9.2 SATA Routing Guidelines

SATA Insertion Loss Budget

Figure 21: SATA Insertion Loss Budge



The Serial ATA source specification provides insertion loss figures only for the SATA cable. There are several cable types defined with insertion losses ranging from 6 dB up to 16 dB. Cross talk losses are separate from material losses in the SATA specification.

The COM Express SATA Insertion loss budgets presented below represent the material losses and do not include cross talk losses. The COM Express SATA Insertion loss budgets are a guideline: module and Carrier Board vendors should not exceed the values shown in the tables below.

Table 32: SATA Insertion Loss Budge

SATA Gen 1 Insertion Loss Budget, 1.5 GHz

Segment	Loss (dB)	Notes
L_A	1.26	Up to 3.0 inches of module trace @ 0.28 dB / GHz / inch
Coupling Caps	0.40	
L_B	0.25	COM Express™ connector at 1.5 GHz measured value
L_C	3.07	Up to 7.2 inches of Carrier Board trace @ 0.28 dB / GHz / inch
L_D	6.00	Source specification cable and cable connector allowance
Total	10.98	

SATA Gen 2 Insertion Loss Budget, 3.0 GHz

Segment	Loss (dB)	Notes
L_A	1.68	Up to 2.0 inches of module trace @ 0.28 dB / GHz / inch
Coupling Caps	0.40	
L_B	0.38	COM Express™ connector at 3.0 GHz measured value
L_C	2.52	Up to 3.0 inches of Carrier Board trace @ 0.28 dB / GHz / inch
L_D	6.00	Source specification cable and cable connector allowance
Total	10.98	

2.9.2.1 General SATA Routing Guidelines

Use the following general routing and placement guidelines when laying out a new design.

- SATA signals must be ground referenced. If changing reference plane is completely unavoidable (that is, ground reference to power reference), proper placement of stitching caps can minimize the adverse effects of EMI and signal quality performance caused by reference plane change. Stitching capacitors are smallvalued capacitors (1 µF or lower in value) that bridge the power and ground planes close to where a high-speed signal changes layers. Stitching caps provide a high frequency current return path between different reference planes. They minimize the impedance discontinuity and current loop area that crossing different reference planes created. The maximum number allowed for SATA to change reference plane is one.
- Route all traces over continuous GND planes, with no interruptions. Avoid crossing over anti-etch if at all possible. Any discontinuity or split in the ground plane can cause signal reflections and should be avoided.
- Minimize layer changes. If a layer change is necessary, ensure that trace matching for either transmit or receive pair occurs within the same layer. Intel recommends to use SATA vias as seldom as possible.
- DO NOT route SATA traces under power connectors, other interface connectors, crystals, oscillators, clock synthesizers, magnetic devices or ICs that use and/or duplicate clocks.
- DO NOT place stubs, test points, test vias on the route to minimize reflection. Utilize vias and connector pads as test points instead.
- For testability, route the TX and RX pairs for a given port on the same layer and close to each other to help ensure that the pairs share similar signaling characteristics. If the groups of traces are similar, a measure of RX pair layout quality can be approximated by using the results from actively testing the TX pair's signal quality.
- Length matching rules are required on SATA differential signals for optimum timing margins, preventing common-mode signals and EMI. Each net within a differential pair should be length matched on a segment-by-segment basis at the point of discontinuity. Total length mismatch must not be more than 20 mils (0.508 mm). Examples of segments might include breakout areas, routes running between two vias, routes between an AC coupling capacitor and a connector pin, etc. The points of discontinuity would be the via, the capacitor pad, or the connector pin. Matching of TX and RX within the same port and between SATA TX and RX pairs from differential ports is not required. When length matching compensation occurs, it should be made as close as possible to the point where the variation occurs.
- DO NOT serpentine to match RX and TX traces; there is NO requirement to match RX and TX traces. In addition, DO NOT serpentine to meet minimum length guidelines on RX and TX traces.
- Recommend keeping SATA traces 20 mils (0.508 mm) from any vias on the motherboard whenever possible.

2.9.2.2 SATA Differential Transitional Via Recommendations

Please refer to 2.3.4.2

2.9.3 SATA Trace Length Guidelines

Figure 22: Topology for SATA

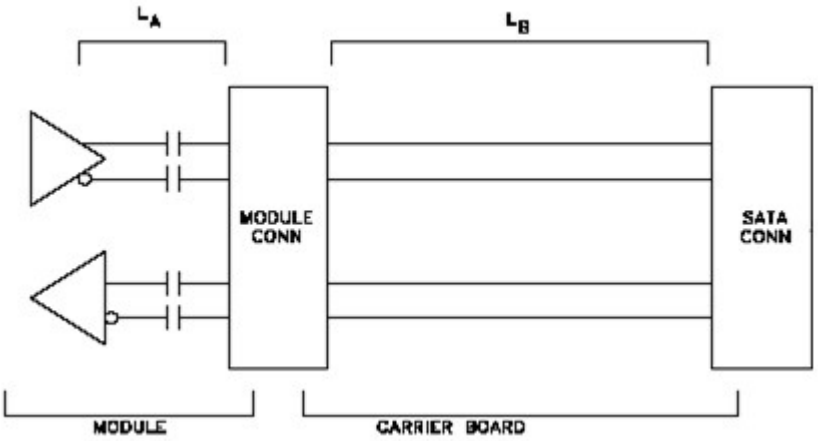


Table 33: SATA Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	SATA	
Differential Impedance Target	90 Ω ±10%	
Single End	50Ω ±10%	
Signal length available for the COM Express Carrier Board	3 inches, a redriver may be necessary for GEN3 signaling rates	
Spacing between RX and TX pairs (inter-pair) (s)	Min. 20 mils	
Spacing between differential pairs and high-speed periodic signals	Min. 50 mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20 mils	
Trace spacing	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	SATA Gen3 to SATA connector: 6" SATA Gen3 to U.2 connector : 6" SATA Gen2 to M.2 connector : 7"	
Length matching	Differential pairs (intra-pair): Max. ±5 mils	
Reference Plane	GND referencing preferred	
Spacing from edge of plane	Min. 15mils	
Carrier Board Via Usage	Try to minimize number of vias.	

Notes:

2.10 LVDS

2.10.1 Signal Definitions

The COM Express Specification provides an optional LVDS interface on the COM Express A-B connector. Module pins for two LVDS channels are defined and designated as LVDS_A and LVDS_B.

Systems use a single-channel LVDS for most displays. Dual LVDS channels are used for very high-bandwidth displays. Single-channel LVDS means that one complete RGB pixel is transmitted per display input clock. Dual-channel LVDS means that two complete RGB pixels are transmitted per display input clock. The two pixels are adjacent along a display line. Dual-channel LVDS does not mean that two LVDS displays can be driven.

Each COM Express LVDS channel consists of four differential data pairs and a differential clock pair for a total of five differential pairs per channel. COM Express Modules and Module chipsets may not use all pairs. For example, with 18-bit TFT displays, only three of the four data pairs on the LVDS_A channel are used, along with the LVDS_A clock. The LVDS_B lines are not used. The manner in which RGB data is packed onto the LVDS pairs (including packing order and color depth) is not specified by the COM Express Specification. This may be Module-dependent.

There are five single-ended signals included to support the LVDS interface: two lines are used for an I2C interface that may be used to support EDID or other panel information and identification schemes. Additionally, there are an LVDS power enable (LVDS_VDD_EN) and backlight control and enable lines (LVDS_BKLT_CTRL and LVDS_BKLT_EN).

Table 34: LVDS Signal Definitions

Signal	Pin#	Description	I/O	Note
LVDS_A0+ LVDS_A0-	A71 A72	LVDS channel A differential signal pair 0 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinO0+/- with 100Ω termination Conn. – RxinO0+/- N/C if not used	O LVDS	
LVDS_A1+ LVDS_A1-	A73 A74	LVDS channel A differential signal pair 1 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinO1+/- with 100Ω termination Conn. – RxinO1+/- N/C if not used	O LVDS	

Signal	Pin#	Description	I/O	Note
LVDS_A2+ LVDS_A2-	A75 A76	LVDS channel A differential signal pair 2 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinO2+/- with 100Ω termination Conn. – RxinO2+/- N/C if not used	O LVDS	
LVDS_A3+ LVDS_A3-	A78 A79	LVDS channel A differential signal pair 3 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinO3+/- with 100Ω termination Conn. – RxinO3+/- N/C if not used	O LVDS	
LVDS_A_CK+ LVDS_A_CK-	A81 A82	LVDS channel A differential clock pair Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RXOC+/- with 100Ω termination Conn. – RXOC+/- N/C if not use	O LVDS	
LVDS_B0+ LVDS_B0-	B71 B72	LVDS channel B differential signal pair 0 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinE0+/- with 100Ω termination Conn. – RxinE0+/- N/C if not used	O LVDS	
LVDS_B1+ LVDS_B1-	B73 B74	LVDS channel B differential signal pair 1 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinE1+/- with 100Ω termination Conn. – RxinE1+/- N/C if not used	O LVDS	
LVDS_B2+ LVDS_B2-	B75 B76	LVDS channel B differential signal pair 2 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinE2+/- with 100Ω termination Conn. – RxinE2+/- N/C if not used	O LVDS	

Signal	Pin#	Description	I/O	Note
LVDS_B3+ LVDS_B3-	B77 B78	LVDS channel B differential signal pair 3 Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RxinE3+/- with 100Ω termination Conn. – RxinE3+/- N/C if not used	O LVDS	
LVDS_B_CK+ LVDS_B_CK-	B81 B82	LVDS channel B differential clock pair Carrier Board: Connect 100Ω @100MHz Common Choke in series to Reciever – RXEC+/- with 100Ω termination Conn. – RXEC+/- N/C if not used	O LVDS	
LVDS_VDD_EN	A77	LVDS flat panel power enable. Carrier Board: Connect to enable control of LVDS panel power circuit. N/C if not used	O 3.3V, CMOS	
LVDS_BKLT_EN	B79	LVDS flat panel backlight enable high active signal Carrier Board: Connect to enable control of LVDS panel backlight power circuit. N/C if not used	O 3.3V, CMOS	
LVDS_BKLT_CTRL	B83	LVDS flat panel backlight brightness control Carrier Board: Connect to brightness control of LVDS panel backlight power circuit. N/C if not used	O 3.3V, CMOS	
LVDS_I2C_CK	A83	DDC I2C clock signal used for flat panel detection and control. Carrier Board: Connect to DDC clock of LVDS panel N/C if not used	O 3.3V, CMOS	
LVDS_I2C_DAT	A84	DDC I2C data signal used for flat panel detection and control. Carrier Board: Connect to DDC data of LVDS panel N/C if not used	I/O 3.3V, OD CMOS	

2.10.1.1 Display Timing Configuration

The graphic controller needs to be configured to match the timing parameters of the attached flat panel display. To properly configure the controller, there needs to be some method to determine the display parameters. Different Module vendors provide differing ways to access display timing parameters. Some vendors store the data in non-volatile memory with the BIOS setup screen as the method for entering the data, other vendors might use a Module or Carrier based EEPROM. Some vendors might hard code the information into the BIOS, and other vendors might support panel located timing via the signals LVDS_I2C_CK and LVDS_I2C_DAT with an EEPROM strapped to 1010 000x. Regardless of the method used to store the panel timing parameters, the video BIOS will need to have the ability to access and decode the parameters. Given the number of variables it is recommended that Carrier designers contact Module suppliers to determine the recommend method to store and retrieve the display timing parameters.

The Video Electronics Standards Association (VESA) recently released DisplayID, a second generation display identification standard that can replace EDID and other proprietary methods for storing flat panel timing data. DisplayID defines a data structure which contains information such as display model, identification information, colorimetry, feature support, and supported timings and formats. The DisplayID data allows the video controller to be configured for optimal support for the attached display without user intervention. The basic data structure is a variable length block up to 256 bytes with additional 256 byte extensions as required. The DisplayID data is typically stored in a serial EPROM connected to the LVDS_I2C bus. The EPROM can reside on the display or Carrier. DisplayID is not backwards compatible with EDID. Contact VESA (www.vesa.org) for more information.

2.10.1.2 Backlight Control

Backlight inverters are either voltage, PWM or resistor controlled. The COM Express specification provides two methods for controlling the brightness. One method is to use the backlight control and enable signals from the CPU chipset. These signals are brought on COM Express LVDS_BKLT_EN and LVDS_BKLT_CTRL. LVDS_BKLT_CTRL is a Pulse Width Modulated (PWM) output that can be connected to display inverters that accept a PWM input. The second method it to use the LVDS I2C bus to control an I2C DAC. The output of the DAC can be used to support voltage controlled inverters. The DAC can be used driving the backlight voltage control input pin of the inverter.

2.10.2 LVDS Routing Guidelines

Route LVDS signals as differential pairs (excluding the five single-ended support signals), with a 100-Ω differential impedance and a 55-Ω, single-ended impedance. Ideally, a LVDS pair is routed on a single layer adjacent to a ground plane. LVDS pairs should not cross plane splits. Keep layer transitions to a minimum. Reference LVDS pairs to a power plane if necessary. The power plane should be well-bypassed.

Length-matching between the two lines that make up an LVDS pair ("intra-pair") and between different LVDS pairs ("inter-pair") is required. Intra-pair matching is tighter than the inter-pair matching.
All LVDS pairs should have the same environment, including the same reference plane and the same number of vias.

2.10.3 LVDS Trace Length Guidelines

Figure 23: Topology for LVDS

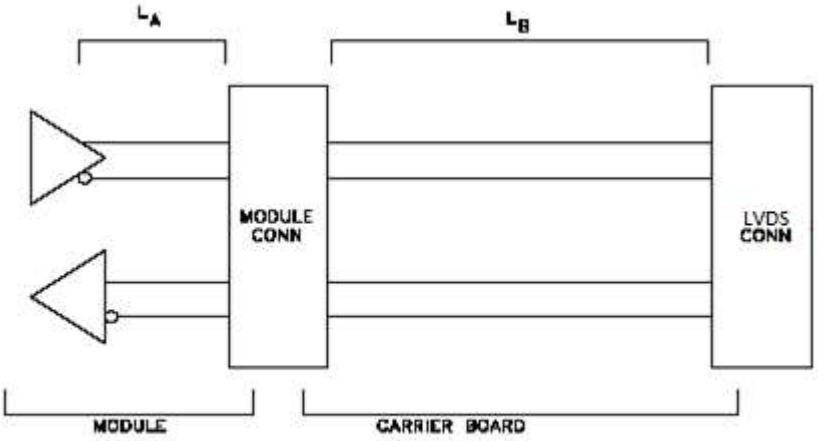


Table 35: LVDS Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	LVDS	
Differential Impedance Target	85 Ω ±10%	
Single End	50Ω ±10%	
Signal length to the LVDS connector available for the COM Express Carrier Board	4"	
Spacing between pair to pairs (inter-pair) (s)	Min. 20 mils	
Spacing between differential pairs and high-speed periodic signals	Min. 30 mils	
Spacing between differential pairs and low-speed non periodic signals	Min. 20 mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	4"	
Length matching	Differential pairs (intra-pair): Max. 5 mils Clock and data pairs (intra-pair): Max. 100 mils data pairs (inter-pair) : Max. 100 mils	
Reference Plane	GND referencing preferred	
Spacing from edge of plane	Min. 40mils	
Carrier Board Via Usage	Max. of 2 vias per line.	

Notes:

2.11 Embedded DisplayPort (eDP) *SOM-5871 is Option function

Embedded DisplayPort (Edp) is a digital display interface standard produced by the Video Electronics Standards Association (VESA) for digital interconnect of Audio and Video.

Embedded DisplayPort defines a standardized display panel interface for internal connections; e.g., graphics interfaces to notebook display panels. It supports advanced power-saving features including seamless refresh rate switching, display panel and backlight control protocol that works through the AUX channel, and Panel Self-Refresh (PSR) feature developed to save system power and further extend battery life in portable PC systems. PSR mode allows the GPU to enter power saving states in between frame updates by including framebuffer memory in the display panel controller.

Embedded DisplayPort is intended to replace LVDS as the interface to flat panel displays integrated into a product. Unlike DisplayPort, embedded DisplayPort does not define a specific connector or pin-out. The COM Express specification shares the LVDS pins with embedded DisplayPort.

2.11.1 eDP Signal Definitions

Edp is available in Type 6 and type 10 pin-outs as an alternative to the LVDS A channel. The Module can provide LVDS only, Edp only or Dual-Mode for both interfaces. Please refer to relevant Module documentation for the supported interfaces.

Table 36: Edp Signal Definitions

Signal	Pin#	Description	I/O	Note
Edp_TX0+ Edp_TX0-	A75 A76	Edp lane 0, TX +/- Carrier Board: Connect AC Coupling Capacitors 75~200 Nf near Edp connector. N/C if not used	O PCIe	1
Edp_TX1+ Edp_TX1-	A73 A74	Edp lane 1, TX +/- Carrier Board: Connect AC Coupling Capacitors 75~200 Nf near Edp connector. N/C if not used	O PCIe	1
Edp_TX2+ Edp_TX2-	A71 A72	Edp lane 2, TX +/- Carrier Board: Connect AC Coupling Capacitors 75~200 Nf near Edp connector. N/C if not used	O PCIe	1

Signal	Pin#	Description	I/O	Note
Edp_TX3+ Edp_TX3-	A81 A82	Edp lane 3, TX +/- Carrier Board:	O PCIe	1

		Connect AC Coupling Capacitors 75~200 Nf near Edp connector. N/C if not used		
Edp_VDD_EN	A77	Edp power enable Carrier Board: Connect to enable control of Edp panel power circuit. N/C if not used	O CMOS	1
Edp_BLKT_EN	B79	Edp backlight enable Carrier Board: Connect to enable control of Edp panel backlight power circuit. N/C if not used	O CMOS	1
Edp_BLKT_CTRL	B83	EDP backlight brightness control Carrier Board: Connect to brightness control of Edp panel backlight power circuit. N/C if not used	O CMOS	1
Edp_AUX+ Edp_AUX-	A83 A84	Edp auxiliary lane +/- Edp_AUX+: Module has integrated PU resistor 100K ohm to 3.3V. Edp_AUX-: Module has integrated PD resistor 100K ohm to GND. Carrier Board: Connect AC Coupling Capacitors 75~200 Nf near Edp connector. Connect to device or Edp connector. N/C if not used.	I/O PCIe	1
Edp_HPD	A87	Detection of Hot Plug / Unplug and notification of the link layer Carrier Board: Connector to device or Edp connector HP pin.	I CMOS	1

Notes:

SOM-5871 default is not support eDP, it is option function. If customers want to use eDP function, please contact Advantech.

2.11.2 eDP Trace Length Guidelines

Figure 24: Topology for eDP

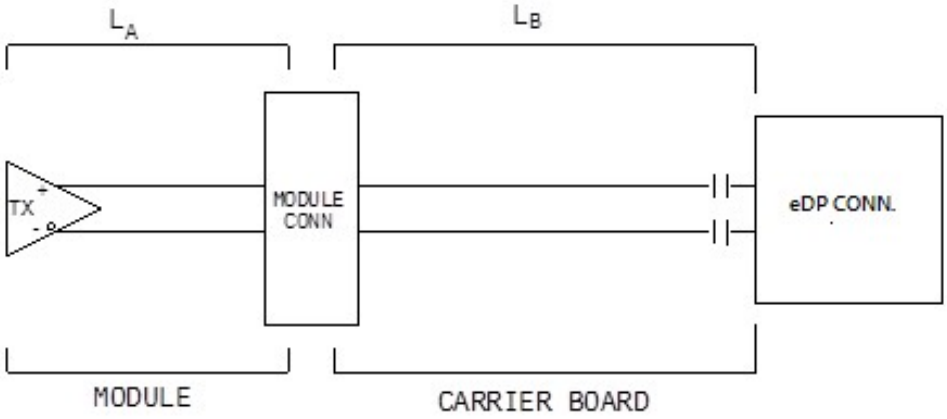


Table 37: DisplayPort Connector / Device Down Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	eDP	2
Differential Impedance Target	85 Ω ±10%	2
Single End	50Ω ±10%	2
Isolation to equivalent pairs	20 mils (MS) and 15 mils (DS)	2
Isolation to other signal groups	20 mils (MS) and 15 mils (DS)	2
Trace spacing	15 mils (MS) and 12 mils (DS)	2
LA	SOM-5871 option function	2
LB	Carrier Board Length	2
Max length of LA+LB	eDP differential pairs to eDP connector: 7"	2
Length matching	Differential pairs (intra-pair): Max. ±5 mils For each channel, match the lengths of the differential pairs (Inter-Pair) to be within a 3-inch window (max length – min length < 3 inch), include AUX channel.	2
Reference Plane	GND referencing preferred. Min 15-mil trace edge-to-major plane edge spacing.	2
Carrier Board Via Usage	Max. 2 vias.	2
AC coupling	Min = 75 nF Max = 200 nF	1, 2

Notes:

1. AC caps are recommended to be placed close to device side (avoid placing AC cpas on mid-bus).
2. SOM-5871 default is not support eDP, it is option function. If customers want to use eDP function, please contact Advantech.

2.12 VGA

2.12.1 VGA Signal Definitions

The interface consists of three analog color signals (Red, Green, Blue); digital Horizontal and Vertical Sync

signals as well as a dedicated I2C bus for Display Data Control (DDC) implementation for monitor capability identification. The corresponding signals can be found on the COM Express Module connector row B.

Table 38: VGA Signal Definitions

Signal	Pin#	Description	I/O	Note
VGA_RED	B89	Red component of analog DAC monitor output, designed to drive a 37.5Ω equivalent load. Carrier Board: Connect to 1. 150Ω±1% resistor to GND 2. 2-pole filter(10 pF,bead 47 Ω @100MHz and 22 pF) 3. ESD diode protection connected to 3.3V or 5V. to VGA Conn Pin 1 RED N/C if not used	O Analog	
VGA_GRN	B91	Green component of analog DAC monitor output, designed to drive a 37.5Ω equivalent load. Carrier Board: Connect to 1. 150Ω±1% resistor to GND 2. 2-pole filter(10 pF,bead 47 Ω @100MHz and 22 pF) 3. ESD diode protection connected to 3.3V or 5V. to VGA Conn Pin 2 GREEN N/C if not used	O Analog	
VGA_BLU	B92	Blue component of analog DAC monitor output, designed to drive a 37.5Ω equivalent load. Carrier Board: Connect to 1. 150Ω±1% resistor to GND 2. 2-pole filter(10 pF,bead 47 Ω @100MHz and 22 pF) 3. ESD diode protection connected to 3.3V or 5V. to VGA Conn Pin 3 BLUE N/C if not used	O Analog	

Signal	Pin#	Description	I/O	Note
VGA_HSYNC	B93	Horizontal sync output to VGA monitor. Carrier Board:	O 3.3V CMOS	

		Connect 1. Bead 150~220Ω@100MHz in series 2. Bypass 22pF to GND. 3. ESD diode protection connected to 3.3V or 5V. to VGA Conn Pin 13 Hsync N/C if not used		
VGA_VSYNC	B94	Vertical sync output to VGA monitor. Carrier Board: Connect 1. Bead 150~220Ω@100MHz in series 2. Bypass 22pF to GND. 3. ESD diode protection connected to 3.3V or 5V. to VGA Conn Pin 14 Vsync N/C if not used	O 3.3V CMOS	
VGA_I2C_CK	B95	DDC clock line (I2C port dedicated to identify VGA monitor capabilities). Carrier Board: Connect to VGA Conn Pin 15 SCL with ESD diode to 5V near connector N/C if not used	O 3.3V CMOS	Level shifter might be necessary
VGA_I2C_DAT	B96	DDC data line. Carrier Board: Connect to VGA Conn Pin 12 SDA with ESD diode to 5V near connector. N/C if not use	I/O 3.3V CMOS	Level shifter might be necessary

2.12.2 VGA Routing Guidelines

2.12.2.1 RGB Analog Signals

The RGB signal interface of the COM Express Module consists of three identical 8-bit digital-toanalog converter (DAC) channels. One each for the red, green, and blue components of the monitor signal. Each

of these channels should have a 150Ω ±1% pull-down resistor connected from the DAC output to the Carrier Board ground. A second 150Ω ±1% termination resistor exists on the COM Express Module itself. An additional 75Ω termination resistor exists within the monitor for each analog DAC output signal. Since the DAC runs at speeds up to 350MHz, special attention should be paid to signal integrity and EMI. There should be a PI-filter placed on each RGB signal that is used to reduce highfrequency noise and EMI. The PI-filter consists of two 10pF capacitors with a 120Ω @ 100MHz ferrite bead between them. It is recommended to place the PI-filters and the terminating resistors as close as possible to the standard VGA connector.

2.12.2.2 HSYNC and VSYNC Signals

The horizontal and vertical sync signals 'VGA_HSYNC' and 'VGA_VSYNC' provided by the COM Express Module are 3.3V tolerant outputs. Since VGA monitors may drive the monitor sync signals with 5V tolerance, it is necessary to implement high impedance unidirectional buffers. These buffers prevent potential electrical over-stress of the Module and avoid that VGA monitors may attempt to drive the monitor sync signals back to the Module. For optimal ESD protection, additional low capacitance clamp diodes should be implemented on the monitor sync signals. They should be placed between the 5V power plane and ground and as close as possible to the VGA connector.

2.12.2.3 DDC Interface

COM Express provides a dedicated I2C bus for the VGA interface. It corresponds to the VESA™ defined DDC interface that is used to read out the CRT monitor specific Extended Display Identification Data (EDID™). The appropriate signals 'VGA_I2C_DAT' and 'VGA_I2C_CK' of the COM Express Module are supposed to be 3.3V tolerant. Since most VGA monitors drive the internal EDID™ EEPROM with a supply voltage of 5V, the DDC interface on the VGA connector must also be sourced with 5V. This can be accomplished by placing a 100kΩ pull-up resistors between the 5V power plane and each DDC interface line. Level shifters for the DDC interface signals are required between the COM Express Module signal side and the signals on the standard VGA connector on the Carrier Board.
Additional Schottky diodes must be placed between 5V and the pull-up resistors of the DDC signals to avoid backward current leakage during Suspend operation of the Module.

2.12.2.4 ESD Protection/EMI

All VGA signals need ESD protection and EMI filters.

2.12.3 VGA Trace Length Guidelines

Figure 25: Topology for VGA

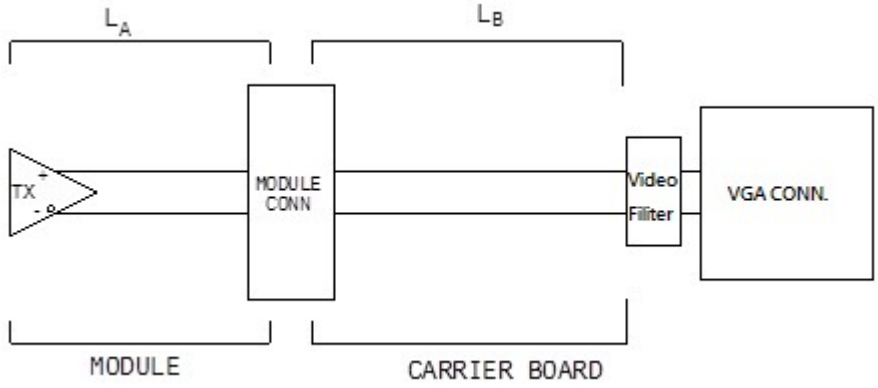


Table 39: VGA Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	VGA	
Single End	50Ω ±10%	
Nominal Trace Space within CRT DAC Signal Group	Min. 15mils	
Spacing to Other Signal Group	Min. 20mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	15.1"	
Length matching	1. Match the trace lengths within R, G, B to ±200 mils. 2. Match the trace lengths of CRT SYNC signals with CRT DAC signals to ±1000 mils	
Reference Plane	GND referencing preferred. Min 20-mil trace edge-to-major plane edge spacing.	

Notes:

1. Video filter must be near VGA connector.

2.13 Digital Audio Interfaces

The COM Express Specification allocates seven pins on the A-B connector to support HD interfaces to audio Codecs on the Carrier Board. The pins are available on all Module types.

The HDA signal level from some chipsets might be 1.5V. Module designers must add any necessary voltage translation circuitry to meet the COM Express 3.3V signaling requirement for HDA signals. The HDA codec on a COM Express Carrier Board *shall* be connected as the primary codec with the codec ID 00 using the data input line 'HDA_SDIN0'. Up to two additional codecs with ID 01 and ID 10 *may* be connected to the COM Express Module by using the other designated signals HDA_SDIN[1:2].

2.13.1 Audio Codec Signal Descriptions

Table 40: Audio Codec Signal Descriptions

Signal	Pin#	Description	I/O	Note
HDA_RST#	A30	CODEC Reset. Carrier Board: HDA - Connect 0 Ω in series to CODEC pin 11 RESET# N/C if not used	O 3.3V Suspend CMOS	
HDA_SYNC	A29	Serial Sample Rate Synchronization. Carrier Board: HDA - Connect 0 Ω in series to CODEC pin 10 SYNC N/C if not used	O 3.3V CMOS	
HDA_BITCLK	A32	24 MHz Serial Bit Clock for HDA CODEC. Carrier Board: HDA - Module has integrated series resistor. Connect 0 Ω in series to CODEC pin 6 BIT_CLK N/C if not used	O 3.3V CMOS	
HDA_SDOUT	A33	Audio Serial Data Output Stream. Carrier Board: HDA - Connect 33-47 Ω in series and PD 10KΩ(NL) to CODEC pin 5 SDATA_OUT N/C if not used	O 3.3V CMOS	
HDA_SDIN0 HDA_SDIN1 HDA_SDIN2	B30 B29 B28	Audio Serial Data Input Stream from CODEC[0:2]. Carrier Board: HDA - Connect 33-47 Ω in series and PD 10KΩ(NL) to CODEC pin 8 SDATA_IN N/C if not used	I 3.3V Suspend CMOS	

The codec on a COM Express Carrier Board is usually connected as the primary codec with the codec ID 00 using the data input line 'HDA_SDIN0'. Up to two additional codecs with ID 01 and ID 10 can be connected to the COM Express Module by using the other designated signals 'HDA_SDIN1' and 'HDA_SDIN2'.

Connect the primary audio codec to the serial data input signal 'HDA_SDIN0' and ensure that the corresponding bit clock input signal 'HDA_BITCLK' is connected to the HDA interface of the COM Express Module.

Clocking over the signal 'HDA_BITCLK' is derived from a 24.576 MHz crystal. The crystal is not required in HDA implementations. This clock also drives the second and the third audio codec if more than one codec is used in the application. For crystal or crystal oscillator requirements, refer to the datasheet of the primary codec.

2.13.2 Audio Routing Guidelines

The implementation of proper component placement and routing techniques will help to ensure that the maximum performance available from the codec is achieved. Routing techniques that should be observed include properly isolating the codec, associated audio circuitry, analog power supplies and analog ground planes from the rest of the Carrier Board. This includes split planes and the proper routing

of signals not associated with the audio section.

The following is a list of basic recommendations:
Traces must be routed with a target impedance of 50Ω with an allowed tolerance of ±15%.
Ground return paths for the analog signals must be given special consideration.
Digital signals routed in the vicinity of the analog audio signals must not cross the power plane split lines.
Locate the analog and digital signals as far as possible from each other.
Partition the Carrier Board with all analog components grouped together in one area and all digital components in another.
Keep digital signal traces, especially the clock, as far as possible from the analog input and voltage reference pins.
Provide separate analog and digital ground planes with the digital components over the digital ground plane, and the analog components, including the analog power regulators, over the analog ground plane. The split between the planes must be a minimum of 0.05 inch wide.
Route analog power and signal traces over the analog ground plane.
Route digital power and signal traces over the digital ground plane.
Position the bypassing and decoupling capacitors close to the IC pins with wide traces to reduce impedance.
Place the crystal or oscillator (depending on the codec used) as close as possible to the codec. (HDA implementations generally do not require a crystal at the codec)
Do not completely isolate the analog/audio ground plane from the rest of the Carrier Board ground plane. Provide a single point (0.25 inch to 0.5 inch wide) where the analog/isolated ground plane connects to the main ground plane. The split between the planes must be a minimum of 0.05 inch wide. Any signals entering or leaving the analog area must cross the ground split in the area where the analog ground is attached to the main Carrier Board ground. That is, no signal should cross the split/gap between the ground planes, because this would cause a ground loop, which in turn would greatly increase EMI emissions and degrade the analog and digital signal quality.

2.13.3 Audio Trace Length Guidelines

Figure 26: Topology for Audio

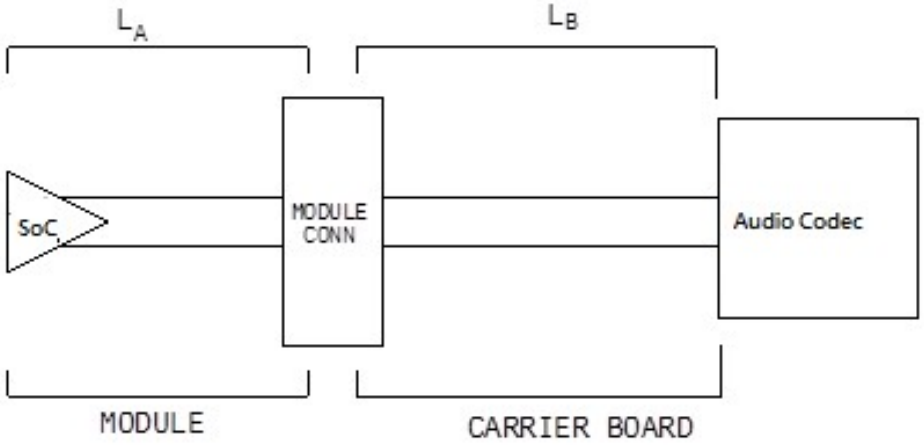


Table 41: Audio Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	Audio	
Single End	50Ω ±15%	
Nominal Trace Space within Audio Signal Group	Min. 15mils	
Spacing to Other Signal Group	Min. 20mils	
Trace spacing	Min. 9mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	11.5"	
Length matching	Data to Clock must be matched within 500mils	
Reference Plane	GND referencing preferred. Min 15-mil trace edge-to-major plane edge spacing.	
Carrier Board Via Usage	Try to minimize number of vias.	

Notes:

2.14 LPC and eSPI Interface *SOM-5871 is not support eSPI.

The Module LPC and eSPI interfaces share connector pins. A Module design *may* support either LPC or eSPI or both, at the Module vendor's discretion. Module pin ESPI_EN# is available for the Carrier to signal to the Module whether LPC or eSPI is to be used. The Carrier *shall* leave the ESPI_EN# unconnected on

the Carrier for LPC operation. The Carrier ***shall*** tie ESPI_EN# to GND for eSPI operation. The Module ***shall*** terminate ESPI_EN# as appropriate to facilitate this.

The LPC bus is a 3.3V bus and eSPI is a 1.8V bus. There is the possibility of a mismatch – an eSPI only Module mated with a LPC only Carrier, or an LPC only Module on an eSPI Carrier. Module designers ***should*** protect the Module eSPI interface against accidental exposure to 3.3V Carrier LPC signals. Carrier designers ***should*** protect a Carrier eSPI interface against accidental exposure to 3.3V Module LPC signals. In both cases, a simple and low cost protection scheme ***may*** be realized with low value in-line series resistors (typically 33 ohms) and BAT54 Schottky diodes on each line. The diode anode is tied to the eSPI device pin and the cathode to the 1.8V supply rail. Ideally, that 1.8V supply rail can sink current. In the event of a mismatch, the offending (Module or Carrier) 3.3V rail is discharged through the series resistor and the Schottky diode to the (Carrier or Module) 1.8V rail and not through the eSPI device.

2.14.1 LPC /eSPI Signal Definition

Table 42: LPC/eSPI Interface Signal Definition

Signal	Pin#	Description	I/O	Note
LPC_SERIRQ	A50	LPC serialized IRQ. Carrier Board: Connect to LPC - SERIRQ N/C if not used	I/O 3.3V CMOS	
ESPI_CS1#		ESPI Mode: eSPI Master Chip Select Outputs Driving Chip Select# A low selects a particular eSPI slave for the transaction. Each of the eSPI slaves is connected to a dedicated Chip Selectn# pin. Carrier Board: Connect to eSPI Device – eSPI_CS1# N/C if not used	O 1.8V Suspend / 1.8V	2

Signal	Pin#	Description	I/O	Note
LPC_FRAME#	B3	LPC frame indicates start of a new cycle or termination of a broken cycle. Carrier Board: LPC - LFRAME# N/C if not used	O 3.3V CMOS	

ESPI_CS0#		ESPI Mode: eSPI Master Chip Select Outputs Driving Chip Select0#. A low selects a particular eSPI slave for the transaction. Each of the eSPI slaves is connected to a dedicated Chip Selectn# pin. Carrier Board: Connect to eSPI Device – eSPI_CS0# N/C if not used	O 1.8V Suspend / 1.8V	2
LPC_AD0 LPC_AD1 LPC_AD2 LPC_AD3	B4 B5 B6 B7	LPC multiplexed command, address and data. Carrier Board: Connect to LPC - LAD0 , LAD1, LAD2, LAD3 N/C if not used	I/O 3.3V CMOS	
ESPI_IO_0 ESPI_IO_1 ESPI_IO_2 ESPI_IO_3		ESPI Mode: eSPI Master Data Input / Outputs These are bi-directional input/output pins used to transfer data between master and slaves. Multiplexed with LPC_AD[0:3] Carrier Board: Connect to eSPI Device – eSPI_IO0, eSPI_IO1, eSPI_IO2, eSPI_IO3, the Carrier ***shall*** have a 33 Ohm series termination. N/C if not used	I/O 1.8V Suspend / 1.8V	2
LPC_DRQ0# LPC_DRQ1#	B8 B9	LPC encoded DMA/Bus master request. Carrier Board: Connect to LPC - LDRQ0#, LDRQ1# N/C if not used	I 3.3V CMOS	Not all Modules support LPC DMA. Contact your vendor for information.
ESPI_ALERT0# ESPI_ALERT1#		ESPI Mode: eSPI pins used by eSPI slave to request service from the eSPI master. Carrier Board: Connect to eSPI Device – eSPI_ALERT0#, eSPI_ALERT1#. N/C if not used	I 1.8V Suspend / 1.8V	2

Signal	Pin#	Description	I/O	Note
ESPI_RESET# /SUS_STAT#	B18	ESPI Mode: eSPI Reset Reset the eSPI interface for both master and slaves. eSPI Reset# is typically driven from eSPI master to eSPI slaves.	O 1.8V Suspend / 1.8V	2

		Carrier Board: Connect to eSPI Device – eSPI_RESET# N/C if not used		
ESPI_EN#	B47	This signal is used by the Carrier to indicate the operating mode of the LPC/eSPI bus. If left unconnected on the carrier, LPC mode (default) is selected. If pulled to GND on the carrier, eSPI mode is selected. This signal is pulled to a logic high on the module through a resistor. The Carrier *should* only float this line or pull it low. Carrier Board: Connect to N/C : LPC mode GND: eSPI mode	I NA CMOS	2
LPC_CLK	B10	LPC clock output 33MHz. Carrier Board: Connect to LPC - LCLK N/C if not used	O 3.3V CMOS	
ESPI_CK		ESPI Mode: eSPI Master Clock Output This pin provides the reference timing for all the serial input and output operations. Carrier Board: Connect to eSPI Device – eSPI_CLK, the Carrier ***shall*** have a 33 Ohm series termination. N/C if not used	O 1.8V Suspend / 1.8V	2

Note:

1. Implementing external LPC devices on the COM Express Carrier Board always requires customization of the COM Express Module's BIOS in order to support basic initialization for those LPC devices. Otherwise the functionality of the LPC devices will not be supported by a Plug&Play or ACPI capable system.

2. SOM-5871 is not support eSPI.

2.14.2 LPC Routing Guidelines

2.14.2.1 General Signals

LPC signals are similar to PCI signals and may be treated similarly. Route the LPC bus as 50 Ω, single-ended signals. The bus may be referenced to ground (preferred), or to a well-bypassed power plane or a combination of the two. Point-to-point (daisy-chain) routing is preferred, although stubs up to

1.5 inches may be acceptable. Length-matching among LPC_AD[3:0], LPC_FRAME# are needed

2.14.2.2 Bus Clock Routing

Route the LPC clock as a single-ended, 50 Ω trace with generous clearance to other traces and to itself. A continuous ground-plane reference is recommended. Routing the clock on a single ground referenced internal layer is preferred to reduce EMI.

The LPC clock implementation should follow the routing guidelines for the PCI clock defined in the COM Express specification and the 'PCI Local Bus Specification Revision 2.3'.

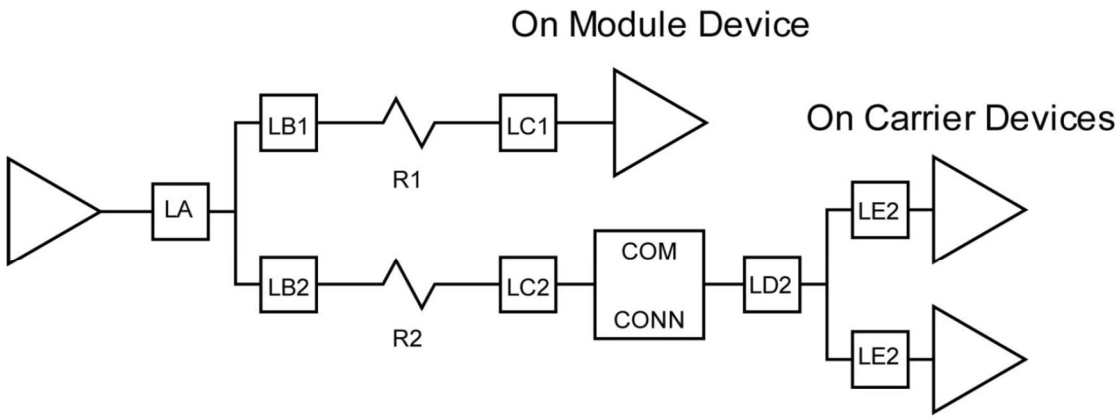
2.14.2.3 Carrier Board LPC Devices

Carrier Board LPC devices should be clocked with the LPC clock provided by the Module interface. If the Carrier Board has two loads on the LPC clock these loads should be connected to the common clock without a buffer. The Carrier Board should not have more than two loads on the LPC clock.

Carrier Board LPC devices should be reset with signal CB_RESET#.

A typical routing topology for a Module LPC device and two Carrier Board LPC devices clock is shown below. This topology is used by Modules that start and stop the LPC clock on the fly. In this case, a buffer cannot be used and all LPC devices must share a common clock.

Figure 27: Typical Routing Topology for a Module LPC Device



LA 500 mils max
LB1 = LB2 = 150 mils max
LC1 = 8.88”+LC2
LC2 = .25” max
LE2 = 1” max
LD2 + LE2 (note 2 instances of LE2) = 8.88”
R1 = R2 = 22Ω

2.14.2.4 eSPI Devices *SOM-5871 is not support eSPI Interface.

At the time of this writing, the use case and design rules for eSPI are still being developed.

Designers of Modules and Carriers are provided with the following guidance:

The maximum trace length for Carrier routed eSPI traces ***shall not*** exceed 4.5”.

Carrier routed ESPI traces *shall* be routed at 50 Ohms.

Carrier routed SPI traces *shall* have 5 mil via to via clearance, 4 mil trace to via clearance, and 10 mil clearance to any other traces.

The Carrier *shall* have a 33 Ohm series termination between 0.5" and 1" of the target device on the ESPI_CK signal.

The Carrier *shall* have a 33 Ohm series termination between 0.5" and 1" of the target device on the ESPI_IO_[0..3] signals.

The Carrier *shall* length match ESPI_CK and ESPI_IO_[0..3] within 250mils.

The Carrier *shall* length match eSPI_CK and ESPI_CS within 100mils.

2.14.3 LPC Trace Length Guidelines

Figure 28: Topology for LPC

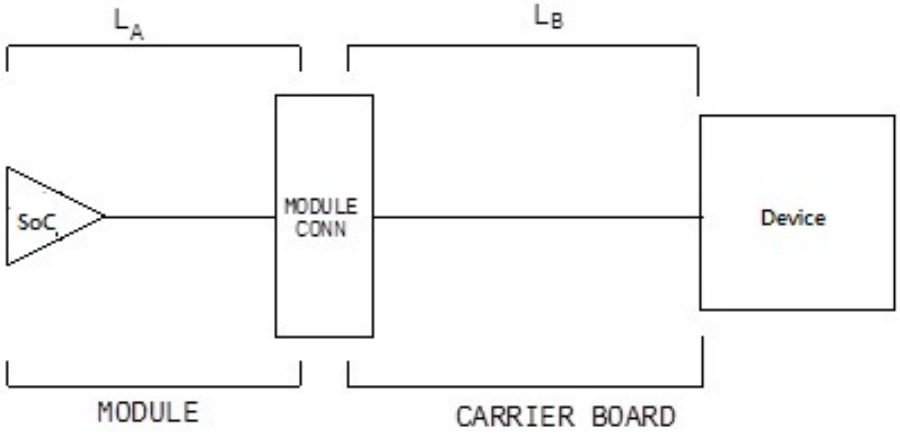


Table 43: LPC Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	LPC	
Single End	50Ω ±15%	
Nominal Trace Space within LPC Signal Group	Min. 15mils	
Spacing to Other Signal Group	Min. 15mils	
Trace spacing	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	13"	
Length matching between single ended signals	Max. 500mils	
Length matching between clock signals	Max. 500mils	
Reference Plane	GND referencing preferred.	
Via Usage	Try to minimize number of vias	

Notes:

2.15 SPI – Serial Peripheral Interface Bus

The SPI bus is used to support SPI-compatible flash devices. The SPI flash device can be up to 16 MB (128 Mb). The SPI bus is clocked at either 20 MHz, 25 MHz, 33 MHz or 50 MHz. SPI devices selected ***should*** support one of these frequencies.

In COM.0 Rev 2, the SPI interface was defined as a 3.3V interface. With COM.0 Rev 3, the SPI interface ***may*** be either 3.3V or 1.8V, as is best for the Module chipset at hand.

2.15.1 SPI Signal Definition

Table 44: SPI Interface Signal Definition

Signal	Pin#	Description	I/O	Note
SPI_CS#	B97	Chip select for Carrier Board SPI – may be sourced from chipset SPI0 or SPI1 Carrier Board: Connect to SPI flash pin 1 Chip Select N/C if not used	O CMOS – 3.3V Suspend	
SPI_MISO	A92	Data in to Module from Carrier SPI Carrier Board: Connect 15~33Ω in series to SPI flash pin 2 Serial Output N/C if not used	I CMOS – 3.3V Suspend	
SPI_MOSI	A95	Data out from Module to Carrier SPI Carrier Board: Connect 33~47 Ω in series to SPI flash pin 5 Serial Input N/C if not used	O CMOS – 3.3V Suspend	
SPI_CLK	A94	Clock from Module to Carrier SPI Carrier Board: Connect 33~47 Ω in series to SPI flash pin 6 Clock N/C if not used	O CMOS – 3.3V Suspend	

Signal	Pin#	Description	I/O	Note
SPI_POWER	A91	Power supply for Carrier Board SPI – sourced from Module – nominally 3.3V. The Module shall provide	O 3.3V	

		a minimum of 100mA on SPI_POWER. Carriers shall use less than 100mA of SPI_POWER. SPI_POWER shall only be used to power SPI devices on the Carrier. Carrier Board: Connect to SPI flash pin 8 VDD N/C if not used	Suspend or 3.3V S0 1.8V Suspend or 1.8V S0	
BIOS_DIS0# / ESPI_SAFS	A34	Selection strap to determine the BIOS boot device. The Carrier should only float these or pull them low, please refer to Table 45: BIOS Selection Straps. Module has integrated PU resistor 4.7Kohm to 3.3V suspend power. Carrier Board: 1 – N/C 0 – PD 1K to GND BIOS options are shown in Table "Effect of the BIOS disable signals"	I CMOS	
BIOS_DIS1#	B88	Selection strap to determine the BIOS boot device. The Carrier should only float these or pull them low, please refer to Table 45: BIOS Selection Straps. Module has integrated PU resistor 4.7Kohm to 3.3V suspend power. Carrier Board: 1 – N/C 0 – PD 1K to GND BIOS options are shown in Table "Effect of the BIOS disable signals"	I CMOS	

Note:

SPI Power

Introducing a SPI_POWER pin is desirable because some Module implementations will have the SPI power domain in power state S0 and others in S5. It is easier for Carrier Board designers to take the Carrier SPI

power from a pin on the Module.

The SPI_POWER voltage level was defined as 3.3V in COM.0 Rev. 2. With COM.0 Rev. 3, the SPI_POWER voltage level *may* be 3.3V or 1.8V. This allows the Carrier SPI interface to operate at the level appropriate for the Module chipset, without the use of level shifters.

Module designs that implement a 1.8V Carrier SPI interface *should* protect themselves against possible exposure to 3.3V Carrier SPI signals.

Carrier designs that implement a 1.8V SPI interface *should* protect themselves against possible exposure to 3.3V Module SPI signals.

Module Vs Carrier Board Pull-ups

There *shall* not be any Carrier Board pull-ups or pull-downs on the five SPI_x signals. All such terminations *shall* be on the Module. The Module designer *shall* determine the correct power domain that these signals are terminated to.

Note: Carrier Board *shall* implement pull-ups to SPI_POWER on the SPI flash pins HOLD# and WP# which are not supported on the COM Express connector.

2.15.2 BIOS Boot Selection

For COM.0 R3, the Module Carrier based BIOS options have been expanded to support eSPI devices. A third pin that affects the BIOS location, named ESPI_EN#, works in conjunction with BIOS_DIS1# and BIOS_DIS0# to define the BIOS boot path. Additionally, the concepts of Master Attached Flash Sharing

(MAFS) and Slave Attached Flash Sharing (SAFS) are introduced.

LPC bus BIOS FWH support is removed in COM.0 R3. SPI and eSPI BIOS options are supported.

SPI Boot Flash Background

Contemporary Intel x86 systems requires that the SPI boot flash to be divided into a number of regions that may include:

- Descriptor
- BIOS code
- Management Engine (ME) code
- GBE parameters
- Platform data

The Descriptor defines where the other regions are in the SPI device(s). The Descriptor is always at the bottom of the first SPI device, the SPI device that is selected by chipset SPI0 chip-select (chipset SPI_CS0#).

The first two regions, the Descriptor and the BIOS, are mandatory. The other regions are optional. The regions may all be packed into the same SPI device, or may be divided between more than one SPI device, although the Descriptor has to be at the bottom of the first SPI device. In most situations, all the SPI regions are packed into a single SPI device that is either on the Module or on the Carrier. Designers may have reasons for dividing the SPI boot flash regions between devices.

COM Express Rev 2 and Rev 3 define a SPI interface on the COM Express connector. The COM Express SPI interface has only one chip select. Chipsets typically have 2 SPI chip selects. Module hardware may steer those chipset chip selects to an on-Module SPI device or devices or to a single off-module SPI device. The chip select steering is defined by the ESPI_EN#, BIOS_DIS1# and BIOS_DIS0# signals.'BIOS Selection Straps' below.

The BIOS Entry point *may* be in SPI0 or SPI1 as determined by the descriptor table in the SPI0 device. The Module *may* have one or two SPI devices. Carrier Boards *may* have zero or one SPI devices.

MAFS and SAFS BIOS Configurations

Master Attached Flash Sharing (MAFS) is defined as the BIOS Flash directly attached to the processor SPI bus.

Slave Attached Flash Sharing (SAFS) is defined as the BIOS Flash being attached behind a board Management Controller (BMC) or Embedded Controller (EC).

MAFS and SAFS configurations apply to both LPC and eSPI enabled configurations. Refer to Figure29 'BIOS Selection LPC Mode' and Figure30 'BIOS Selection eSPI Mode' below.

Please note that some of the features shown in these figures are mutually exclusive.

Figure 29: BIOS Selection LPC Mode

LPC Mode

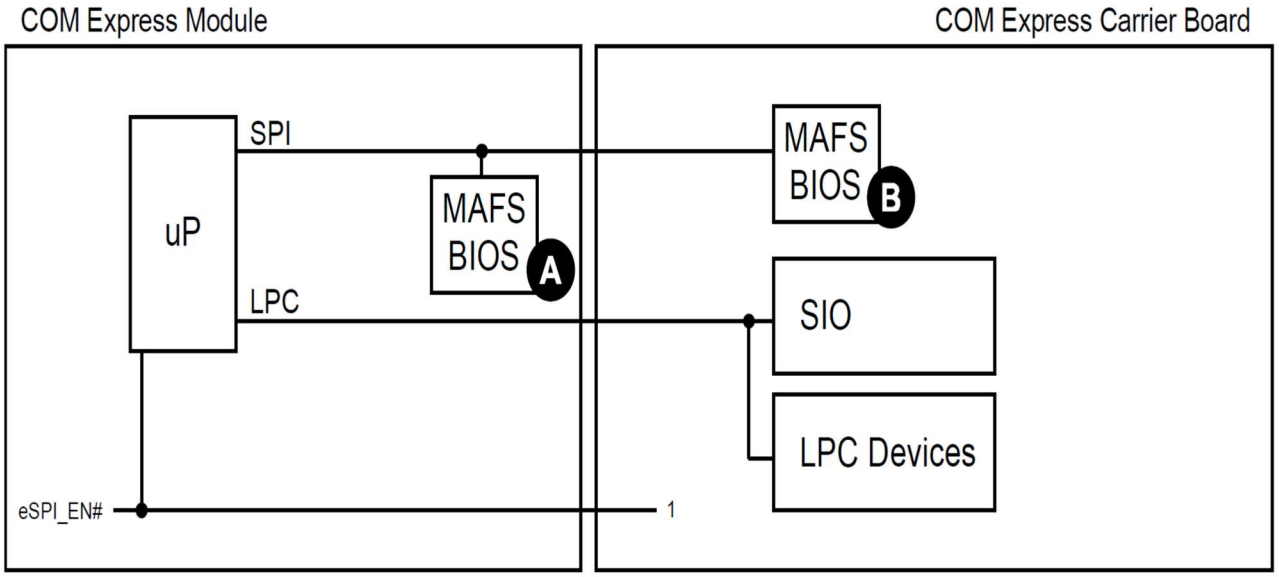
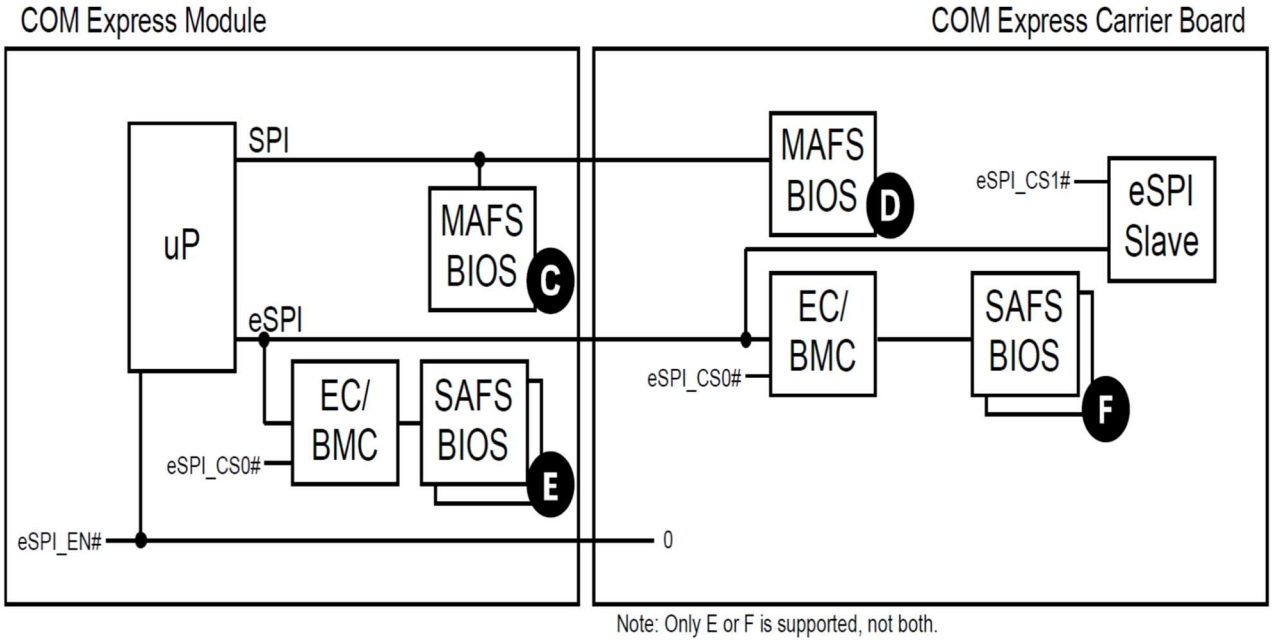


Figure 30: BIOS Selection eSPI Mode

eSPI Mode



Note: Only E or F is supported, not both.

The A and, B notations in the figure29 above and the B, C, D and F notations in figure30 above are referenced in the table and text sections below. Note also that some of the features shown in these

figures are mutually exclusive.

Table 45: BIOS Selection Straps

ESPI_EN#	BIOS_DIS1#	BIOS_DIS0#	Boot Bus	BBS	Chipset ESPI_CS0# Destination	Carrier ESPI_CS0# Pin	Chipset SPI_CS1# Destination	Chipset SPI_CS0# Destination	Carrier SPI_CS# Pin	SPI Descriptor	Ref to Images Above	Notes
1	0	0	SPI	0	-	-	Carrier	Module	SPI1	Module	A	MAFS on Module. LPC bus enabled.
1	0	1	SPI	0	-	-	Module	Carrier	SPI0	Carrier	B	MAFS on Carrier. LPC bus enabled.
1	1	0	-	0	-	-	-	-	High	-	-	Not used – was FWH
1	1	1	SPI	0	-	-	Module	Module	High	Module	A	MAFS on Module. LPC bus enabled.
0	0	0	SPI	0	-	-	Carrier	Module	SPI1	Module	C	MAFS on Module. ESPI bus enabled.
0	0	1	SPI	0	-	-	Module	Carrier	SPI0	Carrier	D	MAFS on Carrier. ESPI bus enabled.
0	1	0	eSPI	1	Module	-	-	-	SPI0	Module	E	SAFS and BMC on Module. ESPI bus enabled.
0	1	1	eSPI	1	Carrier	Chipset ESPI_CS0#	-	-	SPI0	Carrier	E	SAFS and BMC on Carrier. ESPI bus enabled.

The BBS (BIOS Boot Select) is a signal to the chipset that indicates if the system is using a MAFS (Master Attached File Sharing) or SAFS (Slave attached File Sharing) setup. The BBS signal *may* be formed by Module logic looking for ESPI_EN# low and BIOS_DIS1# high.

The ESPI_CS1# line is not used for BIOS boot functions; it may be used to attach a secondary slave device to the eSPI bus, if the chip select is available.

SPI BIOS MAFS Considerations – LPC Enabled

The first four lines in Table 45 above are backwards compatible with the SPI BIOS options described in

COM.0 Rev. 2, except that LPC FWH support is removed in COM.0 Rev 3. The LPC bus is enabled and is available for use on the Module or the Carrier for peripheral devices such as Board Management Controllers (BMC), Embedded Controllers (EC), Super I/O (SIO) or other general purpose devices.

SPI BIOS MAFS Considerations – eSPI Enabled

In an eSPI enabled MAFS system, the BIOS flash is attached to the system SPI bus, either on-Module or off-Module, much as in the LPC enabled MAFS system described above. The eSPI bus replaces the LPC bus for use with peripheral devices such as BMCs, Ecs, SIOs etc. but the BIOS boot path is on the SPI bus.

eSPI BIOS SAFS Considerations – eSPI Enabled

In an eSPI enabled SAFS system, the SPI boot device is located on the far side of a BMC or EC. The system can boot from either a Module SAFS (in Figure 30 above) or a Carrier SAFS (in Figure 30 above). The BIOS boot traffic is routed through the BMC or EC to the system eSPI bus and on to the chipset.
It is possible for both a Module and a Carrier SAFS to be present in a system, but only one can be enabled. This is accomplished by routing the ESPI_CS0# signal to the Module or the Carrier, but never both. This is by definition of the eSPI specification. A second ESPI_CS1# is available to select eSPI slave devices. Slave devices can be on the Module or the Carrier.
Two eSPI alert pins are provided. Additional alert pins are permitted by the eSPI specification through alert pin sharing on the EC/BMC or by signal tunneling.

2.15.3 SPI Routing Guidelines

NA

2.15.4 SPI Trace Length Guidelines

Figure 31: Topology for SPI

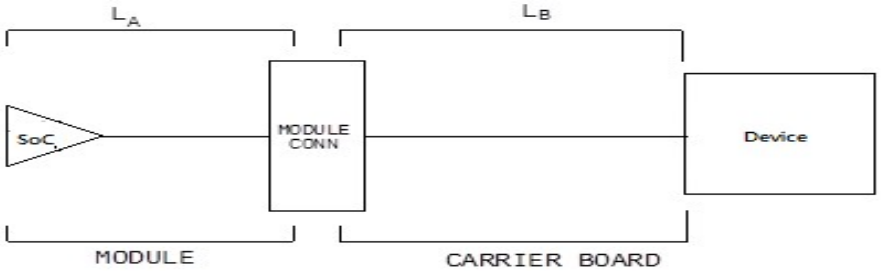


Table 46: SPI Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	SPI	
Single End	50Ω ±15%	
Nominal Trace Space within SPI Signal Group	Min. 10mils	
Spacing to Other Signal Group	Min. 15mils	
Trace spacing	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	5"	
DATA to CLK Maximum Pin to Pin Length Mismatch	Max. 500mils	
Via Usage	Try to minimize number of vias	

Notes:

2.16 General Purpose I2C Bus Interface

The I2C (Inter-Integrated Circuit) bus is a two-wire serial bus originally defined by Philips. The bus is used for low-speed (up to 400kbps) communication between system ICs. The bus is often used to access small serial EEPROM memories and to set up IC registers. The COM Express Specification defines several I2C interfaces that are brought to the Module connector for use on the Carrier. Some of these interfaces are for very specific functions (VGA, LVDS, and DDIX), one interface is the SMBus used primarily for

management and one other interface is a general purpose I2C interface. Since COM.0 Rev. 2.0 this interface should support multi-master operation. This capability will allow a Carrier to read an optional Module EEPROM before powering up the Module.

Revision 1.0 of the specification placed the I2C interface on the non-standby power domain. With this connection, the I2C interface can only be used when the Module is powered on. Since the I2C interface is used to connect to an optional Carrier EEPROM and since it is desirable to allow a Module based board controller access to the optional Carrier EEPROM before the Module is powered on, revision 2.0 of this specification changes the power domain of the I2C interface to standby-power allowing access during power down and suspend states. There is a possible leakage issue that can arise when using a R2.0 Module with a R1.0 Carrier that supports I2C devices. The R1.0 Carrier will power any I2C devices from the non-standby power rail. A R2.0 Module will pull-up the I2C clock and data lines to the standby-rail through a 2.2K resistor. The difference in the power domains on the Module and Carrier can provide a leakage path from the standby power rail to the non-standby power rail.

Vendor interoperability is given via EAPI – Embedded Application Programming Interface, which allows and easier interoperability of COM Express Modules.

2.16.1 Signal Definitions

The general purpose I2C Interface is powered from 3.3V suspend rail. The I2C_DAT is an open collector line with a pull-up resistor located on the Module. The I2C_CK has a pull-up resistor located on the Module. The Carrier should not contain pull-up resistors on the I2C_DAT and I2C_CK signals. Carrier based devices should be powered from 3.3V suspend voltage. The use of main power line for a Carrier I2C device will require a bus isolator to prevent leakage to other I2C devices on 3.3V power.

At this time, there is no allocation of I2C addresses between the Module and Carrier. Carrier designers will need to consult with Module providers for address ranges that can be used on the Carrier.

Table 47: General Purpose I2C Interface Signal Descriptions

Signal	Pin#	Description	I/O	Pwr Rail	Note
I2C_CK	B33	General Purpose I2C Clock output. Module has integrated PU resistor 2.2K ohm to 3.3V suspend power.	I/O OD CMOS	3.3V Suspend	

		Carrier Board: 3.3VSB I2C device - Connect to SCL of I2C device. 3.3V I2C device - Connect 3.3V isolation circuit controlled by COME pin B24 PWR_OK to SCL of I2C device. 5VSB I2C device - Connect 5VSB Level Shifter to SCL of I2C device. 5V I2C device – Connect an 5V isolation circuit controlled by COME pin B24 PWR_OK to SCL of I2C device. N/C if not used			
I2C_DAT	B34	General Purpose I2C data I/O line. Module has integrated PU resistor 2.2K ohm to 3.3V suspend power. Carrier Board: 3.3VSB I2C device - Connect to SDA of I2C device. 3.3V I2C device - Connect 3.3V isolation circuit controlled by COME pin B24 PWR_OK to SDA of I2C device 5VSB I2C device - Connect 5VSB Level Shifter to SDA of I2C device 5V I2C device - Connect an 5V isolation circuit controlled by COME pin B24 PWR_OK to SDA of I2C device N/C if not used	I/O OD CMOS	3.3V Suspend	

Note:

2.16.2 I2C Routing Guidelines

NA

2.16.3 I2C Trace Length Guidelines

Figure 32: Topology for I2C

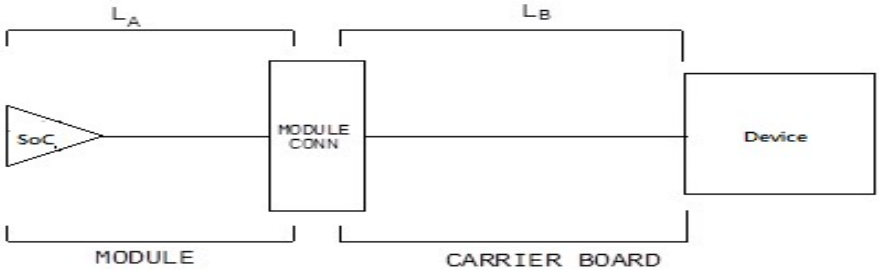


Table 48: I2C Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	I2C	
Single End	50Ω ±15%	
Nominal Trace Space within I2C Signal Group	Min. 10mils	
Spacing to Other Signal Group	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	NA	
Length Mismatch	+/- 500 mils	
Via Usage	Try to minimize number of vias	

Notes:

2.16.4 Connectivity Considerations

The maximum amount of capacitance allowed on the Carrier General Purpose I2C bus lines (I2C_DAT, I2C_CK) is specified by Advantech's Module. The Carrier designer is responsible for ensuring that the maximum amount of capacitance is not exceeded and the rise/fall times of the signals meet the I2C bus specification. As a general guideline, an IC input has 8pF of capacitance, and a PCB trace has 3.8pF per inch of trace length.

2.17 System Management Bus (SMBus)

The SMBus is primarily used as an interface to manage peripherals such as serial presence detect (SPD) on RAM, thermal sensors, PCI/PCIe devices, smart battery, etc. The devices that can connect to the SMBus can be located on the Module and Carrier. Designers need to take note of several implementation issues to ensure reliable SMBus interface operation. The SMBus is similar to I2C. I2C devices have the potential to lock up the data line while sending information and require a power cycle to clear the fault condition. SMBus devices contain a timeout to monitor for and correct this condition. Designers are urged to use

SMBus devices when possible over standard I2C devices. COM Express Modules are required to power SMBus devices from Early Power in order to have control during system states S0-S5. The devices on the Carrier Board using the SMBus are normally powered by the 3.3V main power. To avoid current leakage between the main power of the Carrier Board and the Suspend power of the Module, the SMBus on the Carrier Board must be separated by a bus switch from the SMBus of the Module. However, if the Carrier Board also uses Suspend powered SMBus devices that are designed to operate during system states S3-S5, then these devices must be connected to the Suspend powered side of the SMBus, i. e. between the COM Express Module and the bus switch. Since the SMBus is used by the Module and Carrier, care must be taken to ensure that Carrier based devices do not overlap the address space of Module based devices. Typical Module located SMBus devices and their addresses include memory SPD (serial presence detect 1010 000x, 1010 001x), programmable clock synthesizes (1101 001x), clock buffers (1101 110x), thermal sensors (1001 000x), and management controllers (vendor defined address). Contact Advantech for information on the SMBus addresses used.

2.17.1 SMB Signal Definitions

Table 49: SMB Signal Definitions

Signal	Pin#	Description	I/O	Pwr Rail	Note
SMB_CK	B13	System Management Bus bidirectional clock line. Module has integrated PU resistor 1K ohm to 3.3V suspend power. Carrier Board: 3.3VSB SMBus device - Connect to SMBCLK of SMBus device.	I/O OD CMOS	3.3V Suspend rail	

		3.3V SMBus device - Connect 3.3V isolation circuit controlled by COME pin B24 PWR_OK to SMBCLK of SMBus device. 5VSB SMBus device - Connect 5V Level Shifter to SMBCLK of SMBus device. 5V SMBus device - Connect 5V isolation circuit controlled by COME pin B24 PWR_OK to SMBCLK of SMBus device N/C if not used.			
SMB_DAT	B14	System Management bidirectional data line. Module has integrated PU resistor 1K ohm to 3.3V suspend power. Carrier Board: 3.3VSB SMBus device - Connect to SMBDAT of SMBus device. 3.3V SMBus device - Connect 3.3V isolation circuit controlled by COME pin B24 PWR_OK to SMBDAT of SMBus device. 5VSB SMBus device - Connect 5V Level Shifter to SMBDAT of SMBus device. 5V SMBus device - Connect 5V isolation circuit controlled by COME pin B24 PWR_OK to SMBDAT of SMBus device N/C if not used.	I/O OD CMOS	3.3V Suspend rail	
SMB_ALERT#	B15	System Management Bus Alert. Module has integrated PU resistor 2.2K ohm to 3.3V suspend power. Carrier Board: Connect to SMBALERT# of SMBus device. N/C if not used.	I CMOS	3.3V Suspend Rail	

Note:

2.17.2 SMB Routing Guidelines

The SMBus should be connected to all or none of the PCIe/PCI devices and slots. A general recommendation is to not connect these devices to the SMBus.

The maximum load of SMBus lines is limited to 3 external devices. The maximum capacitance on the Carrier Board shall not exceed 100pF. Please contact Advantech if more devices are required.

Do not connect Non-Suspend powered devices to the SMBus unless a bus switch is used to prevent back feeding of voltage from the Suspend rail to other supplies.

Contact Advantech for a list of SMBus addresses used on the Module. Do not use the same address for Carrier located devices.

2.17.3 SMB Trace Length Guidelines

Figure 33: Topology for SMB

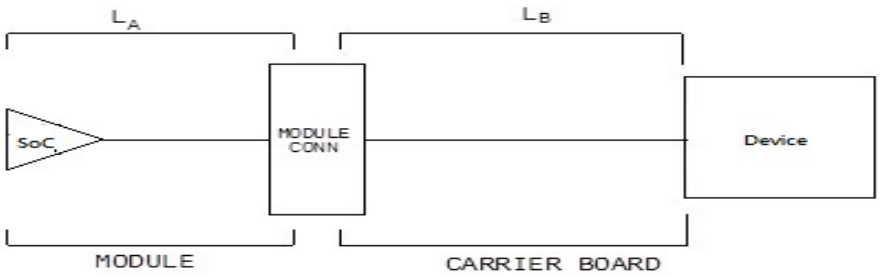


Table 50: SMB Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	SMB	
Single End	50Ω ±15%	
Nominal Trace Space within SMB Signal Group	Min. 10mils	
Spacing to Other Signal Group	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	ASAP	
Length Mismatch	+/- 500 mils	
Via Usage	Try to minimize number of vias	

Notes:

2.18 General Purpose Serial Interface

Advantech provides two serial ports on Type 6 COM Express Modules.

Two TTL compatible two wire asynchronous serial ports are available on Module Types. This feature was introduced in COM.0 Revision 2 and uses pins on the A-B connector that have been re-claimed from the A-B VCC_12V pool. As such, it is possible that if a Type 6 Module is deployed in an R1.0 Carrier Board designed for Module Types 1,2,3,4,5 then the Module TTL level serial pins may be exposed to the 12V supply, and Module designers must plan for this. Similarly, an R1.0 Module deployed on an R2.0 Carrier may bridge 12V to the serial pins and Carrier designers must plan for this. These pins are designated SER0_TX, SER0_RX, SER1_TX and SER1_RX. Data out of the Module is on the _TX pins. Hardware

handshaking and hardware flow control are not supported.

Any of the Module asynchronous serial ports, if implemented on an Intel X86 architecture Module platform, should be I/O mapped serial ports that are register compatible with the National Semiconductor 16550 UARTs that were used in the PC AT architecture.

The Module asynchronous serial ports are intended for general purpose use and for use with debugging software that make use of the "console redirect" features available in many operating systems. The Module asynchronous serial ports should not be implemented as USB peripherals, as such implementations are generally not useful for low level debug purposes.

2.18.1 Serial interface Signal Definitions

Table 51: Serial interface Signal Definitions

Signal	Pin#	Description	I/O	Note
SER0_TX	A98	Transmit Line for Serial Port 0 Carrier Board: SER0_TX need pull down 4.7K ohm, refer to Figure 36 connect to Device - TXD COM DB-9 port - TxIN of Serial Transceiver and	O CMOS	

		TxOUT to DB-9 pin 3 TXD N/C if not used.		
SER0_RX	A99	Receive Line for Serial Port 0 Carrier Board: Connect to Device - RXD COM DB-9 port - TxOUT of Serial Transceiver and TxIN to DB-9 pin 2 RXD N/C if not used	I CMOS	
SER1_TX	A101	Transmit Line for Serial Port 1 Carrier Board: SER1_TX need pull down 4.7K ohm, refer to Figure 36 Device - TXD COM DB-9 port - TxIN of Serial Transceiver and TxOUT to DB-9 pin 3 TXD N/C if not used.	O CMOS	
SER1_RX	A102	Receive Line for Serial Port 1 Carrier Board: Connect to Device - RXD COM DB-9 port - TxOUT of Serial Transceiver and TxIN to DB-9 pin 2 RXD N/C if not used	I CMOS	

Note:

2.18.2 Serial interface Routing Guidelines

NA

2.18.3 Serial interface Trace Length Guidelines

Figure 34: Topology for Serial interface

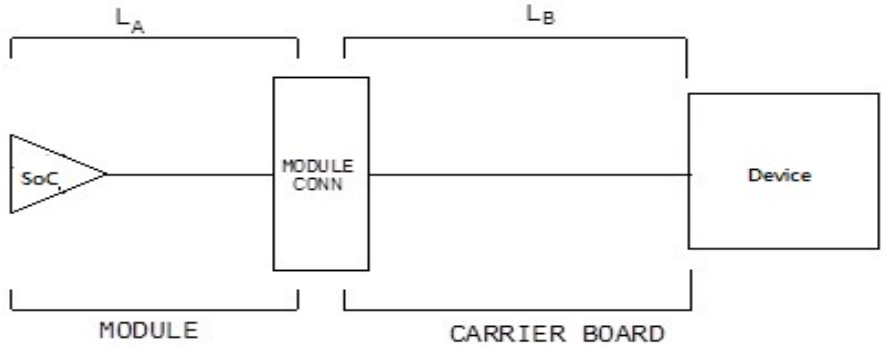


Table 52: Serial interface Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	Serial interface	
Single End	50Ω ±15%	
Nominal Trace Space within Serial Signal Group	Min. 10mils	
Spacing to Other Signal Group	Min. 15mils	
LA	Please see the SOM-5871 Layout Checklist	
LB	Carrier Board Length	
Max length of LA+LB	NA	
Length Mismatch	NA	
Via Usage	Try to minimize number of vias	

Notes:

2.19 CAN Interface

CAN Bus Operation Over SER1 Lines The SER1_TX and SER1_RX asynchronous serial port lines defined for COM.0 Types 6, 7 and 10 *may* be used alternatively to carry CMOS 3.3V logic level CAN (Controller Area Network) bus signals from a COM Express Module based CAN protocol controller. The
CAN bus is an asynchronous, message based protocol widely used in the automotive and industrial control sectors. It is defined by ISO 11519, ISO 11898, and SAEJ2411. Data rates on a CAN bus *may* be as

high as 1 MBit/s, although lower rates in the range from 10 kBit/s to 125 kBit/s are more common. The achievable data rates are dependent on the protection scheme used see Section 2.20.1.1 for further information.

Use of the CAN bus in a COM Express system requires a CAN bus transceiver on the Carrier Board to interface to the CAN physical layer. CAN bus transceivers are available from NXP, Texas Instruments, Linear Technology, and others.
Data from the COM Express Module based CAN controller to the Carrier Board CAN transceiver is carried on Module line SER1_TX. Data from the Carrier Board CAN transceiver to the COM Express Module based CAN controller is carried on Module line SER1_RX. The Carrier Board CAN transceiver converts the logic level CAN protocol TX and RX signals from the Module into a differential half duplex line per the CAN specification.

How the SER1 asynchronous lines are shared with CAN bus operation is Module vendor Advantech specific. An Advantech *may* choose to use the SER1 TX and RX lines to support asynchronous serial port operation, or CAN bus operation, or both, or neither. Module build option(s) or software controlled muxing implementations *may* be used.

CAN bus is a vehicle bus standard designed to allow controllers and devices to communicate with each other without a host computer. CAN bus is a message-based protocol, designed specifically for automotive applications but now also used in other areas such as industrial automation and medical equipment.

Development of CAN bus started originally in 1983. The protocol was officially released in 1986 at the Society of Automotive Engineers (SAE) congress in Detroit, Michigan. The first CAN controller chips, produced by Intel and Philips, came on the market in 1987. In 1991 the CAN 2.0 specification was published.

Since 2008 CAN bus has been mandatory in any US vehicle in the OBD-II car diagnostic port. It is also used extensively in industrial automation.

2.19.1 CAN interface Signal Definitions

Table 53: CAN interface Signal Definitions

Signal	Pin#	Description	I/O	Note
CAN_TX	A101	Transmit Line for CAN Carrier Board: CAN_TX need pull down 4.7K ohm, refer to Figure 35.	O CMOS	1

		Check your CAN transceiver application notes.		
CAN_RX	A102	Receive Line for CAN Carrier Board: Check your CAN transceiver application notes.	I CMOS	1

Note:

1. SOM-5871 don't support CAN Bus by default, it is option function. If customers want to use CAN Bus function, please contact Advantech.

2.19.2 CAN interface Routing Guidelines

It should be routed as a differential pair signal with 120 Ohm differential impedance. The end points of CAN bus should be terminated with 120 Ohms or with 60 Ohms from the CAN_H line and 60 Ohms from the CAN_L line to the CAN Bus reference voltage. Check your CAN transceiver application notes for further details on termination.

2.19.3 CAN interface Trace Length Guidelines

Figure 35: Topology for CAN interface

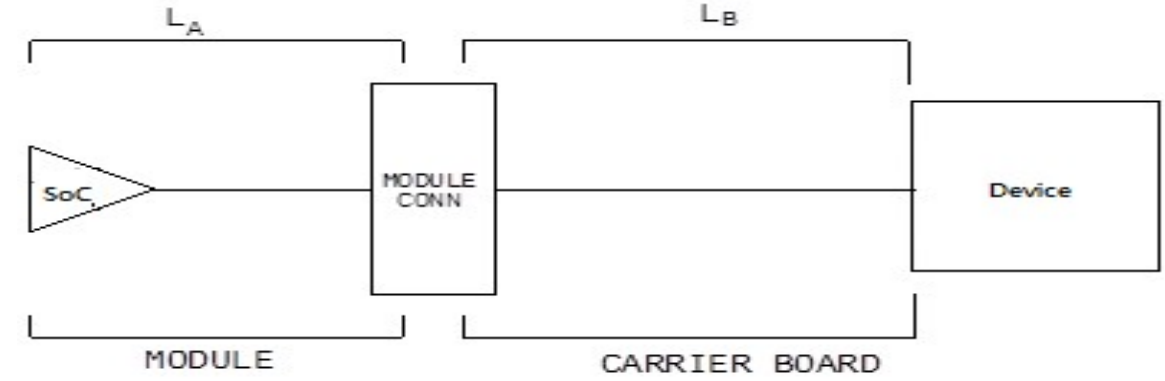


Table 54: CAN interface Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	CAN interface	1
Single End	50Ω ±15%	
Nominal Trace Space within CAN Signal Group	Min. 10mils	
Spacing to Other Signal Group	Min. 15mils	
LA	ASAP	
LB	Carrier Board Length	
Max length of LA+LB	NA	
Length Mismatch	NA	
Via Usage	Try to minimize number of vias	

Notes:

1. SOM-5871 default is serial interface. BOM option with CAN interface.

2.20 Miscellaneous Signals

2.20.1 Miscellaneous Signals

Table 55: Miscellaneous Signal Definitions

Signal	Pin#	Description	I/O	Note

TYPE0# TYPE1# TYPE2#	C54 C57 D57	The Type pins indicate the COM Express pin-out type of the Module. To indicate the Module's pin-out type, the pins are either not connected or strapped to ground on the Module. The Carrier Board has to implement additional logic, which prevents the system to switch power on, if a Module with an incompatible pin-out type is detected.	O 5V PDS	For TYPE6 module. TYPE0#: NC TYPE1#: NC TYPE2#: GND
TYPE10#	A97	Indicates to the Carrier Board that a Type 10 Module is installed. Indicates to the Carrier Board, that a Rev 1.0/2.0 Module is installed. TYPE10# NC Pin-out R2.0 PD Pin-out Type 10 pull down to ground with 47k 12V Pin-out R1.0		
SPKR	B32	Output used to control an external FET or a logic gate to drive an external PC speaker. Carrier Board: Connect to Speaker circuit. N/C if not used	O 3.3V CMOS	
BIOS_DIS0#	A34	Selection straps to determine the BIOS boot device. Module has integrated PU resistor 4.7 Kohm to 3.3V suspend power. The Carrier should only float these or pull them low.	I 3.3V CMOS	See Section 2.15 'SPI – Serial Peripheral Interface Bus'
BIOS_DIS1#	B88	Selection straps to determine the BIOS boot device. Module has integrated PU resistor 4.7 Kohm to 3.3V suspend power. The Carrier should only float these or pull them low.	I 3.3V CMOS	See Section 2.15 'SPI – Serial Peripheral Interface Bus'

Signal	Pin#	Description	I/O	Note
WDT	B27	Output indicating that a watchdog time-out event has occurred. Carrier Board:	O 3.3V CMOS	

		Connect to Watchdog trigger input. N/C if not used		
LID#	A103	LID switch. Low active signal used by the ACPI operating system for a LID switch. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: R1.0/R2.x Module both - Connect protection scheme referred to Figure 35 to LID button. R2.x Module only - Connect to LID button. N/C if not used.	I 3.3V Suspend CMOS OD	
SLEEP#	B103	Sleep button. Low active signal used by the ACPI operating system to bring the system to sleep state or to wake it up again. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: R1.0/R2.x Module both - Connect protection scheme referred to Figure 35 to Sleep button. R2.x Module only - Connect to Sleep button. N/C if not used	I 3.3V Suspend CMOS OD	
FAN_PWMOUT	B101	Fan speed control. Uses the Pulse Width Modulation (PWM) technique to control the fan' s RPM. Carrier Board: R1.0/R2.x Module both - Connect protection scheme referred to Figure 35 to FAN connector pin 2 PWMOUT via Smart FAN circuit R2.x Module only - PD 4.7KΩ to GND and connects to FAN connector pin 2 PWMOUT via Smart FAN circuit N/C if not used	O 3.3V CMOS OD	

Signal	Pin#	Description	I/O	Note
FAN_TACHIN	B102	Fan tachometer input for a fan with a two pulse output. Carrier Board:	I 3.3V CMOS OD	

		R1.0/R2.x Module both - Connect protection scheme referred to Figure 35 to FAN connector pin 3 TACHIN via Smart FAN circuit R2.x Module only - Connect to FAN connector pin 3 TACHIN via Smart FAN circuit N/C if not used		
TPM_PP	A96	Trusted Platform Module (TPM) Physical Presence pin. Active high. TPM chip has an internal pull down. This signal is used to indicate Physical Presence to the TPM. Carrier Board: Physical Absence - N/C Physical Presence - PU 1KΩ to 3.3V N/C if not used	I 3.3V CMOS	
VCC_RTC	A47	Real-time clock circuit power input. Nominally +3.0V		

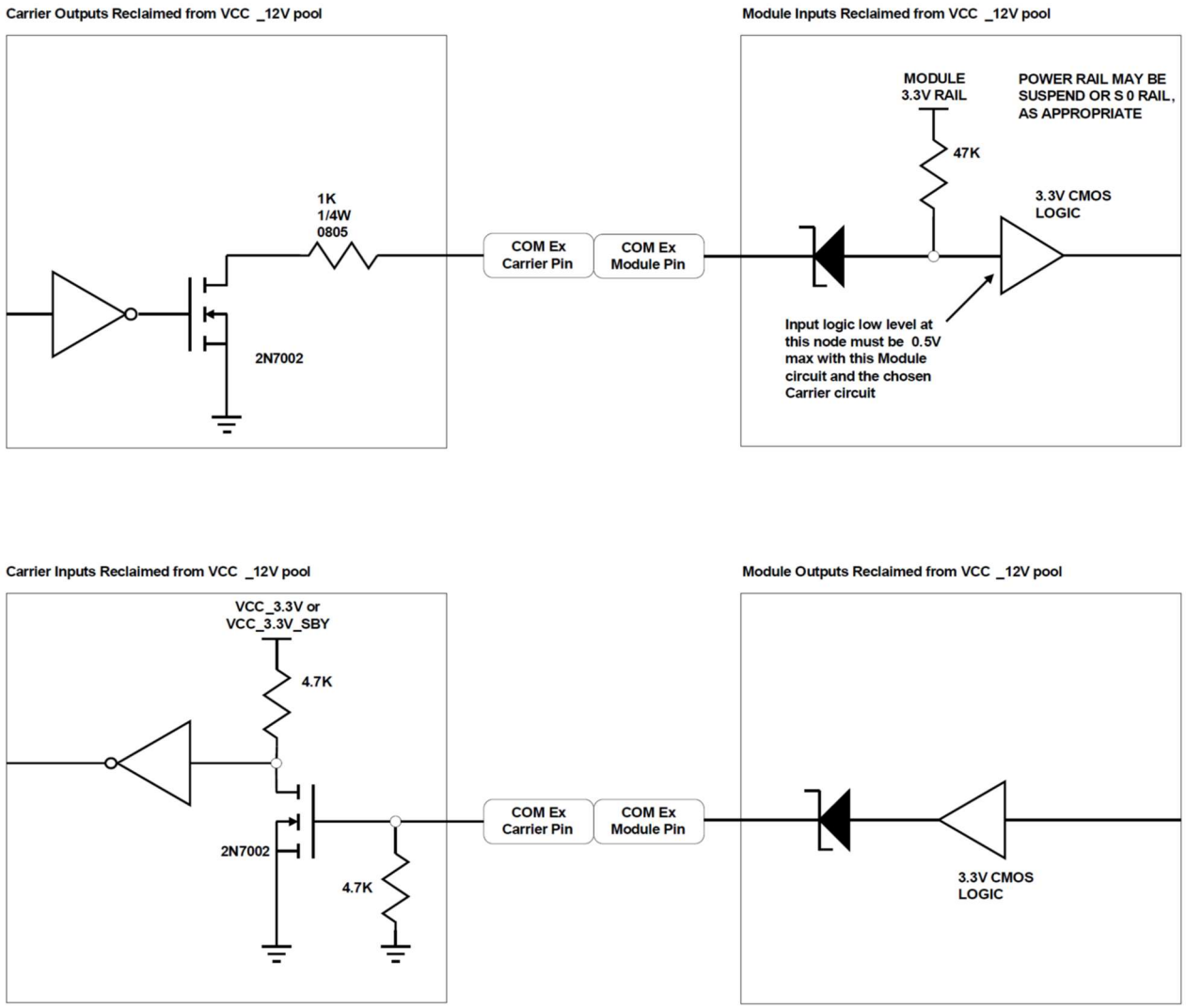
Note:

2.20.1.1 Logic Level Signals on Pins Reclaimed from VCC_12

Carrier Board logic level inputs and outputs that are implemented on pins reclaimed from the VCC_12V pool shall be protected against protracted accidental exposure to 12V. The protection scheme shown in the left side of the Figure 35 below may be used. Any scheme that is used shall be able to pull the

reclaimed Module input low enough such that Module CMOS input logic sees a maximum voltage of 0.5V for a logic low, as indicated in the figure.

Figure 36: Protecting Logic Level Signals on Pins Reclaimed from VCC_12V



2.20.2 Power Management Signals

Table 56: Power Management Signal Definitions

Signal	Pin#	Description	I/O	Note
PWRBTN#	B12	Power button low active signal used to wake up the	I 3.3V	

		system from S5 state (soft off). This signal is triggered on the falling edge. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: ATX - Connect to Power Button or SIO Power Button output pin (Active low) AT - N/C N/C if not used	Suspend / 3.3V CMOS	
SYS_RESET#	B49	Reset button input. Active low request for Module to reset and reboot. May be falling edge sensitive. For situations when SYS_RESET# is not able to reestablish control of the system, PWR_OK or a power cycle may be used. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to Reset button N/C if not used	I 3.3V Suspend / 3.3V CMOS	
CB_RESET#	B50	Reset output signal from Module to Carrier Board. This signal may be driven low by the Module to reset external components located on the Carrier Board. Carrier Board: Connect to reset pin of devices except PCI slots or devices. N/C if not used.	O 3.3V Suspend / 3.3V CMOS	

Signal	Pin#	Description	I/O	Note
PWR_OK	B24	Power OK status signal generated by the ATX power supply to notify the Module that the DC operating voltages are within the ranges required for proper	I 3.3V CMOS	

		operation. Carrier Board: Connect to power good pin of main power supply ATX - PW-OK pin 8 of ATX power connector connects 3.3V level shifter to COME PWR_OK. AT - PG pin P8.1 of AT power connector connects 3.3V level shifter to COME PWR_OK. Other - PWROK of 12V power generator circuit connects 3.3V level shifter to COME PWR_OK. N/C is not allowed, if the system is ATX mode. N/C if not used.		
SUS_STAT#	B18	Suspend status signal to indicate that the system will be entering a low power state soon. It can be used by other peripherals on the Carrier Board as an indication that they should go into power-down mode. Carrier Board: Connect to LPCPD# of LPC device. N/C if not used.	O 3.3V Suspend / 3.3V CMOS	
SUS_S3#	A15	S3 Sleep control signal indicating that the system resides in S3 state (Suspend to RAM). This signal can be used to control the ATX power supply via the *'PS_ON#'* signal. Module has integrated PD resistor 100Kohm to GND. Carrier Board: Connect to SLP_S3# (Suspend To RAM) of LPC device or SIO. N/C if not used.	O 3.3V Suspend / 3.3V CMOS	
SUS_S4#	A18	S4 Sleep control signal indicating that the system resides in S4 state (Suspend to Disk). Carrier Board: Connect to SLP_S4# (Suspend To Disk) of LPC device or SIO. N/C if not used.	O 3.3V Suspend / 3.3V CMOS	1

Signal	Pin#	Description	I/O	Note
SUS_S5#	A24	S5 Sleep Control signal indicating that the system resides in S5 State (Soft Off). Carrier Board: Connect to SLP_S5# (Soft Off) of LPC device or SIO.	O 3.3V Suspend / 3.3V CMOS	1

		N/C if not used.		
WAKE1#	B67	General purpose wake-up signal. Module has integrated PU resistor 4.7K ohm to 3.3V suspend power. Carrier Board: Connect to PME# of SIO N/C if not use	I 3.3V Suspend CMOS	
BATLOW#	A27	Battery low input. This signal may be driven low by external circuitry to signal that the system battery is low. It also can be used to signal some other external power management event. Module has integrated PU resistor 10K ohm to 3.3V suspend power. Carrier Board: Connect to BATLOW# of Smart Battery. N/C if not used.	I 3.3V Suspend CMOS	

Note:

1. SLP_S4# in AMD V1000 platform design does support. Thus, SOM-5871 uses SLP_S5# as COM Express pin SUS_S4#.

2.20.3 Thermal Protection Interface

Table 57: Thermal Management Signal Definitions

Signal	Pin#	Description	I/O	Note
THRM#	B35	Thermal Alarm active low signal generated by the external hardware to indicate an over temperature situation. This signal can be used to initiate thermal throttling. Carrier Board: Connect to THRM# output of Hardware Monitor. N/C if not used.	I 3.3V CMOS	1
THRMTRIP#	A35	Thermal Trip indicates an overheating condition of the processor. If *'THRMTRIP#'* goes active the system immediately transitions to the S5 State (Soft Off). Module has integrated PU resistor 8.2K ohm to 3.3V. Carrier Board: Connect to THERMTRIP# input of devices. N/C if not used.	O 3.3V CMOS	

Note:

1. Carrier board doesn't need to pull down.

2.20.4 Miscellaneous Signals Routing Guidelines

NA

2.20.5 Miscellaneous Signals Trace Length Guidelines

NA

2.21 Rapid Shutdown

COM Express Modules ***may*** support rapid shutdown. On a Module equipped with rapid shutdown, the assertion of the RAPID_SHUTDOWN input will cause the internal power supply regulators on the Module to be disabled, and for all residual voltages on the internal power supply rails to be discharged through crowbar circuits.

Modules supporting rapid shutdown ***shall*** specify the power rail discharge behavior, including discharge time constants and end-of-discharge voltages. An example of such a specification is “all internal power supply rails must decay to 37% of initial value within 300uS of RAPID_SHUTDOWN assertion, and to a voltage below 1.5V within 2mS of RAPID_SHUTDOWN assertion.”
A rapid shutdown implementation also requires supporting circuitry on the Carrier Board. Upon the assertion of RAPID_SHUTDOWN, the 12V (main) input power to the Module ***shall*** be removed by Carrier board circuitry and the input power pins ***shall*** be externally clamped to ground though a crowbar circuit located on the Carrier Board. This clamping circuit ***shall*** maintain a maximum resistance of 1 ohm and ***shall*** be active for a minimum of 2mS following the rise of RAPID_SHUTDOWN. The Module ***shall*** be designed with sufficiently low input capacitance to allow the input discharge specification to be met with a 1 ohm discharge resistance.
The Module design ***should*** prevent overheating or damage to any Module circuitry in the event that RAPID_SHUTDOWN is asserted without the removal of input power. This condition could occur due to malfunction of the Carrier Board support circuitry, or if the RAPID_SHUTDOWN signal is inadvertently asserted when the Module is installed on a Carrier Board that does not implement rapid shutdown support circuitry.
In some system implementations the power for both the Module and Carrier Board will fail shortly after the RAPID_SHUTDOWN signal is asserted. Therefore, the driving source for the RAPID_SHUTDOWN pin typically must charge an RC circuit on the Module that maintains crowbar assertion for several milliseconds following power failure. In order to charge this RC circuit rapidly, the RAPID_SHUTDOWN signal ***shall*** be sourced from a source impedance of 50 ohms or less. The RAPID_SHUTDOWN signal also ***shall*** have a rise time of <=1us and ***shall*** have a duration of >=20 uS. If the same RAPID_SHUTDOWN signal source is used to drive the Carrier Board input clamp circuitry, then any additional load from that Carrier Board circuitry needs to be considered in its design.

2.21.1 Rapid Shutdown Signal

Table 58: Rapid Shutdown Signal

Signal	Pin#	Description	I/O	Note
RAPID_SHUTDOWN	C67	Trigger for Rapid Shutdown. Must be driven to 5V though a <=50 ohm source impedance for ≥ 20 µs. Module has integrated PD resistor 10K ohm to GND. Carrier Board: N/C if not used.	I 3.3V CMOS 5.0V Suspend/5.0V	

Note:

2.21.2 Rapid Shutdown Signal Routing Guidelines

NA

2.21.3 Rapid Shutdown Signal Trace Length Guidelines

NA

2.22 General Purpose Input Output (GPIO)

GPI and GPO pins *may* be implemented as GPIO (Module specific). GPI and GPO pins *may* be implemented as SDIO (refer to Section 2.23 'SDIO'). If SDIO is supported the BIOS **may** be used to set the default state (SDIO or GPIO) of the GPIO.

2.22.1 GPIO Signals

Table 59: GPIO Signals, Pin Types, and Descriptions

Signal	Pin#	Description	I/O	Note
GPO0 / SDIO_CLK GPO1 / SDIO_CMD GPO2 / SDIO_WP GPO3 / SDIO_CD#	A93 B54 B57 B63	General Purpose Outputs for system specific usage. EC has internal pull up 75Kohm on the module. Carrier Board: Connect to GPO[3..0] N/C if not used	O 3.3V CMOS	1
GPI0 / SDIO_DAT0 GPI1 / SDIO_DAT1 GPI2 / SDIO_DAT2 GPI3 / SDIO_DAT3	A54 A63 A67 A85	General Purpose Input for system specific usage. EC has internal pull up 75Kohm on the module. Carrier Board: Connect to GPI[3..0] N/C if not used	I 3.3V CMOS	1

Note:

1. SOM-5871 support GPIO only.

2.22.2 Rapid Shutdown Signal Routing Guidelines

NA

2.22.3 Rapid Shutdown Signal Trace Length Guidelines

NA

2.23 SDIO *SOM-5871 is not support SDIO Interface.

Support for an SDIO interface is optional and added in R2.0. The SDIO signals are piggybacked on the

existing COM.0 General Purpose IO (GPIO) signals (refer to Section 2.22 'General Purpose Input Output'). An EEPROM bit is added so that the Carrier Board can define if the GPIO are used as GPIO or SDIO. The Module *may* use this information to allow boot from SDIO. Modules that support SDIO over GPIO *shall* implement the pin mapping based on the table below.

2.23.1 SDIO Signals

Table 60: SDIO Signals, Pin Types, and Descriptions

Signal	Pin#	Description	I/O	Note
GPO0 / SDIO_CLK	A93	SDIO Clock. With each cycle of this signal a one-bit transfer on the command and each data line occurs. This signal has maximum frequency of 48 MHz. Maps to GPO0. Carrier Board: Connect to SD Card CLK pin. N/C if not used.	O 3.3V CMOS	1
GPO1 / SDIO_CMD	B54	SDIO Command/Response. This signal is used for card initialization and for command transfers. During initialization mode this signal is open drain. During command transfer this signal is in push-pull mode. Maps to GPO1. Carrier Board: Connect to SD Card CMD pin. N/C if not used.	O 3.3V CMOS	1
GPO2 / SDIO_WP	B57	SDIO Write Protect. This signal denotes the state of the write-protect tab on SD cards. Maps to GPO2; used as an input when used for SD card support. Carrier Board: Don't pull up or pull down resistors on Carrier board. Connect to SD Card WP pin. N/C if not used.	I 3.3V CMOS	1
GPO3 / SDIO_CD#	B63	SDIO Card Detect. This signal indicates when a SDIO/MMC card is present. Maps to GPO3; used as an input when used for SD card support. Carrier Board: Connect to SD Card CD pin. N/C if not used.	I 3.3V CMOS	1

Signal	Pin#	Description	I/O	Note

GPI0 / SDIO_DAT0	A54	SDIO Data lines. These signals operate in push-pull mode. Maps to GPI0. Carrier Board: Connect to SD Card DAT0 pin. N/C if not used.	IO 3.3V CMOS	1
GPI1 / SDIO_DAT1	A63	SDIO Data lines. These signals operate in push-pull mode. Maps to GPI1. Carrier Board: Connect to SD Card DAT1 pin. N/C if not used.	IO 3.3V CMOS	1
GPI2 / SDIO_DAT2	A67	SDIO Data lines. These signals operate in push-pull mode. Maps to GPI2. Carrier Board: Connect to SD Card DAT2 pin. N/C if not used.	IO 3.3V CMOS	1
GPI3 / SDIO_DAT3	A85	SDIO Data lines. These signals operate in push-pull mode. Maps to GPI3. Carrier Board: Connect to SD Card DAT3 pin. N/C if not used.	IO 3.3V CMOS	1
SD_PWR_EN / RSVD	A86	SD card power enable pin from Module to Carrier Board. Active high. Carrier Board: Connector to SD power controller enable pin. N/C if not used	O 3.3V CMOS	1

Note:

1. SOM-5871 is not support SDIO Interface.

2.23.2 SDIO Topology Guidelines

NA

2.23.3 SDIO Signals Trace Length Guidelines

Figure 37: Topology for SDIO

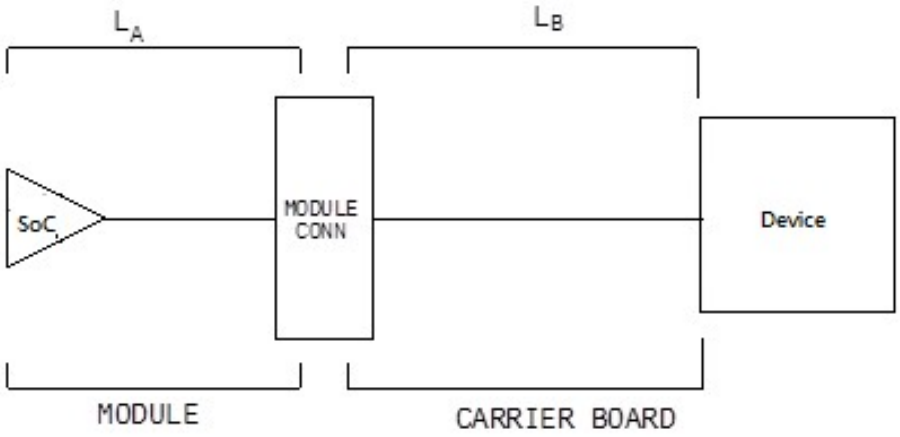


Table 61: SDIO Trace Length Guidelines

Parameter	Main Route Guidelines	Notes
Signal Group	SDIO	1
Single End	50Ω ±15%	1
Trace Spacing for Clock signal	Min. 15mils	1
Trace Spacing between Data Signals	Min. 10mils	1
LA	Please see the SOM-5899 Layout Checklist	1
LB	Carrier Board Length	1
Max length of LA+LB	7"	1
SD Card Clock and SD Card Data /CMD Signal Length Mismatch	Max. 100mils	1
Reference Plane	GND referencing preferred.	1
Via Usage	Try to minimize number of vias	1

Notes:

1. SOM-5871 is not support SDIO Interface.

3 Power

3.1 General Power requirements

COM Express calls for the Module to be powered by a single 12V power rail, with a +/-5% tolerance. The Mini format Modules are specified in COM.0 Rev. 2.1 to support a power input range of 4.75V to 20.0V. Advantech offer a wide range input even on Compact and Basic Modules. COM Express Modules may consume significant amounts of power – 25 to 50W is common, and higher levels are allowed by the standard. Close attention must be paid by the Carrier Board designer to ensure adequate power delivery. Details are given in the sections below.

If Suspend functions such as Suspend-to-RAM, Suspend-to-disk, wake on power button press, wake on USB activity, etc. are to be supported, then a 5V Suspend power source must also be provided to the Module. If Suspend functions are not used, the Module VCC_5V_SBY pins should be left open. On some Modules, there may be a slight power efficiency advantage to connecting the Module VCC_5V_SBY rail to VCC_5V rather than leaving the Module pin open.

Please contact Advantech for further details. Carrier Boards typically require other power rails such as 5V, 3.3V, 3.3V Suspend, etc. These may be derived on the Carrier Board from the 12V and 5V Suspend rails.

3.2 ATX and AT Power Sequencing Diagrams

A sequence diagram for an ATX style boot from a soft-off state (S5), initiated by a power button press, is shown in Figure 38 below.

A sequence diagram for an AT style boot from the mechanical off state (G3) is shown in Figure 39 below . In both cases, the VCC_12V, VCC_5V and VCC_3V3 power lines should rise together in a monotonic ramp with a positive slope only, and their rise time should be limited. Please refer to the ATX specification for more details.

Figure 38: ATX Style Power Up Boot – Controlled by Power Button

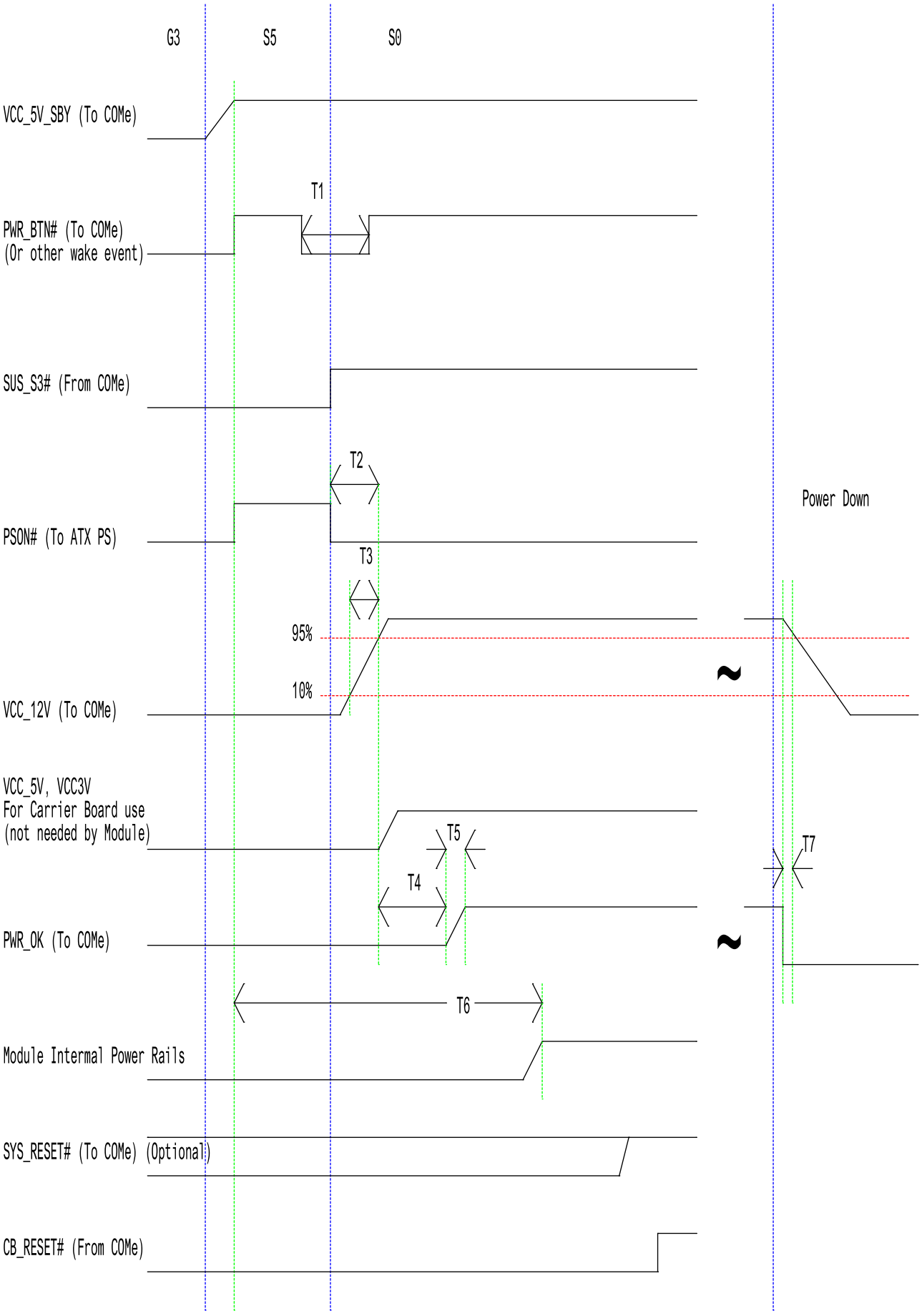


Figure 39: AT Style Power Up Boot

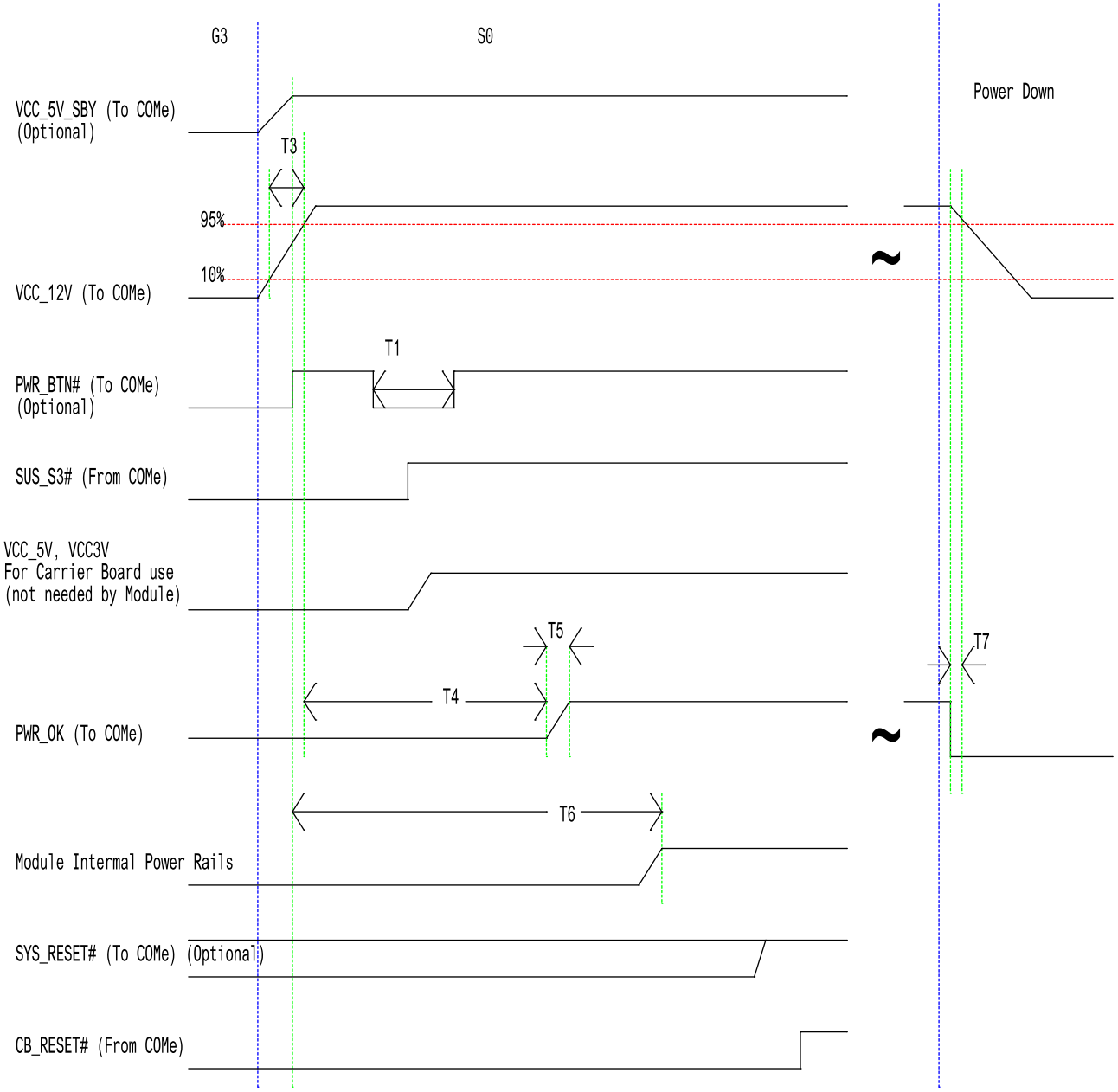


Table 62: Power Management Timings

Sym	Description	Min	Max
T1	Power Button	16ms	
T2	The power-on time is defined as the time from when PS_ON# is pulled low to when the VCC_12V, VCC_5V and VCC_3V output.		500ms
T3	VCC_12V rise time from 10% to 95%	0.1ms	20ms
T4	PWR_OK delay	100ms	
T5	PWR_OK rise time		10ms
T6	See Note 1		
T7	Power-down warning	1ms	

Note:

1. There is a period of time (T6 in Figure 38 and Figure 39 above) during which the Carrier Board circuits have power but the COM Express Module main internal power rails are not up. This is because almost

all COM Express internal rails are derived from the external VCC_12V and there is a non-zero start-up time for the Module internal power supplies.

Carrier Board circuits should not drive any COM Express lines during the T6 interval except for those identified in the COM Express Specification as being powered from a Suspend power rail. Almost all such signals are active low. Such signals, if used, should be driven low by open drain Carrier Board circuits to assert them. Pull-ups, if present, should be high value (10K to 100K) and tied to VCC_5V_SBY.

The line PWR_OK may be used during the T6 interval to hold off a COM Express Module boot. Sometimes this is done, for example, to allow a Carrier Board device such as an FPGA to be configured before the Module boots.

The deployment of Carrier Board pull-ups on COM Express signals should be kept to a minimum in order to avoid back-driving the COM Express signal pins during this interval.
Carrier Board pull-ups on COM Express signal pins are generally not necessary – most signals are pulled up if necessary on the Module.

3.3 Design Considerations for Carrier Boards containing FPGAs/CPLDs

Very often, the Carrier Board will contain custom FPGA or other programmable devices which require the loading of program code before they are usable. The Carrier Board designer needs to take the necessary precautions to ensure that his Carrier Board logic is up and running before the Module starts. Conflicts

can occur if the Module is powered on and allowed to run before devices on the Carrier Board are fully programmed and initialized. A typical example is an FPGA which includes a PCIe device. Such devices must be initialized and ready before the chipset on the Module performs link training and before the BIOS code performs enumeration of PCI devices. The Module should therefore be prevented from starting before Carrier Board devices are ready.

One method to achieve this is to delay assertion of the PWR_OK# signal to the Module until the Carrier Board initialization process has completed. Note that during the phase when the Carrier Board is powered and the Module is not powered there is potential for back drive voltages from the carrier to the Module.

Another possibility is to use the SYS_RESET# signal to delay Module start-up. However, depending on the Module implementation and the chipset used, SYS_RESET# may only be a falling edge triggered signal and not a low active signal as was originally intended. In that case, asserting SYS_RESET# may not hold the Module in the reset state. Also, PCIe link training will occur regardless of the reset signal state for some chipsets.

Please refer to the COM.0 R2.1 specification (Power and System Management section) for more details and check the Module provider's documentation for their implementations of these signals.

4 Electrical Characteristics

4.1 Absolute Maximum Ratings

Table 63: Absolute Maximum Ratings

SOM-5871		MIN	MAX	UNIT
Power	VIN	8.5 (9-5%)	20 (19+5%)	V
	VSB	4.75 (5-5%)	5.25 (5+5%)	V
	RTC Battery	2.4	3.3	V

4.2 DC Characteristics

Table 64: Input Power Consumption of 8.5V / 12V / 20V

AMD Ryzen Embedded V1807B with Radeon Vega Gfx 3.35GHz					
Power Plane	Maximum Power Consumption				
Symbol	S0		S3	S5	G3
+VIN (+12V)	TAT turbo CPU80%+GPU100%	66.3483W			
	Idle	11.1397W			
+VIN (+8.5V)	TAT turbo CPU80%+GPU100%	64.3225W			
	Idle	12.1444W			
+VIN (+20V)	TAT turbo CPU80%+GPU100%	67.1199W			
	Idle	14.9999W			
+V5SB_CB			1.3363W	0.9558W	
RTC Battery					4.82uA

Note:S0 power consumption is Intel TAT tools setting CPU-All 100% Turbo On.

4.3 Inrush Current

Table 65: Inrush Current

Power Plane	Maximum	
Symbol	G3 to S5	S5 to S0
+V5SB_CB	1.04382A	
+VIN (+12V)		5.1954 A
+VIN (+8.5V)		7.3966 A
+VIN (+20V)		3.26578 A